Competitive Persuasion with a Noisy Judge

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Abstract

Can a firm choosing between two agents gain from judging them with error? I show that she can when the agents, each good or bad, design their own evidence. Noiseless judging fully rewards even a narrow lead, so agents conceal bad news; suitably shaped noise devalues a narrow lead, so concealing pays less. For a broad class of noise, the equilibrium is unique: each agent sometimes concedes that he is bad and otherwise sends one pooled signal; for a high prior both reveal fully, which noiseless judging never induces. Moderate noise turns out to be best: it curbs concealment but never overturns a difference between the signals agents send. Within this class, the best judge beats a noiseless one unless agents are very likely bad and attains the first best at priors of at least one half. Beyond this class, among rules of the signal gap, only a linear one attains the first best for every prior, but it leaves agents indifferent among all signals; knowing whether agents are promising or unpromising suffices for a unique fully revealing equilibrium.

Keywords: flexible information disclosure, competition, information design

JEL Classification: C72, D82, D83

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1. Introduction

A capital committee must fund one of two projects this quarter. Each sponsor decides how rigorous a pilot to run before the meeting, knowing that the committee's verdict will turn partly on things that have nothing to do with the projects: who presents second, how long the meeting has run. The committee would like to see past all of that. This paper asks whether it should want to.

Such imperfect verdicts, biased or merely arbitrary, are well recognized. In Scharfstein and Stein (2000), divisional rent-seeking leads to "socialist" internal capital markets, in which weaker divisions are subsidized by stronger ones. In Prendergast and Topel (1996), evaluators who lack objective performance measures act on personal preferences, favoring some subordinates beyond their true performance. And Pier et al. (2018) find essentially no agreement among experienced reviewers rating the same NIH grant applications, all of which were eventually funded: ratings of the same application were no more similar than ratings of different ones. The usual response is to engineer such imperfections away, through more reviewers, more structured evaluations, and more objective criteria. A few research funders have nonetheless moved the other way: they now fund by lot among proposals that clear a quality bar, citing reviewer bias, reviewer burden, and the hope of bolder proposals (Fang and Casadevall, 2016; Heyard et al., 2022; Röbbecke and Simon, 2023). Only the hope of bolder proposals concerns how applicants respond, and none concerns what the competitors choose to reveal about themselves, the channel I study.

The question is whether a decision-maker should eliminate all noise from her evaluation when the competitors know that it is noisy and adapt their reports to it. It arises whenever three features come together. First, a single prize is at stake, with no price attached: the decision-maker cannot pay more for the better competitor, and a close second earns nothing. Second, the competitors decide what and how much to

reveal about themselves after learning how the decision-maker will choose between them. Third, the decision-maker can make the verdict more or less accurate, so the competitor with the stronger evidence need not always win. I call the decision-maker *the firm*, a noisy judge when her verdict carries error, and the competitors *agents*. Each agent is either a good or a bad type and simultaneously commits, at no cost, to a signal, that is, to a distribution of the posterior probability that he is good; the firm observes the resulting posterior with a random error and selects the agent with the higher score. For many standard noise distributions and equal priors, this is also what she would do on seeing only the scores, so committing to the rule costs her nothing there.

My key observation is that noise of a suitable shape weakens each agent's incentive to conceal. An agent's chance of winning depends only on how far he looks ahead of his rival, in the same way for both agents; I call this dependence the firm's selection rule, and every such rule is generated by some noise. A noiseless firm always selects the agent who looks better, so her rule is a step: a narrow lead is as good as a wide one, and an agent revealed as bad loses unless his rival is too. Each agent therefore spreads his posterior to land just above his rival's, blending his bad type with his good one: Boleslavsky and Cotton (2015) show that both agents reveal only partially, randomizing over a continuum of posteriors, and the firm obtains less than 80% of her first best whenever agents are good with probability below one third.

With noise, the rule instead rises gradually with the lead. A narrow lead is worth little, conceding bad news no longer means certain defeat, and the shape of the noise decides whether an agent gains more from gambling on being revealed as good or from pooling his evidence, and so how much information reaches the firm. The price is accuracy: a rule near one half is close to a coin flip, so the firm passes over the agent who looks better more often, and noise trades selection accuracy for information.

An example. Suppose that each agent is equally likely to be good and bad. Under

the noiseless rule, each agent's posterior is uniformly distributed on $[0, 1]$, so the firm sometimes selects a bad agent over a good one and obtains about 89% of her first best. Now suppose the firm scores each agent by his signal plus an independent draw from $U[0, 1]$. Against a rival who reveals his type, an agent's chance of winning is linear in his own signal, so revealing his type is a best reply, if not the only one; against any other signal it is not, and, since the agents' game is constant-sum, full revelation by both agents is the unique equilibrium. Since a revealed good type always outscores a revealed bad one, the noise only decides between agents of the same type, and the firm attains her first best.

Main results. The example relies on uniform noise and an even prior; the results below show how far this extends. The first set of results, in Section 3, characterizes the equilibrium when the difference between the two evaluation errors has a concave density with a single peak over possible signal gaps, which I call a *well S-shaped* selection rule; normal and logistic noise qualify if its variance is large enough. Under such a rule, whatever his rival reveals, an agent's chance of winning first accelerates, then slows as his own signal rises, so below some signal he gains from gambling and above it from playing safe; concavity preserves this shape when the rival randomizes. Each agent thus uses a *bad news* signal: he reveals the bad type with some probability and otherwise sends one pooled signal, with full revelation and pooling as edge cases.

The equilibrium is unique whenever the priors are equal, and for every pair of priors unless the density is linear next to its peak. In the symmetric case, a single statistic governs it, the elasticity of the normalized selection rule at the pooled signal: an agent concedes bad news with unconditional probability equal to its inverse, and both agents reveal fully once the prior is high enough. With asymmetric priors (Section 6), each agent reveals more when his rival is better, and when one agent is much better, the favorite reveals nothing while the underdog reveals fully.

The second set of results, in Section 4, concerns the firm's utility with equal priors. Does the firm gain from better agents? Under every well S-shaped rule she does, although better agents may reveal less, since an agent's own strength can outweigh his rival's; when the density of the noise is convex instead, agents reveal good news, and better agents can hurt her. With asymmetric priors, a better agent can hurt the firm only where both agents concede bad news. Within the class, the firm's best rule is generated by i.i.d. uniform shocks and sits exactly at its edge. It beats the noiseless benchmark once agents are good with probability above roughly 21%, and delivers the first best once they are at least as likely good as bad; a continuum of types changes this.

This does not mean that any noise is worth keeping. Scaling the uniform shocks shows the trade-off: more noise leaves the signals unchanged and only blurs selection, while less noise lowers her utility and pushes the equilibrium toward the noiseless one once signals stop being binary. The firm does best at the least noise that keeps signals binary, which lies at or beyond the class's edge. There, in the equilibrium I construct, noise never overturns a difference between signals agents send, so a moderately noisy judge beats both a noiseless and an excessively blurry one.

The third set of results, in Section 5, asks when the firm can attain her first best with any rule. With binary types and equal priors, she needs full revelation and a rule that always picks a revealed good agent over a revealed bad one. For a known prior, the rules that implement full revelation have rather vague structure, so I ask which rules do so robustly. If agents may be promising or unpromising – with priors on either side of one half – only one rule depending on the gap between signals works: the linear one, which leaves agents indifferent among all signals, so robustness to the prior costs uniqueness. If she knows whether agents are promising or unpromising, a rich family of rules makes full revelation the unique equilibrium. For promising agents, i.i.d. shocks can generate some of these rules; for unpromising agents, they

cannot, since under i.i.d. shocks full revelation is never an equilibrium once an agent is more likely bad than good.[1]

Section 7 collects extensions. Beyond the well S-shaped class, the equilibrium stays unique, with at most two signals per agent, under a weaker condition on the noise; further from the class, agents randomize over more signals. With a continuum of types, each agent still reveals his type below a threshold and pools all types above it, but under a convex prior CDF, the noiseless rule attains the first best, which no noisy rule can, and under a concave one, the ranking depends on how dispersed the prior is. A head start for one agent keeps the evaluation noiseless but breaks anonymity, induces both agents to concede bad news, and outperforms every well S-shaped rule at low priors.

1.1 Literature review

This paper contributes to the Bayesian persuasion literature (Kamenica and Gentzkow, 2011; Kamenica, 2019), in particular to its strand on competing senders. Most results of Gentzkow and Kamenica (2017) require a Blackwell-connected information environment, which fails when each sender can only reveal information about himself, as here, and their comparison with collusion is vacuous, since the agents' game is constant-sum. Each agent's best reply solves a single-sender problem with an S-shaped value function, so it is an upper censorship as characterized by Kolotilin et al. (2022), whose result also gives the equilibrium under a general prior. To a single agent, the rule acts like a receiver with a privately known threshold (Kolotilin et al., 2017), here the rival's signal plus noise, and for a single sender, making a bell-shaped threshold density more diffuse leads to more information (Curello and Sinander, 2025). Here, the threshold's distribution is an equilibrium object, set by the rival's signal.

[1]These limits are specific to rules of the signal gap: comparing squared signals, which no additive noise generates, makes full revelation dominant at every prior.

The closest paper is Boleslavsky and Cotton (2015), part of the literature on grading policies (Ostrovsky and Schwarz, 2010; Chan et al., 2007; Popov and Bernhardt, 2013; Bar et al., 2012): two schools design grades, and an evaluator hires the graduate with the higher posterior. I add evaluation noise to this game, and its shape decides what agents reveal. Under the noiseless rule, the equilibrium is unique, posteriors are uniform on an interval, possibly with atoms at 0 and 1, and schools never reveal fully. Informativeness increases with the number of senders, both in this setting, where it becomes full in the limit (Au and Kawai, 2020), and when senders also compete in prices (Hwang et al., 2026). In search markets, whether competition raises informativeness depends on what sellers observe and how buyers search (Board and Lu, 2018; Au and Whitmeyer, 2023; He and Li, 2023).

An imperfect evaluation can also benefit the evaluator by strengthening investment incentives, as strategic grading does in Boleslavsky and Cotton (2015) and a noisy signal of universities' relative quality does in Boleslavsky and Li (2026). Here the gain comes through information instead: a suitably noisy evaluation makes signals more informative. In this, noise plays a role familiar from other settings: noise in prices preserves the incentive to acquire information that fully revealing prices would destroy (Grossman and Stiglitz, 1980), and noise in a cheap-talk channel makes the receiver less responsive to each message, which curbs the sender's exaggeration and improves communication (Blume et al., 2007). Closer in spirit, an organization that can fund only one of two proposals induces more informative evidence than an unconstrained one and is better off unless both are ex-ante strong (Boleslavsky and Cotton, 2018), and moderate attention costs can make a rationally inattentive consumer select the better of two disclosed products more often than when attention is free (Jain and Whitmeyer, 2026). In both, the gain comes from a constraint on the receiver; I instead treat the evaluation error as a primitive of the selection rule, whose shape determines

the equilibrium and how much information the firm extracts.

My normative results also relate to the idea that observing an agent more closely can hurt the principal once the agent responds to it (Prat, 2005), and that a receiver can extract more information by committing to use it less: random costs of approval make a single sender indifferent among signals and elicit full revelation, at the price of burned utility (Tsakas et al., 2021), and when agents manipulate their data at a cost, so do a less responsive allocation rule, a test that passes some noncompliant agents, and a score that underweights some features (Frankel and Kartik, 2022; Perez-Richet and Skreta, 2022; Ball, 2025). Extra noise in an already noisy channel can benefit the sender and sometimes both players (Tsakas and Tsakas, 2021), and coarse grades can inform the public better by inducing participation (Harbaugh and Rasmusen, 2018). Here signals are costless and flexible, and noise works only through competition, by changing the curvature of each agent's value function; it costs the firm nothing when it decides only between agents of the same type. Closest to the firm's commitment, Gao and Luo (2026) let a designer commit to a mechanism mapping each sender's experiment and posterior into a random action: robust mechanisms depend only on posteriors and induce full revelation, and with competing senders the efficient allocation is robustly implementable. Bergemann et al. (2023) show that a receiver uncertain about the sender's preferences optimally commits to a quota. Here the firm commits only to a noisy comparison of the signals, which makes robustness costly (Section 5) and, with a single agent and an absolute standard, caps her payoff (Subsection 7.6). A single receiver who could commit to any decision rule would need no noise, but a firm comparing agents lacks the absolute standard that makes it unnecessary. Randomized choice also serves other purposes: breaking ties in school choice (Abdulkadiroğlu and Sönmez, 2003; Erdil and Ergin, 2008), saving on costly signaling (Hoppe et al., 2009), economizing on verification (Border and Sobel, 1987), deterring gaming (Ederer

et al., 2018), and keeping improper reasons out of selection (Stone, 2011; Fang and Casadevall, 2016).

The agents' game is also a contest in which each agent chooses a distribution with a fixed mean, and noise gives it a difference-form success function (Lazear and Rosen, 1981; Hirshleifer, 1989; Che and Gale, 2000), the counterpart of the ratio form of Tullock (1980), with more noise playing the role of lower decisiveness. When contestants choose the riskiness of their performance, limiting competition helps a principal select the better one (Hvide and Kristiansen, 2003; Fang and Noe, 2022), and a noisy selection rule is a new instrument with the same effect. In effort contests, a noisy contest can also be optimal (Wang, 2010; Morgan et al., 2022; Letina et al., 2023), and the shape of the noise matters (Drugov and Ryvkin, 2020); see Fu and Wu (2019) for a survey.

2. Model

Agents. There are 2 agents and the firm (from here on she). Each agent's type θ_i is drawn independently from $\{0, 1\}$, and let p_i be the probability of $\theta_i = 1$. From here on, I refer to p_i as a *prior*. For notational simplicity, if $p_1 = p_2$, I refer to them simply as p.

Information and Timing. Agents play a simultaneous-move game in which, before the realization of his type, each agent i commits to a signal structure. Then, after the realization of each agent's type θ_i, the firm receives a signal s_i according to that structure. Since the prior is binary, I can re-identify the reported signal with its posterior mean $s_i := \mathbb{E}[\theta_i|s_i]$, and hence WLOG agent i chooses a distribution of posterior means G_i (from here on signal). Let $\mathcal{G}(p_i)$ be the set of feasible signals for a prior p_i. Since the prior is binary, $\mathcal{G}(p_i) := \{G_i : supp(G_i) \subseteq [0, 1], \int_0^1 s_i dG_i(s_i) = p_i\}$, see Gentzkow and Kamenica (2016).

Selection. Agents select signal structures knowing the firm's *selection rule*: a function $v : [-1, 1] \mapsto [0, 1]$, such that $v(x) - 0.5$ is odd, increasing and continuous on $[-1, 1]$.[2] If the firm observes signals s_1, s_2, the respective probabilities of agents $1, 2$ being selected are $(v(s_1 - s_2), v(s_2 - s_1))$, and since $v(x) - 0.5$ is odd, $v(s_1 - s_2) + v(s_2 - s_1) \equiv 1$ for all s_1, s_2, so that the firm always selects exactly one agent. The oddness of $v(x) - 0.5$ also means that the selection rule treats the agents anonymously; Subsection 7.5 discusses biased rules. The selection rule is assumed to be exogenously given to the firm for most of the sections; however, I endogenize the choice of $v(x)$ in Section 5.

Noisy evaluation. Such selection rules arise naturally when the firm judges the agents with noise. Suppose that, instead of observing s_i without noise, the firm observes agent i's *score* $s_i + \varepsilon_i$, where ε_i is a random shock, independent of s_i, s_j, reflecting the idiosyncrasies of whoever is judging – a particular committee member's mood, the order in which proposals were read, an unmodeled personal rapport. The noise is additive on the scale of the firm's payoff, the posterior probability that an agent is good; noise on another scale, such as log-odds, would give rules that depend on more than the gap between signals, which I leave aside. Taking the scores at face value, the firm selects the agent with the higher score. Suppose that $\xi := \varepsilon_j - \varepsilon_i$ has a continuous distribution, symmetric around 0; symmetry holds, for instance, whenever the shocks are exchangeable, i.e., $(\varepsilon_i, \varepsilon_j)$ and $(\varepsilon_j, \varepsilon_i)$ have the same distribution. Then she selects agent i with probability

$$\mathbb{P}(s_i + \varepsilon_i > s_j + \varepsilon_j) = \mathbb{P}(\xi < s_i - s_j) = v(s_i - s_j),$$

where $v(x)$ is the CDF of ξ restricted to $[-1, 1]$, and $v(x)$ is a selection rule: it is increasing as a CDF, continuous as ξ has no atoms, and $v(x) - 0.5$ is odd as ξ is

[2] Throughout, increasing and decreasing are meant weakly; strict monotonicity is stated explicitly.

symmetric. Conversely, every selection rule arises in this way, for example from $\varepsilon_j = -\varepsilon_i = \xi/2$, where ξ is symmetric with CDF $v(x)$ on $[-1, 1]$. These shocks are exchangeable but not independent, and independent shocks generate only some selection rules (Corollary 6). I usually give $v(x)$ explicitly, and otherwise say that $v(x)$ *is generated by* a distribution of $(\varepsilon_i, \varepsilon_j)$ with these properties.

Example 1. If $v(x)$ is generated by ε_i i.i.d. Gumbel or normal, then $v(x)$ is the CDF of a logistic or normal distribution, respectively.

Payoffs. WLOG, each agent receives 1 if selected and 0 otherwise. If the firm selects agent i, she receives θ_i, so her ex-ante expected utility equals the reported signal s_i. The shocks ε_i that may generate the selection rule do not enter the firm's utility.

Commitment. The firm commits to her selection rule. The commitment need not be deliberate: some noise is unavoidable and undetectable, so the firm cannot filter it out, and everyone understands how it shapes the verdict. Nor, for the leading noise distributions, is it a restriction: if the shocks are i.i.d. with a log-concave density, such as normal, Gumbel or uniform, and both agents use the same signal, as in the equilibrium for equal priors, then by the monotone likelihood ratio property the firm's expected payoff $\mathbb{E}[s_i|s_i + \varepsilon_i]$ is non-decreasing in the score, so selecting the higher score is optimal even without commitment. For asymmetric priors, and for rules that i.i.d. shocks cannot generate or that break anonymity, commitment can bind.

Agents' Equilibrium. Since θ_i is unobservable to agent i, he should simply maximize the ex-ante probability of being selected when choosing a signal structure $G_i(s_i)$. Hence, an equilibrium of the agents' informational game is a tuple (G_1^*, G_2^*), where

$$G_i^* \in \arg\max_{G_i \in \mathcal{G}(p_i)} \int_0^1 \int_0^1 v(s_i - s_j) dG_j^*(s_j) dG_i(s_i).$$

For any priors and selection rule, an equilibrium exists by Sion's minimax theorem, since the game is constant-sum, as $v(x) - 0.5$ is odd, payoffs are linear and continuous in G_i, and $\mathcal{G}(p_i)$ is a compact convex subset of a topological vector space (Proposition 1 in Kleiner et al. (2021)). Moreover, all equilibria are pairs of max-minimizer strategies and form a product set.

Interpretation. The model is deliberately abstract and says nothing about a price: agent i's type θ_i is simply the value the firm realizes from picking i, with no wage or other transfer attached to that choice. This fits any setting in which a decision-maker allocates one indivisible prize between two competing options and cannot pay more to secure a better one. My running example is **internal capital budgeting**: a firm's capital committee (*the firm*) must fund exactly one of two project proposals this cycle, and each project's sponsor (*agent i*) controls how much hard evidence – the scope of a pilot, the rigor of a feasibility study, which numbers are led with – to place in front of the committee. The same primitives fit a **grant panel** choosing between two proposals, and **grading**, as in Boleslavsky and Cotton (2015): a university chooses a grading policy for a graduate who competes for a single place and is evaluated by an admissions office. In this reading, the grading policy is the signal, the prior p_i is the strength of the school's cohort, and the noise is the office's judgment of what a transcript leaves out, such as interviews and essays; the school sets its policy before it learns how good its graduates are, which is the commitment the model assumes. I use *the firm* and *agent* as generic labels and switch between these readings wherever a concrete example helps. The binary prior simplifies the analysis and sharpens the results; Subsection 7.1 relaxes it.

Under any of these readings, the signal $G_i(s_i)$ is the design of the hard information about the project, applicant or student – the scope and pre-registration of a pilot study,

the preliminary results in a grant proposal, a school's grading policy – summarized by the distribution of the posterior s_i that the agent is good. Equivalently, as in Boleslavsky and Cotton (2015), each type generates a more primitive random report – a grade, a pilot outcome, a preliminary result – whose distribution satisfies the monotone likelihood ratio property, so that, by Bayes' rule, the posterior increases in the report. I read θ_i as fit with what the firm needs, which an agent can hardly assess; what he controls, before any result exists, is how much evidence there will be. If he knew more than the firm, this would be a signaling problem, which I do not study.

The selection rule $v(x)$ translates the two signals into the ex-ante probability of being picked when the firm selects the agent with the higher score. The shocks ε_i that generate $v(x)$ are imperfections of the kind documented in the introduction: idiosyncrasies of the evaluator that are uncorrelated with the substance of the signal and with the firm's realized payoff, but still move her decision. An agent may also have outcome-relevant characteristics that are uncorrelated with his signal and influence the firm's decision, but I set these aside for two reasons. First, the agents care only about the selection rule, not about whether the shock also affects the firm's payoff. Second, comparing the firm's utility across selection rules is easier to interpret when the shock is pure evaluation noise.

3. Well S-Shaped Selection Rules

In this section, I study the agents' game under the additional assumption that the selection rule $v(x)$ has a concave density with a unique mode. This is sufficient for an equilibrium with binary support strategies to exist for all $p \in (0, 1)$, and for it to be unique when the priors of the agents are equal. I then study the symmetric equilibrium: how signals change with the priors, when full revelation is an equilibrium, and which characteristic of $v(x)$ drives the equilibrium.

3.1 Preliminary definitions and lemmas

I begin with the key definition of this and the following sections.

Definition 1. A function $v : [-1, 1] \to [0, 1]$ is called *well S-shaped* if $v(x) - 0.5$ is odd and increasing, and $v'(x)$ is concave on $[-1, 1]$ and uniquely maximized at $x = 0$.

Since $v(x) - 0.5$ is odd, $v'(x)$ is even, so a concave $v'(x)$ is always maximized at $x = 0$. Requiring this maximum to be unique only rules out densities that are flat around 0, including the linear selection rule. Without this requirement, the equilibrium may fail to be unique: if $v(x)$ is linear on some $[-b, b]$, then for priors (p, p) with $p \leq \frac{b}{2}$, both $(\{p\}; \{p\})$ and $(\{0, 2p\}; \{0, 2p\})$ are equilibria. If $v(x)$ is generated by ε_i, ε_j, well S-shapedness means that ξ has a concave density on $[-1, 1]$ with a unique mode, which holds for the distributions of Example 1 if their tails are wide enough.

Example 2. For each of the two classes from Example 1, there is a threshold $\overline{V}$ s.t. $v(x)$ is well S-shaped if $\mathrm{Var}[\varepsilon_i] \geq \overline{V}$. So is the rule generated by i.i.d. $\varepsilon_i \sim U[0, u]$ with $u \geq 1$, although its triangular density is linear on $[0, 1]$.

Every result for equal priors holds on the whole well S-shaped class, including the firm's best rule, generated by $U[0, 1]$ shocks, whose density is linear next to its mode. For unequal or non-binary priors, uniqueness also requires that $v'(x)$ is *not linear near* 0, i.e., not linear on any $[0, \varepsilon]$, which makes every best reply unique (Lemma 1).

Let $\overline{v}(s_i, G_j) := \int_0^1 v(s_i - s_j) dG_j(s_j)$, often written $\overline{v}(s_i)$, be the *value function* of agent i. By choosing a signal G_i, agent i solves a Bayesian persuasion problem as in Kamenica and Gentzkow (2011):

$$\max_{G_i \in \mathcal{G}(p_i)} \int_0^1 \overline{v}(s_i, G_j) dG_i(s_i) \tag{BP}$$

I call G_i *an optimal signal* for $(\overline{v}(s_i), p_i)$ if it solves (BP), and *the optimal signal* if, in addition, no other G_i solves it.

To find an optimal signal, I use the *concavification* $\overline{v}^{cv}(s_i)$ of $\overline{v}(s_i)$ for a fixed G_j, the pointwise lowest concave function majorizing $\overline{v}(s_i)$. Then G_i is an optimal signal for $\overline{v}(s_i)$ if and only if it is optimal for $\overline{v}^{cv}(s_i)$ and its support is contained in the *concavification support* $supp(\overline{v}^{cv}) := \{s_i : \overline{v}(s_i) = \overline{v}^{cv}(s_i)\}$. For the concave $\overline{v}^{cv}(s_i)$, this is simple: if it is linear on some interval around p_i, any $G_i \in \mathcal{G}(p_i)$ supported in that interval is optimal; otherwise, the optimal signal has $supp(G_i) = \{p_i\}$ (pooling), see Kamenica and Gentzkow (2011).

A well S-shaped $v(x)$ helps because if $v'(x)$ is concave on $[-1, 1]$, then $\overline{v}'(s_i)$ is concave on $[0, 1]$ for every G_j, which gives $\overline{v}(s_i)$ the following shape.

Definition 2. $\overline{v}(s)$ is called *S-shaped* on $[a, b]$, if there exists some $c \in [a, b]$ s.t. $\overline{v}(s)$ is convex on $[a, c]$ and concave on $[c, b]$.

This is the standard definition of an S-shaped curve. As $v(x)$ is continuously differentiable, so is $\overline{v}(s_i)$, and then $\overline{v}(s_i)$ is S-shaped if and only if $\overline{v}'(s_i)$ is quasi-concave. Since $\overline{v}'(s_i)$ is even concave, $\overline{v}(s_i)$ is S-shaped for every well S-shaped $v(x)$ and every G_j. As quasi-concave functions are not closed under addition, there is no simple extension of the well S-shaped class for which $\overline{v}(s_i)$ is always S-shaped. Characterizing the equilibrium, however, needs this shape only against the rival's equilibrium signals. Using this, Proposition 12 goes beyond the class, giving a weaker sufficient condition for a unique bad news equilibrium, though one that is less explicit.

So under a well S-shaped $v(x)$, an agent's marginal gain from a higher signal (posterior mean) is concave, and against any signal G_j, $\overline{v}(s_i)$ has at most two regions of different convexity. Hence its concavification consists of at most two parts, which strongly restricts the equilibrium signals, characterized in Lemma 1 (ii).

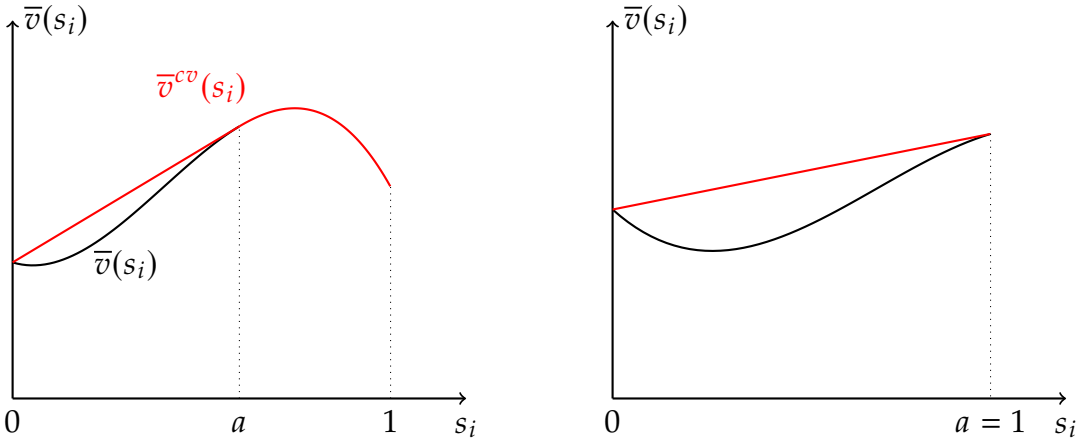
Lemma 1. *If $v(x)$ is well S-shaped, then*

(i) $\overline{v}(s_i, G_j)$ is S-shaped on $[0, 1]$ for all $p_j \in (0, 1)$ and $G_j \in \mathcal{G}(p_j)$.

(ii) An optimal signal for $(\overline{v}(s_i), p_i)$ *is* $\{0, a\}$ *if* $p_i < a$ *and* $\{p_i\}$ *otherwise, where* $a = \max \arg\max_{s_i \in [0,1]} slp(s_i)$, $slp(s_i) := \frac{\overline{v}(s_i) - \overline{v}(0)}{s_i}$ *and* $slp(0) := \overline{v}'(0)$. *If* $v'(x)$ *is not linear near* 0, *it is the optimal signal.*

Remark 1. I identify a signal G_i with its support whenever $|supp(G_i)| \leq 2$.

Figure 1: Illustration of Lemma 1 (ii)



(a) $supp(G_i) = \{0, a\}$ or $\{p_i\}$ (b) $supp(G_i) = \{0, 1\}$

This is not the image of $\overline{v}(s_i)$, *but of* $\overline{v}(s_i) - bs_i$ *for some* $b > 0$ *for clarity. Since* $cav(f(x) + bx) = cav(f(x)) + bx$ *for all* x, b, *this does not influence the optimal signal.*

Lemma 1 thus restricts an agent's optimal signal to a very limited class, the *bad news* signals: the agent reveals the low type with some probability and otherwise sends a pooled signal, with full revelation ($\{0, 1\}$) and pooling ($\{p_i\}$) as the edge cases. Binary support itself is not special: two-point signals are the pure strategies of an agent, every other signal is their mixture, and against any continuous value function some signal with at most two support points is a best reply. What a well S-shaped $v(x)$ adds is that, unless pooling is optimal, such a best reply has 0 in its support, and that it is unique if $v'(x)$ is not linear near 0. Moreover, the point a depends only on the rival's signal, and the agent's own prior only determines whether he pools at p_i or plays $\{0, a\}$. The concavity of the density is what makes agents concede bad news: under the mirror-image, reverse S-shaped rules of Subsection 7.2, they reveal good news instead.

The first claim was shown above. For the second, the concavification of an S-shaped $\overline{v}(s_i)$ coincides with the secant from $\overline{v}(0)$ to $\overline{v}(a)$ on $[0, a]$ and with $\overline{v}(s_i)$ on $[a, 1]$. Hence $\{0, a\}$ is an optimal signal if $a > p_i$, and $\{p_i\}$ if $a \leq p_i$, and if $v'(x)$ is not linear near 0, these are **the** optimal signals. Figure 1 illustrates this claim.

By the Bayesian constraint, positive scaling of $v(x)$ and adding a linear function to it do not change the concavification support, and hence do not change equilibria for any (p_1, p_2). I represent each such equivalence class of selection rules by the normalization

$$\tilde{v}(x) := \frac{v(x) - 0.5 - x \cdot v'(0)}{|v(1) - 0.5 - v'(0)|}$$

It satisfies $\tilde{v}(0) = \tilde{v}'(0) = 0$ and $\tilde{v}(1) = -1$, which simplifies the exposition, and it is well defined for every well S-shaped $v(x)$: as $v'(x)$ is uniquely maximized at $x = 0$, $v(1) - 0.5 = \int_0^1 v'(x)dx < v'(0)$.

The key characteristic of $\tilde{v}(x)$ is its elasticity $\mathcal{E}_{\tilde{v}}(x) = \frac{x\tilde{v}'(x)}{\tilde{v}(x)}$. Since $\tilde{v}(0) = 0$, $\mathcal{E}_{\tilde{v}}(a)$ is the ratio of the slopes of the tangent to $\tilde{v}(x)$ at $x = a$ and of the secant from 0 to a, so it measures how curved $\tilde{v}(x)$ is on $[0, a]$. Hence it governs the curvature of $\overline{v}(s_i)$ and the location of the support point a from Lemma 1, as the results below show.

3.2 Main equilibrium results

3.2.1 Full revelation

A natural first question is whether the firm can ever get the first best information, full revelation by both agents, in equilibrium. Under a well S-shaped $v(x)$, the answer is simple: both priors must be high enough.

Definition 3. For a given well S-shaped $v(x)$, let $\overline{p} := 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)} = 1 - \frac{1}{|\tilde{v}'(1)|}$.

Proposition 1. *Let $v(x)$ be well S-shaped. Then both agents playing full revelation is an equilibrium if and only if $p_1, p_2 \geq \overline{p}$, and then it is the unique equilibrium unless $v'(x)$ is*

linear near 0 *and* $p_i > p_j = \overline{p}$ *for some* $i \neq j$.

Competing with a sufficiently good rival induces an agent to reveal fully, and the proof of the condition $p_1, p_2 \geq \overline{p}$ is built on exactly that. If agent j plays $\{0, 1\}$, the normalized value function of agent i is

$$\overline{v}(s_i) = (1 - p_j)\tilde{v}(s_i) + p_j\tilde{v}(s_i - 1) = (1 - p_j)\tilde{v}(s_i) - p_j\tilde{v}(1 - s_i).$$

By Lemma 1, it is S-shaped, so $\{0, 1\}$ is optimal if and only if the secant from 0 to 1 is weakly flatter than the tangent at $x = 1$ (see Figure 1b):

$$\overline{v}(1) - \overline{v}(0) \leq \overline{v}'(1) \Leftrightarrow \tilde{v}(1) \leq (1 - p_j)\tilde{v}'(1) \Leftrightarrow p_j \geq 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)}.$$

For uniqueness, as the game is constant-sum, all equilibria form a product set, so any signal played in equilibrium is a best reply to full revelation. An S-shaped $\overline{v}(s_i)$ can touch its secant from 0 to 1 at some $s \in (0, 1)$ only by coinciding with it on $[s, 1]$, which non-linearity of $v'(x)$ near 0, or a tangent at 1 strictly steeper than the secant ($p_j > \overline{p}$), rules out, and then full revelation is the only best reply. For $p = \overline{p}$, the best reply need not be unique, but uniqueness follows from Theorem 1.

Remark 2. Uniqueness can fail when both conditions of the exception hold. For $v(x) = 0.5 + x - 0.5x|x|$, $v'(x)$ is linear near 0 and $\overline{p} = 0.5$; for $(p_1, p_2) = (0.75, 0.5)$, agent 1's value function against $\{0, 1\}$ is linear, and $\{0, 1\}$ remains a best reply to any $G_1 \in \mathcal{G}(0.75)$, since $\mathbb{E}_{G_1}[s_1^2] \geq 0.5$. So $(G_1; \{0, 1\})$ is an equilibrium for every such G_1.

Proposition 1 contrasts with the noiseless benchmark[3] of Boleslavsky and Cotton (2015), in which schools never use a fully informative grading policy. This is not a feature of well S-shaped rules alone: Section 5 shows that full revelation can be reached

[3]The noiseless rule $v(x) = 0.5 + 0.5 \cdot sign(x)$ is not a selection rule, as it is discontinuous at 0; Appendix D gives its equilibria.

by selection rules far beyond this class.

Since $\tilde{v}'(x)$ is concave with $\tilde{v}'(0) = 0$, and $\tilde{v}(x)$ is the (oriented) area under it, $\mathcal{E}_{\tilde{v}}(x) \geq 2$ for every well S-shaped $v(x)$, and $\mathcal{E}_{\tilde{v}}(1) = 2$ only if $\tilde{v}(x) = -x^2$ on $[0,1]$ (Lemma 9). Hence $\overline{p} \geq 0.5$, with equality only in that case.

Corollary 1. *For any well S-shaped $v(x)$, $(\{0,1\};\{0,1\})$ is never an equilibrium if $p_1 < 0.5$ or $p_2 < 0.5$, and, unless $\tilde{v}(x) = -x^2$ on $[0,1]$, also if $p_1 = 0.5$ or $p_2 = 0.5$.*

In other words, unless $\tilde{v}(x) = -x^2$ on $[0,1]$, in which case $\overline{p} = 0.5$, full revelation requires both agents to be more likely good than bad.

3.2.2 Symmetric equilibrium

For equal priors, Lemma 1 and Proposition 1 characterize the equilibrium on the well S-shaped class. Unless it is full revelation, it is a bad news equilibrium where the probability of conceding bad news ($s_i = 0$) is inversely proportional to the elasticity of $\tilde{v}(x)$ at the pooled signal a.[4]

Theorem 1. *Let $v(x)$ be well S-shaped and the priors (p,p). If $p \geq \overline{p}$, the unique equilibrium is $(\{0,1\};\{0,1\})$. If $p < \overline{p}$, the unique equilibrium is $(\{0,a\};\{0,a\})$, where a solves $p = a(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)})$, and $\mathbb{P}(s_i = 0) = \frac{1}{\mathcal{E}_{\tilde{v}}(a)}$.*

The intuition is simple. Against $\{0,a\}$, an agent's normalized value function is

$$\overline{v}(s_i) = \mathbb{P}(s_i = 0)\tilde{v}(s_i) + \mathbb{P}(s_i = a)\tilde{v}(s_i - a) = \frac{a-p}{a}\tilde{v}(s_i) + \frac{p}{a}\tilde{v}(s_i - a),$$

where the second term $\tilde{v}(s_i - a) = -\tilde{v}(a - s_i)$ reflects $\tilde{v}(s_i)$ on $[0,a]$. For $\{0,a\}$ to be a best reply, the secant from $\overline{v}(0)$ to $\overline{v}(a)$ must be tangent at $s_i = a$. Since $\tilde{v}'(0) = 0$, the

[4]The characterization extends to costly signals. If agent i pays $\kappa \operatorname{Var}_{G_i}[s_i]$, the equilibrium is still unique and of the same form, except that it is pooling for a high enough cost; what matters is the cost relative to the curvature of the rule, $\lambda = \frac{\kappa}{|v(1)-0.5-v'(0)|}$ (Proposition 16 in Appendix C).

second term has zero derivative at a, so $\overline{v}'(a) = \frac{a-p}{a}\tilde{v}'(a)$, while the secant has the slope $\frac{\tilde{v}(a)}{a}$, the same for both terms. Hence, in equilibrium,

$$\frac{a-p}{a}\tilde{v}'(a) = \frac{\tilde{v}(a)}{a} \Leftrightarrow \frac{a-p}{a} = \frac{1}{\mathcal{E}_{\tilde{v}}(a)} \Leftrightarrow p = a\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right).$$

Such a is unique, continuous and strictly increasing in p: the pooled signal rises with the prior. Indeed, $p = a - \frac{\tilde{v}(a)}{\tilde{v}'(a)}$ is the point where the tangent to $\tilde{v}(x)$ at $x = a$, $y = \tilde{v}(a) + (x-a)\tilde{v}'(a)$, crosses the x-axis. Since $\tilde{v}(x)$ is strictly concave and decreasing on $[0,1]$ with $\tilde{v}(0) = 0$, this point is continuous and strictly increasing in a, and so is its inverse $a(p)$. As $\mathbb{P}(s_i = 0) = \frac{a-p}{a}$, this also gives the inverse elasticity result.

Uniqueness needs one more step, since $\{0, a\}$ need not be the unique best reply to itself: if $\tilde{v}(x)$ is quadratic on $[0, 2p]$, every signal supported there is a best reply to $\{0, 2p\}$. But then $\{0, 2p\}$ is a best reply to none of them except itself, so, as equilibria form a product set, the equilibrium is still unique.

3.2.3 Pooling

Pooling is the other edge case of a bad news signal, and, unlike full revelation, it is never played by both agents.

Lemma 2. *Let $v(x)$ be well S-shaped. Then $(\{p_1\}; \{p_2\})$ is never an equilibrium.*

Against pooling, an agent's value function $v(s_i - p_j)$ is convex on $[0, p_j]$ and concave on $[p_j, 1]$, so pooling is a best reply only for the agent with the strictly higher prior. Still, pooling can be approached in the limit: for $\tilde{v}_n(x) = -x^n$ on $[0,1]$, Theorem 1 gives $a = \frac{np}{n-1} \to p$. More generally, $a \to p$ whenever the elasticity of $\tilde{v}(x)$ diverges uniformly on $(0,1)$; Lemma 11 in Appendix A states this formally.

3.3 Informativeness of the signals

This subsection studies how the prior and the selection rule affect the informativeness of the signals in the symmetric bad news equilibrium, which is unique for every well S-shaped $v(x)$ by Theorem 1.

3.3.1 Informativeness and change in prior

For a fixed p, a higher a means a more informative bad news signal. In equilibrium, a rises with p, but this does not imply a more informative signal: if a rises only a little relative to p, the signal may become less informative. To compare signals for different a and p, I represent them as experiments $\pi_i : \{0,1\} \to [0,1]$ and use the Blackwell order (see Blackwell (1953)).

A bad news experiment sends $s_i = 0$ or $s_i = a$ if $\theta_i = 0$, and always $s_i = a$ if $\theta_i = 1$, so it is characterized by the conditional probability of revealing the bad type, $\frac{\mathbb{P}(s_i=0)}{\mathbb{P}(\theta_i=0)}$. A higher probability gives a more informative experiment, since a lower one is obtained by garbling. Hence, I say that agent *i reveals more for* p_{i1} *than for* p_{i2} if $\frac{\mathbb{P}(s_{i1}=0)}{1-p_{i1}} > \frac{\mathbb{P}(s_{i2}=0)}{1-p_{i2}}$ for the corresponding equilibrium strategies G_{i1}, G_{i2}. By Theorem 1, this measure depends on a single variable:

$$\mathbb{P}(s_i = 0) = \frac{1}{\mathcal{E}_{\tilde{v}}(a)} \text{ and } p = a\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right) \text{ hence } \frac{\mathbb{P}(s_i = 0)}{1-p} = \frac{1}{1 + (\mathcal{E}_{\tilde{v}}(a) - 1)(1-a)}.$$

This gives the following lemma.

Lemma 3. *Let $v(x)$ be well S-shaped and $(\{0,a\};\{0,a\})$ the equilibrium for $p \in (0,\overline{p})$. Then informativeness strictly increases in the prior at p if and only if $(\mathcal{E}_{\tilde{v}}(x) - 1)(1-x)$ strictly decreases at $x = a$.*

Corollary 2. *Let $v(x)$ be well S-shaped with $\mathcal{E}_{\tilde{v}}(x)$ non-increasing. Then in equilibrium agents reveal more for p_2 than for p_1 whenever $0 < p_1 < p_2 \leq \overline{p}$.*

Some algebra shows that Corollary 2 applies to the natural examples.

Example 3. For each of the two classes from Example 1 and every $\text{Var}[\varepsilon_i] \geq \overline{V}$, $\mathcal{E}_{\tilde{v}}(x)$ is non-increasing, so in equilibrium agents reveal more as p increases.

However, the elasticity of a well S-shaped rule can vary flexibly, and ex-ante better agents may then reveal less.

Example 4. There exists a well S-shaped $v(x)$ with $\tilde{v}(x) = -\frac{x^2}{16(1-x)^2}$ on $[0, 0.5]$ s.t. in equilibrium agents reveal less for p_2 than for p_1 if $0 < p_1 < p_2 \leq 0.375$.

3.3.2 Informativeness and change in selection rule

Next, fix the prior and change the selection rule. If $\tilde{v}(x)$ has a constant elasticity, a higher elasticity lowers $\mathbb{P}(s_i = 0)$ for a fixed p, and hence a, making the signal less informative. The next corollary extends this argument to any two selection rules.

Corollary 3. *Let $v_e(x), v_n(x)$ be well S-shaped, let the priors be (p, p) with $p \in (0, \min(\overline{p}_e, \overline{p}_n))$, and let $(\{0, a_i\}; \{0, a_i\})$ be the equilibrium for v_i, $i \in \{e, n\}$. If $\tilde{v}_e(x)$ is more elastic than $\tilde{v}_n(x)$, i.e. $\mathcal{E}_{\tilde{v}_e}(x) > \mathcal{E}_{\tilde{v}_n}(x)$ for all $x \in (0, 1)$, then $a_e < a_n$.*

So a more elastic selection rule suppresses information, and within the well S-shaped class, the most informative signal is induced by the rules with $\mathcal{E}_{\tilde{v}}(x) = 2$, i.e. $\tilde{v}(x) = -x^2$ on $[0, 1]$. For the natural examples, this reads intuitively in terms of noise.

Example 5. For each of the two classes from Example 1, $\mathcal{E}_{\tilde{v}}(x)$ increases in $\text{Var}[\varepsilon_i]$ on $[\overline{V}, +\infty)$ for every $x \in (0, 1)$, so noisier shocks lead to less information in equilibrium.

3.3.3 Selection reversals

Informativeness also governs how often noise makes the firm select the agent with the lower posterior. For $p < \overline{p}$, posteriors differ with probability $\frac{2}{\mathcal{E}_{\tilde{v}}(a)}\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right)$, and the firm then selects the agent with $s_i = 0$ with probability $1 - v(a)$. If $\mathcal{E}_{\tilde{v}}(x)$ is

non-decreasing, the first factor does not increase in p, as $\frac{1}{\mathcal{E}_{\tilde{v}}(a)} \leq \frac{1}{2}$, and the second decreases, so such reversals become rarer. Under the condition of Corollary 2, which makes informativeness increase in p, the two factors move in opposite directions, and either can dominate. For the two classes from Example 1, which satisfy it by Example 3, a direct computation shows that reversals still become rarer, but for $v(x) = 0.5 + 0.6x - 0.1e^{5-\frac{5}{x}}$ on $[0, 1]$, the probability of a reversal increases in p for $p < 0.36$.

4. Utility of the firm

This section turns to the normative side and studies the firm's utility under a well S-shaped selection rule: whether it increases in the prior in the symmetric equilibrium, which rule in the class maximizes it, and, outside the class, the optimal level of noise. Section 5 develops this further by allowing any selection rule.

4.1 Monotonicity with a change in the prior

It is natural to expect the firm to benefit from ex-ante better agents. But better agents do not always reveal more, and this effect could outweigh the higher prior.

For a well S-shaped $v(x)$ and priors (p, p), let (G, G) be the unique equilibrium and $u_F(p) := \mathbb{E}_G[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)]$ the firm's expected utility.

Proposition 2. *Let $v(x)$ be well S-shaped. Then $u_F(p)$ is strictly increasing in p.*

So, despite possibly decreasing informativeness, a firm that uses a well S-shaped selection rule is never harmed by ex-ante better agents. The result is not immediate: the equilibrium signals for different priors need not be ranked by first-order stochastic dominance or in the Blackwell order, and under a noisy selection rule the firm's payoff need not be monotone or convex in the signals. However, the change from p to $p' > p$ splits into two steps within the bad news class. First, keeping the prior p, the pooled

signal rises from $a(p)$ to $a(p')$: this spread reveals the bad type more often and widens the gap between the signals, so the firm selects the worse agent less often. Second, keeping the pooled signal $a(p')$, the prior rises to p', which moves mass from 0 to $a(p')$, and on $\{0, a\}$ the firm's payoff is monotone. Both steps benefit the firm. This logic relies on the pooled signal moving away from the revealed one as p increases. Under the reverse S-shaped selection rules of Subsection 7.2, it moves toward it, and there exist $p_1 < p_2$ s.t. $u_F(p_1) > u_F(p_2)$ in the unique equilibrium.

4.2 Maximal utility in the well S-shaped class

I now depart from an exogenous $v(x)$: for a fixed p, the firm chooses the selection rule to maximize $u_F(p)$, for instance, by choosing between committee structures, evaluation protocols or review panels. By changing the biases that shape her selection rule, she changes the equilibrium signals and her utility.

Two questions arise. First, what is the best equilibrium utility within the well S-shaped class for each p, and which rules attain it? Second, how does it compare with the noiseless selection rule? The answer to the latter is not obvious: in the noiseless equilibrium of Boleslavsky and Cotton (2015), an agent with some probability uses a uniformly distributed posterior, which is less informative than the binary signal under a well S-shaped rule, so noise trades selection accuracy for more information.

The natural examples show an important pattern.

Example 6. For each of the two classes from Example 1, $u_F(p)$ decreases in $\mathrm{Var}[\varepsilon_i]$ on $[\overline{V}, +\infty)$ for every $p \in (0, 1)$, so noisier shocks lower the firm's equilibrium utility.

So within each of these classes, the best rule lies at the "edge" of the well S-shaped class: the lowest level of noise that keeps $v(x)$ well S-shaped. The next theorem gives a tight upper bound on the firm's utility over the whole class and the rules that attain it.

Theorem 2. *Let $v(x)$ be well S-shaped. Then*

(i) $u_F(p) \le u_F^{ws}(p) = p + \min(2p, 1) \cdot p(1-p)$.

(ii) *If* $p \ge 0.5$, *then* $u_F(p) = u_F^{ws}(p)$ *if and only if* $p \ge \overline{p}$ *and* $v(1) = 1$.

(iii) *If* $p < 0.5$, *then* $u_F(p) = u_F^{ws}(p)$ *if and only if* $v(x) = 0.5 + x - 0.5x|x|$.

Since the first best utility is $u_F^*(p) = 2p - p^2 = p + p(1-p)$, the best well S-shaped rule captures the share $\min(2p, 1)$ of the first best gain over random selection.

Corollary 4. *Let* $v(x) = 0.5 + x - 0.5x|x|$. *Then*

(i) $u_F(p) = u_F^{ws}(p)$ *for all* $p \in (0, 1)$;

(ii) $u_F(p)$ *is strictly higher than under the noiseless rule if and only if* $p > \frac{3-\sqrt{3}}{6} \approx 0.21$.

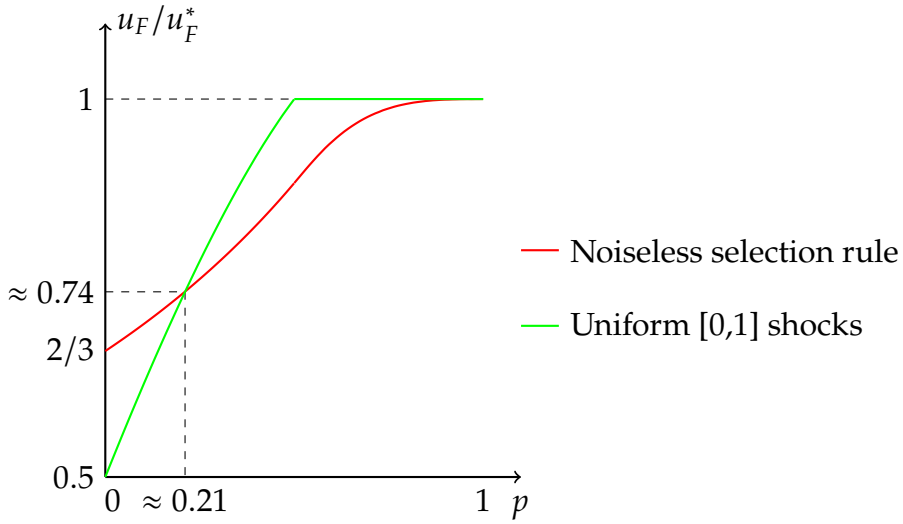


Figure 2: Losses from the best well S-shaped selection rules

So the selection rule $v^*(x) = 0.5 + x - 0.5x|x|$ is optimal within the well S-shaped class for every p, and uniquely so for $p < 0.5$. It has a triangular density and is generated by two independent $\varepsilon_i \sim U[0, 1]$. Its optimality has a simple source. In the equilibrium $(\{0, a\}; \{0, a\})$, the firm's utility is $p + \mathbb{P}(s_1 \ne s_2) \cdot a \cdot \big(v(a) - 0.5\big)$: she obtains p from a random pick and gains only when the signals differ, by their gap times her

excess probability of selecting the higher one. This gain rises with the pooled signal a and, at a given a, with the accuracy $v(a)$, and $v^*(x)$ is best on both counts. It induces the highest possible pooled signal, $a = \min(2p, 1)$, and since a concave density on $[0, 1]$ declines no faster, in proportion, than the triangular density $1 - x$, and its mass there is at most 0.5, $v^*(x)$ is also the most accurate well S-shaped rule at every gap: $v(x) \leq v^*(x)$ on $[0, 1]$. So $v^*(x)$ lies at the far edge of the class, as in the natural examples, and there the firm does not trade sharper selection against more information; this trade-off arises only beyond the edge.

Figure 2 shows the share of the first best utility $u_F^*(p)$ attained by $v^*(x)$ and by the noiseless rule. By Corollary 4, $v^*(x)$ outperforms the noiseless rule if and only if $p > \frac{3-\sqrt{3}}{6} \approx 0.21$, and attains the first best for $p \geq 0.5$: the extra information it induces outweighs the cost of sometimes selecting a bad agent over a good one, most of all at high priors, where it induces full revelation.

4.3 The optimal level of noise

Theorem 2 leaves open what happens beyond the edge of the well S-shaped class, where the noise is too small for the density of the shock difference to be concave. The selection rules generated by i.i.d. $\varepsilon_i \sim U[0, u]$, which have a triangular density on $[-u, u]$, cover both sides of this edge. For $u \geq 1$, $v(x)$ is well S-shaped with $\tilde{v}(x) = -x^2$ on $[0, 1]$, and $u = 1$ gives $v^*(x)$; for $u < 1$, $v'(x)$ is not concave on $[0, 1]$, and as $u \to 0$, $v(x)$ converges pointwise to the noiseless selection rule. For every u and p, a symmetric equilibrium can be constructed explicitly, and it has binary support if and only if $u \geq 2\min(p, 1 - p)$; for smaller u, each agent randomizes over more than two posteriors.[5] The construction is technical and left to Subsection 7.4; the next proposition concerns only the informativeness of the signals and the firm's utility.

[5]For $u < 1$, I do not prove uniqueness, but the equilibria of the discretized game converge to this one.

Proposition 3. *Let $v(x)$ be the CDF of a triangular distribution on $[-u, u]$, $u \in (0, 1]$, and let $u_F(p, u)$ be the firm's utility in the equilibrium (G_u, G_u) constructed in Proposition 13.*

(i) If $p \leq 0.5$, G_u is strictly increasing in u in the Blackwell order on $(0, 2p]$, and $G_u = \{0, 2p\}$ for $u \in [2p, 1]$. If $p > 0.5$, $G_u = \{0, 1\}$ for $u \in [2 - 2p, 1]$.

(ii) If $p \leq 0.5$, $u_F(p, u)$ is uniquely maximized at $u = 2p$ and strictly decreasing on $[2p, 1]$. If $p > 0.5$, $u_F(p, u)$ equals the first best $u_F^*(p) = 2p - p^2$ if and only if $u \in [2 - 2p, 1]$.

(iii) As $u \to 0$, G_u converges weakly to the equilibrium signal for the noiseless rule.

The proposition considers only $u \leq 1$: for $u \geq 1$, $v(x)$ is well S-shaped with $\tilde{v}(x) = -x^2$ on $[0, 1]$, so the equilibrium signal does not change with u, while $v(x) = 1 - \frac{(u-x)^2}{2u^2}$ decreases in u on $(0, 1]$, and so does the firm's utility.

Part (i) shows that for $p \leq 0.5$, less noise only suppresses information. For $p > 0.5$ and $u < 2 - 2p$, this ranking breaks down. For instance, for $p = \frac{2}{3}$, G_u puts $(\frac{1}{5}, \frac{1}{5}, \frac{3}{5})$ on $\{0, \frac{1}{3}, 1\}$ for $u = \frac{1}{3}$ and $(\frac{4}{21}, \frac{4}{21}, \frac{13}{21})$ on $\{0, \frac{1}{4}, 1\}$ for $u = \frac{1}{2}$. The latter reveals the good type more often but the bad type less often, and since a mean-preserving spread on $[0, 1]$ cannot lower the probabilities of 0 and 1, neither is more informative. The same holds for any two $u \in [1 - p, 2 - 2p)$ if $p \geq \frac{5}{9}$ (see the proof of Proposition 3). Still, on this range, the probability of revealing the good type and the variance increase in u.

Part (ii) shows that the firm's utility is maximized at the least noise at which agents play a binary signal, $\{0, 2p\}$ or $\{0, 1\}$, and less noise, which breaks the binary structure, strictly lowers it. So the optimal level of noise is intermediate: it trades off information from the agents against accuracy of selection by the firm. Moreover, at least within this family, the firm never benefits from non-binary signals, so going far beyond the well S-shaped class may be unnecessary.

Part (ii) also compares the best rule in this family with the noiseless rule at every prior. For $p \leq 0.5$, the noise $u = 2p$ gives $v(2p) = 1$, so the firm selects the higher of two

different signals for sure and noise changes only what agents reveal. Her utility is then $\frac{3p}{2}$, against $\frac{4p}{3}$ under the noiseless rule and $u_F^{ws}(p)$ under the best well S-shaped rule; as $p \to 0$, this is three quarters of the first best, against two thirds. For $p > 0.5$, every $u \in [2-2p, 1]$ attains the first best. So the right amount of noise beats the noiseless rule at every prior, whereas $v^*(x)$ does so only for $p > 0.21$.

Part (iii) shows that there is no discontinuity at zero noise: as $u \to 0$, $v(x)$ converges pointwise to $0.5 + 0.5 \cdot sign(x)$, and the equilibrium to the noiseless one of Boleslavsky and Cotton (2015). Simulations show the same for logistic and normal noise.

Two caveats, however, point to the next section. First, the optimal noise is tied to the prior: for $p \leq 0.5$, more noise than $u = 2p$ blurs selection and less breaks the binary signals, so a firm that misjudges p loses either way. Second, for $u < 1$ I do not prove that the equilibrium is unique, so this comparison rests on the equilibrium of Proposition 13. A firm that does not know the priors exactly thus needs a design that is robust to them, which is the subject of Section 5.

5. Designing the Selection Rule

Subsection 4.2 found the best well S-shaped rule, which outperforms the noiseless rule for sufficiently high priors and attains the first best for all priors $p \geq 0.5$. The next question is which $v(x)$ maximizes the firm's utility if she can choose any selection rule, and which properties of $v(x)$ are necessary or sufficient for optimality.

I say that $v(x)$ is *revelation-accurate* if $v(1) = 1$, i.e. whenever agent i is revealed to be good and agent j bad ($s_i = 1$, $s_j = 0$), the firm selects agent i with probability 1. This is a simple necessary condition for optimality: if $v(1) < 1$, an affine modification of $v(x)$ leads to the same equilibria but gives the firm a strictly higher utility.

I also say that $v(x)$ *implements a fully revealing equilibrium (uniquely) for* (p_1, p_2) if, for these priors, full revelation by both agents is an (the unique) equilibrium.

The firm's maximal equilibrium utility[6] under a given selection rule and her first best utility, both for symmetric priors, are defined as follows.

Definition 4. For a selection rule $v(x)$ and priors (p, p), let E be the set of all equilibria. Then the maximal expected utility of the firm among all equilibria is $\overline{u}_F(p) := \max_{(G_1^*, G_2^*) \in E} \mathbb{E}_{G_1^*, G_2^*}[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)]$.

Definition 5. Let $\mathcal{V}$ be the set of all selection rules. Then the first best expected utility of the firm is $u_F^*(p) := \sup_{G_1, G_2 \in \mathcal{G}(p), v \in \mathcal{V}} \mathbb{E}_{G_1, G_2}[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)]$.

The first result shows that, for symmetric agents, a fully revealing equilibrium and revelation accuracy are not only sufficient, but also necessary for the first best.

Lemma 4. *For every selection rule $v(x)$, $\overline{u}_F(p) \leq u_F^*(p) = 2p - p^2$, with equality if and only if $v(x)$ implements a fully revealing equilibrium for (p, p) and is revelation-accurate.*

The proof is simple. For priors (p, p),

$$u_F^*(p) = \max_{G_1, G_2 \in \mathcal{G}(p)} \mathbb{E}_{G_1, G_2}[\max(s_1, s_2)].$$

As $\max(s_1, s_2)$ is convex and piecewise linear, the maximum is attained if and only if at least one agent plays full revelation. As all equilibria form a symmetric product set, if full revelation and some other signal form an equilibrium, so do two full revelations. Since $s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1) \leq \max(s_1, s_2)$, $\overline{u}_F(p) < u_F^*(p)$ if $v(x)$ does not implement a fully revealing equilibrium, and in a fully revealing equilibrium, the firm gets the first best if and only if $v(x)$ is revelation-accurate. The result generally fails for $p_1 \neq p_2$: the set of equilibria is then not symmetric and may contain only equilibria in which exactly one agent reveals fully.

[6]The maximum is well defined: by the continuity of $v(x)$, each agent's set of max-minimizers is compact, and so is the set of all equilibria.

For every $p \in (0,1)$, the selection rules that implement a fully revealing equilibrium form a non-empty class, described by an infinite system of linear inequalities:

$$(1-p)\hat{v}(x) - p\hat{v}(1-x) \leq (x-p)\hat{v}(1) \quad \forall x \in (0,1), \quad \text{where } \hat{v}(x) := v(x) - 0.5.$$

Except for $p = 0.5$, this class has no simple structure, and the assumption that the firm knows the priors exactly is restrictive. Instead, suppose that the firm only knows that the priors lie in some exogenous set S. This robust approach leads to sharper results.

First, although a fully revealing equilibrium is not necessary for the first best under asymmetric priors, this approach covers this case too.

Lemma 5. *Let $S \subseteq (0,1)$. A selection rule $v(x)$ implements a fully revealing equilibrium for all $(p_1, p_2) \in S^2$ if and only if it does so for all (p,p) with $p \in S$.*

The *only if* direction is straightforward; as for *if*, suppose that $v(x)$ implements a fully revealing equilibrium for some (p,p) with $p \in S$, and let agent j with prior $p_j = p$ play full revelation. Agent i's value function does not depend on his own prior, and since full revelation is a best reply for $p_i = p$, its concavification is the secant from 0 to 1, so full revelation is a best reply for every $p_i \in (0,1)$. Hence, if $v(x)$ implements a fully revealing equilibrium for all (p,p) with $p \in S$, then it does so for all $(p_1, p_2) \in S^2$.

Second, robustly implementing full revelation strongly restricts $v(x)$.

Proposition 4. *Let a selection rule $v(x)$ implement a fully revealing equilibrium for all $(p_1, p_2) \in S^2$, where $\inf S < 0.5 < \sup S$. Then*

(i) $v(x)$ implements a fully revealing equilibrium for all (p,p) with $p \in (0,1)$;

(ii) $v(x) = 0.5 + ax$, where $a \in [0, 0.5]$.

Corollary 5. *Let $v(x)$ be a selection rule with $\overline{u}_F(p) = u_F^*(p)$ for $p \in \{p_1, p_2\}$, where $p_1 < 0.5 < p_2$. Then $v(x) = 0.5 + 0.5x$.*

To see why, note that by Lemma 5, the system of inequalities above holds for every $p \in S$. Adding it at x and at $1-x$ with weights $1-p$ and p gives $(1-2p)\big(\hat{v}(x)-x\hat{v}(1)\big) \leq 0$ for all $x \in (0,1)$. So a prior in S below 0.5 rules out any upward deviation of $\hat{v}(x)$ from the linear function $x\hat{v}(1)$, and a prior in S above 0.5 rules out any downward one. Hence, $v(x) = 0.5 + x(v(1) - 0.5)$ is linear, so it implements full revelation for all priors, and any $a = v(1) - 0.5 \in [0, 0.5]$ is valid.

So a firm that is unsure about the priors can robustly attain her first best only with $v(x) = 0.5 + 0.5x$. But this rule has a crucial issue. Since $v(x)$, and hence $\overline{v}(s_i)$, is linear, every pair of signals is an equilibrium, and the firm need not get the first best: robustness comes at the cost of complete indifference. This indifference is also fragile: if signals are costly, any positive cost makes pooling the unique equilibrium (Proposition 17 in Appendix C).

The issue disappears under slightly less robustness. Suppose that the firm knows whether both agents are more or less likely to be good than bad. The rules that implement a fully revealing equilibrium in either case are described by the deviation of $\hat{v}(x)$ from its secant from 0 to 1.

Definition 6. Let $v(x)$ be a selection rule, $a := v(1) - 0.5$, and $g(x) := v(x) - 0.5 - ax$, the deviation of $v(x)$ from its secant on $[0,1]$. Then $v(x)$ is *balanced* if $g(x) = g(1-x)$ on $(0,1)$; *positively* [*negatively*] *balanced* if also $g(x) \geq 0$ [≤ 0] there; and *strictly* so if $g(x) > 0$ [< 0].

Remark 3. Like $\tilde{v}(x)$, $g(x)$ removes a linear part of $v(x)$, but it subtracts the secant from 0 to 1 rather than the tangent at 0. So, for every selection rule, equilibria depend on $v(x)$ only through $g(x)$, and revelation accuracy only through a. A well S-shaped $v(x)$ is positively balanced if and only if $\tilde{v}(x) = -x^2$ on $[0,1]$.

Balance alone is exactly the condition for a fully revealing equilibrium at $p = 0.5$, as shown after Proposition 5; the sign of $g(x)$ then determines the half-interval of priors

on which full revelation extends. For a selection rule, $a \in [0, 0.5]$ and $g(0) = g(1) = 0$ hold automatically, and for a balanced $v(x)$, the monotonicity of $v(x)$ is equivalent to $|g(x) - g(y)| \leq a|x - y|$.

Proposition 5. *Let $v(x)$ be a selection rule.*

(i) $v(x)$ implements a fully revealing equilibrium $\forall (p_1, p_2) \in [0.5, 1)^2$ $\left[\forall (p_1, p_2) \in (0, 0.5]^2\right]$ if and only if it is positively [negatively] balanced.

(ii) If $v(x)$ is strictly positively [negatively] balanced, it implements a fully revealing equilibrium uniquely $\forall (p_1, p_2) \in (0.5, 1)^2$ $\left[\forall (p_1, p_2) \in (0, 0.5)^2\right]$.

Part (i) gives necessary and sufficient conditions for implementing full revelation for all priors on a half-interval. They impose a specific structure: $v(x)$ can deviate from the universal implementation $v(x) = 0.5 + ax$ ($v(1) = 1$ again implies $a = 0.5$) only by a function that is symmetric around 0.5 on $[0, 1]$, non-negative for high priors and non-positive for low ones.

Indeed, the conditions hold for all $p \in (0, 0.5]$ if and only if they hold for $p = 0$ and $p = 0.5$, and the latter gives the symmetry. If agent j with $p = 0.5$ reveals fully, agent i's normalized value function is odd around $s_i = 0.5$:

$$\overline{v}(s_i) = 0.5\hat{v}(s_i) - 0.5\hat{v}(1 - s_i) = -\overline{v}(1 - s_i) \; \forall s_i \in [0, 1].$$

Hence $\{0, 1\}$ is a best reply only if $\overline{v}(s_i)$ is linear:

$$\overline{v}(s_i) = 0.5\hat{v}(s_i) - 0.5\hat{v}(1 - s_i) = (s_i - 0.5)\hat{v}(1) \Leftrightarrow \hat{v}(s_i) - s_i\hat{v}(1) = \hat{v}(1 - s_i) - (1 - s_i)\hat{v}(1).$$

That is, the deviation $g(s_i)$ of $\hat{v}(s_i)$ from the secant from 0 to 1 must be symmetric around 0.5: $g(s_i) = g(1 - s_i)$ for all $s_i \in [0, 1]$, i.e., $v(x)$ must be balanced.

Part (ii) gives a sufficient condition for full revelation to be the unique equilibrium. As all equilibria form a product set, it suffices that $\{0, 1\}$ is each agent's unique best reply to itself. This is also why $p = 0.5$ is excluded: there, any $v(x)$ that implements full revelation makes $\overline{v}(s_i)$ linear, so the best reply is not unique. The condition is not necessary, but since it is only a slight strengthening of (i), the room for other rules with a unique fully revealing equilibrium is small, and checking uniqueness for them requires showing that no signal supported on $\{s_i : g(s_i) = 0\}$ is a max-minimizer. Unlike the linear rule, a strictly positively or negatively balanced rule also withstands costly signals if $g(x)$ has a non-zero slope at 0: full revelation then remains the unique equilibrium for all priors in its half-interval that are farther from 0.5 than a margin proportional to the cost (Proposition 17).

A firm facing a high prior needs a positively balanced rule, and some subclasses of these rules can be generated by i.i.d. shocks. A firm facing a low prior, however, needs a negatively balanced rule, and additive evaluation noise cannot deliver one: by the next result, which extends Corollary 1 beyond the well S-shaped class, i.i.d. shocks rule out full revelation as soon as one prior is below 0.5.

Corollary 6. *Let $v(x)$ be a selection rule with $v'(0) > v'(1)$. Then $(\{0, 1\}; \{0, 1\})$ is never an equilibrium if $p_1 < 0.5$ or $p_2 < 0.5$. In particular, this holds for every $v(x)$ generated by i.i.d. shocks with a square-integrable density.*

Both limitations, indifference under full robustness and the failure for low priors, are specific to rules that depend only on the difference between the signals, as those generated by additive evaluation noise do. A more general anonymous rule avoids them: if the firm selects agent i with probability $0.5 + 0.5(s_i^2 - s_j^2)$, each agent maximizes $\mathbb{E}[s_i^2]$, so full revelation is strictly dominant for every prior, and as this rule always selects the good type over the bad one, the firm attains the first best for all priors in the unique equilibrium. I do not pursue such rules, which additive noise cannot generate.

6. Asymmetric Priors

This section characterizes the equilibrium for asymmetric priors under a well S-shaped selection rule with $v'(x)$ not linear near 0. Without loss of generality let $p_2 \leq p_1$. By Lemma 1, every best reply is unique and takes one of three forms: pooling $\{p_i\}$, full revelation $\{0,1\}$, or bad news $\{0,c\}$ with $p_i < c < 1$. In general, an agent's best reply depends on his rival's whole signal, but in the following example it depends only on the rival's prior, which makes the equilibrium explicit.

Example 7. Let $\tilde{v}(x) = -x^3$. Agent i's best reply is $\{0, \min(\frac{3p_j}{2}, 1)\}$, or $\{p_i\}$ if $p_i \geq \frac{3p_j}{2}$. So for $p_2 \leq p_1$, the unique equilibrium is $(\{0,1\};\{0,1\})$ if $p_2 \geq \frac{2}{3}$, and otherwise

(i) $(\{p_1\};\{0,1\})$ if $p_1 \geq \max(\frac{3p_2}{2}, \frac{2}{3})$;

(ii) $(\{p_1\};\{0, \frac{3p_1}{2}\})$ if $\frac{3p_2}{2} \leq p_1 < \frac{2}{3}$;

(iii) $(\{0, \frac{3p_2}{2}\};\{0,1\})$ if $\frac{2}{3} \leq p_1 < \frac{3p_2}{2}$;

(iv) $(\{0, \frac{3p_2}{2}\};\{0, \frac{3p_1}{2}\})$ if $p_1 < \min(\frac{3p_2}{2}, \frac{2}{3})$.

The example shows what asymmetry does. Each agent would ideally pool at one and a half times his rival's prior. When the priors are close and low enough, both concede bad news, and each pooled signal is set by the rival: the underdog, facing the better rival, pools higher and so reveals more. As the favorite's lead grows, the signal at which he would pool falls below his own prior, so once $p_1 \geq \frac{3p_2}{2}$ he pools at his prior. The underdog, facing a rival who reveals nothing, gambles harder and reveals fully once $p_1 \geq \frac{2}{3}$. In the example, the $\frac{2}{3}$ threshold is the same whether the favorite pools or concedes bad news, because an agent's best reply depends only on his rival's prior. In general, it depends on the rival's signal, so the underdog's threshold for full revelation differs, and the boundaries are no longer multiples of the priors. The same logic still

produces the same four regions, with boundaries defined by best replies to the three forms, listed below with their values in Example 7.

	definition	*Example 7*
$\overline{p}$	$1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)}$: full revelation for $p_1, p_2 \geq \overline{p}$	$\frac{2}{3}$
$b(p_1)$	agent 2's best reply to $\{p_1\}$ is $\{0, b(p_1)\}$ if $p_2 < b(p_1)$, else $\{p_2\}$	$\min(\frac{3p_1}{2}, 1)$
p^*	smallest p_1 with $b(p_1) = 1$	$\frac{2}{3}$
$a(p_2)$	agent 1's best reply to $\{0, 1\}$ is $\{0, a(p_2)\}$ if $p_1 < a(p_2)$, else $\{p_1\}$	$\frac{3p_2}{2}$
$\check{p}_2(p_1)$	largest p_2 for which $\{p_1\}$ is agent 1's best reply to $\{0, b(p_1)\}$	$\frac{2p_1}{3}$
$\hat{p}_1(p_2)$	smallest p_1 for which $\{0, 1\}$ is agent 2's best reply to $\{0, a(p_2)\}$	$\frac{2}{3}$

Proposition 6. *Let $v(x)$ be well S-shaped with $v'(x)$ not linear near 0, and let $p_1, p_2 \in (0, 1)$ with $p_2 \leq p_1$. Then the equilibrium is unique, and it is $(\{0, 1\}; \{0, 1\})$ if $p_2 \geq \overline{p}$, and otherwise*

(i) $(\{p_1\}; \{0, 1\})$ *if* $p_1 \geq \max\{p^*, a(p_2)\}$;

(ii) $(\{p_1\}; \{0, b(p_1)\})$ *if* $p_1 < p^*$ *and* $p_2 \leq \check{p}_2(p_1)$;

(iii) $(\{0, a(p_2)\}; \{0, 1\})$ *if* $\hat{p}_1(p_2) \leq p_1 < a(p_2)$;

(iv) $(\{0, a\}; \{0, b\})$ *with* $p_1 < a \leq b < 1$ *otherwise, where* $a = b$ *if and only if* $p_1 = p_2$.

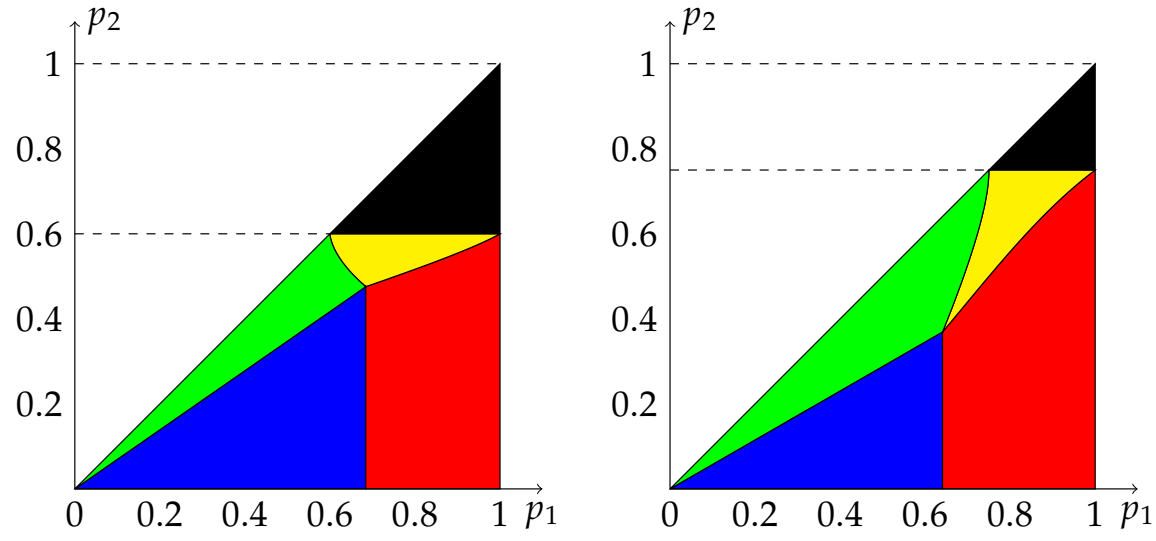
The equilibrium strategies are continuous in (p_1, p_2).

Figure 3 depicts these regions. In Example 7, regions (i)–(iv) are exactly the cases listed, and $\overline{p} = p^*$: it is the knife-edge case in panel (c), between $\overline{p} < p^*$ and $\overline{p} > p^*$ in panels (a) and (b). The boundary $a(\cdot)$ is monotone, but $\hat{p}_1(\cdot)$ and $\check{p}_2(\cdot)$ need not be: in panel (d), neither is monotone. Non-linearity of $v'(x)$ near 0 cannot be dropped from Proposition 6: without it, the equilibrium need not be unique, and signals need not take these forms (Remark 4).

Remark 4. Let $v(x) = v^*(x)$, for which $v'(x)$ is linear on $[0, 1]$, and $(p_1, p_2) = (0.4, 0.3)$. Then agent 2 plays $\{0, 0.6\}$ in every equilibrium, and agent 1's value function against it

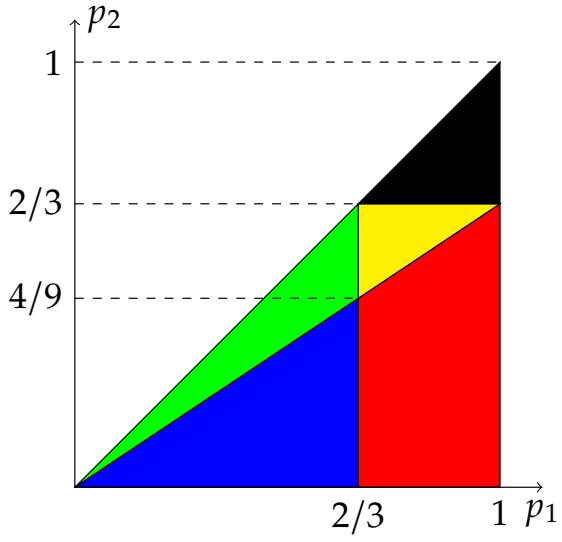
is linear on $[0, 0.6]$. So agent 1's equilibrium signals are exactly those with support in $[0, 0.6]$ and $\mathbb{E}[s_1^2] = 0.18$, which keep $\{0, 0.6\}$ a best reply of agent 2, such as $\{0, 0.45\}$, $\{0.3, 0.6\}$ and their mixtures.

Figure 3: Equilibria for asymmetric priors, with $\tilde{v}(x)$ given on $[0, 1]$

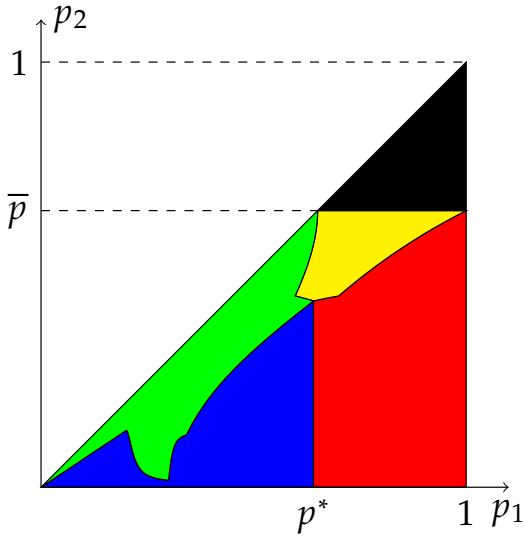


(a) $\tilde{v}(x) = -x^{2.5}$: $\overline{p} = 0.6 < p^* \approx 0.68$

(b) $\tilde{v}(x) = -x^4$: $\overline{p} = 0.75 > p^* \approx 0.64$



(c) $\tilde{v}(x) = -x^3$: $\overline{p} = p^* = \frac{2}{3}$



(d) $\tilde{v}(x) = -(x^3 + 60\max(x - 0.3, 0)^2)/30.4$: $p^* \approx 0.641 < \overline{p} \approx 0.651$

Colors are for the corresponding equilibrium type: black – $(\{0, 1\}; \{0, 1\})$; *red* – $(\{p_1\}; \{0, 1\})$; *green* – $(\{0, a\}; \{0, b\})$; *blue* – $(\{p_1\}; \{0, b\})$; *yellow* – $(\{0, a\}; \{0, 1\})$.

Every pair of forms except pooling by both agents (Lemma 2) is played for some priors, and the agent with the lower prior never pools at his prior. For a fixed $v(x)$, however, not every such pair is played: full revelation is a best reply to $\{0, c\}$ only if $c \geq p^*$, so $(\{0, c\}; \{0, 1\})$ is an equilibrium only if $c \geq p^*$. In regions (ii) and (iii), an agent's bad news signal depends only on his rival's prior. More generally, it is the rival who forces an agent to reveal.

Proposition 7. *Let $v(x)$ be as in Proposition 6. Then the equilibrium signal of agent i is non-decreasing in p_j, and strictly increasing whenever agent i plays $\{0, a\}$ with $p_i < a < 1$.*

An agent's own prior usually has the opposite effect: in Example 7, his high signal does not depend on p_i, so he reveals less as p_i increases. This effect need not be monotone, though, as the next example shows.

Example 8. Let $v(x)$ be well S-shaped with $\tilde{v}(x) = -x^{2.5}$ on $[0, 1]$, and let $p_j = 0.6$. Then agent i's equilibrium signal is non-monotone in p_i.

Region (i) has a simple explanation. Against full revelation, agent 1's value function $(1 - p_2)v(s_1) + p_2 v(s_1 - 1)$ is close to $v(s_1)$ for low p_2, so it is concave except near 0, and pooling at p_1 is optimal. Against pooling, agent 2's value function $v(s_2 - p_1)$ is convex on $[0, p_1]$ and concave on $[p_1, 1]$, and for high p_1, its concavification is supported on $\{0, 1\}$ despite the concave part. So an agent who is nearly sure to win plays safe, while one who is likely to lose gambles; under a well reverse S-shaped selection rule, the roles are reversed.

For the firm, the monotonicity of Proposition 2 extends to asymmetric priors everywhere except in region (iv) of Proposition 6.

Proposition 8. *Let $v(x)$ be as in Proposition 6 and $p_2 \leq p_1$. In the interior of the full revelation region and of each of the regions (i)–(iii), both partial derivatives of u_F are positive.*

Example 9. Let $v(x) = 0.5 + (20x - x^{20})/38$ on $[0, 1]$, so that $\tilde{v}(x) = -x^{20}$ and $v(1) = 1$, and let $p_1 = 0.924$. Then u_F strictly decreases in p_2 on $[0.839, 0.873]$, where the equilibrium is in region (iv): once agent 2 stops revealing fully, his high signal b decreases in p_2.

For the agents, monotonicity in the priors is trivial under equal priors, as each is selected with probability 0.5. Under asymmetric priors, since the game is constant-sum, each agent's equilibrium payoff $u_i(p_1, p_2) = \mathbb{E}_{G_1,G_2}[v(s_i - s_j)]$ is well defined for any selection rule, whether or not the equilibrium is unique.

Lemma 6. *For any selection rule, $u_i(p_1, p_2)$ is non-decreasing in p_i and non-increasing in p_j. If $v(x)$ is strictly increasing, in particular if it is well S-shaped, $u_i(p_1, p_2)$ is strictly increasing in p_i and strictly decreasing in p_j.*

7. Discussion

This section discusses the main assumptions by relaxing them one at a time. Subsection 7.1 relaxes the binary prior. Subsections 7.2–7.4 leave the well S-shaped class: Subsection 7.2 studies its mirror image, Subsection 7.3 describes equilibria around its boundary, and Subsection 7.4 constructs the equilibrium for the triangular selection rules of Subsection 4.3. Subsection 7.5 relaxes the anonymity of the selection rule. Subsection 7.6 removes the rival.

7.1 Non-binary priors

Every result up to this point assumes a binary prior. That assumption buys tractability, but it is not innocuous: the two-point distribution on $\{0, 1\}$ is the *largest* element in the convex order among all distributions on $[0, 1]$ with mean p, so it gives agents the richest possible set of feasible signals. This subsection relaxes it.

7.1.1 Censorship equilibrium

Under a well S-shaped rule, an agent's value function is S-shaped against every signal of his rival, so if the prior has a strictly positive density, Theorem 1 of Kolotilin et al. (2022) applies: the optimal signal reveals types below some θ^* and pools all types above it into one signal. The same censorship logic applies to a binary prior in the quantile space. Let $F^{-1}(x)$ be its quantile function:

$$F^{-1}(x) = \begin{cases} 0, & x \in [0, 1-p) \\ 1, & x \in [1-p, 1] \end{cases}.$$

Then a censorship with quantile $q < 1-p$ reveals $\theta_i = 0$ with probability q (below the quantile) and otherwise sends the pooled signal $s_i = \frac{(1-p-q)\cdot 0 + p\cdot 1}{1-p-q+p} = \frac{p}{1-q}$ (above the quantile), while a censorship with quantile $q \geq 1-p$ is full revelation.

With a positive density $f(x)$, the equilibrium is again a symmetric upper censorship:

Proposition 9. *Suppose that $v(x)$ is well S-shaped and $F(x)$ has a strictly positive density $f(x)$ on $[0,1]$. Then $(G(x), G(x))$ is an equilibrium, unique if $v'(x)$ is not linear near 0, where*

$$G(x) = \begin{cases} F(x) & , x \leq \theta^* \\ F(\theta^*) & , x \in (\theta^*, m^*) \\ 1 & , x \geq m^* \end{cases}$$

with $m^* = \mathbb{E}[\theta | \theta \geq \theta^*]$ and where $\theta^* \in (0,1)$ solves

$$\int_0^{\theta^*} [\tilde{v}(m^* - s) - \tilde{v}(\theta^* - s) - (m^* - \theta^*)\tilde{v}'(m^* - s)] f(s) ds + \tilde{v}(m^* - \theta^*)(1 - F(\theta^*)) = 0 \quad (1)$$

For the best well S-shaped rule from Corollary 4, the equation collapses entirely.

Corollary 7. *Let $v(x) = 0.5 + x - 0.5x|x|$. Then θ^* in Proposition 9 solves $F(\theta^*) = 0.5$.*

So this rule induces median censorship as an equilibrium, whatever the prior is.

Step 2 of the proof of Proposition 9 does not use the density, so the median censorship remains an equilibrium whenever F is continuous at its median.

7.1.2 The noiseless benchmark

With a binary prior the only restriction on a signal G is that its mean equal p, but with a general prior the feasible set becomes smaller: by Gentzkow and Kamenica (2016), G is attainable if and only if it is a mean-preserving contraction of F, i.e. $\mathbb{E}_G[\phi] \leq \mathbb{E}_F[\phi]$ for every convex ϕ. This substantially changes the noiseless benchmark and the interpretation of Subsection 4.2. Here I count the noiseless rule $v(x) = 0.5 + 0.5 \cdot sign(x)$ among selection rules, with $\underline{\theta} = \min supp(F)$, $\overline{\theta} = \max supp(F)$.

Proposition 10. *Let F be atomless. Then the noiseless selection rule implements a fully revealing equilibrium if and only if F is convex on $[\underline{\theta}, \overline{\theta}]$, and then it is the unique equilibrium.*

Both directions rest on the fact that, against full revelation, the deviator's value function is F itself. If F is convex, then $\mathbb{E}_G[F] \leq \mathbb{E}_F[F] = 0.5$ for every feasible G, so no deviation beats full revelation; otherwise some interval of types can be pooled into its mean at a profit, and if F has a density, these are types where it decreases.

With a general prior, the first best is $\mathbb{E}[\max(\theta_1, \theta_2)]$, and the comparison in Subsection 4.2 reverses sharply for convex priors.

Corollary 8. *Let F be atomless. A selection rule attains the first best in some equilibrium if and only if it is the noiseless selection rule and F is convex on $[\underline{\theta}, \overline{\theta}]$.*

The first best requires both agents to reveal fully and the firm never to select the worse of two distinct types. A well S-shaped rule always fails the second requirement, while under the noiseless rule the first one holds exactly when F is convex.

For concave priors the equilibrium structure is also simple: a suitable uniform distribution is always feasible, precisely because the density is decreasing.

Proposition 11. *Let F be concave on its support $[\underline{\theta}, \overline{\theta}]$, possibly with an atom at $\underline{\theta}$, and with mean p. Then both agents playing $G = U[\underline{\theta}, 2p - \underline{\theta}]$ is the unique equilibrium of the noiseless game, and the firm's equilibrium utility is $u_F = \frac{4p-\underline{\theta}}{3}$.*

Under the noiseless selection rule, the firm's payoff depends on the prior only through its mean and its lower support point, so, for a given lower support point, it does not grow as the prior becomes more dispersed. The first best does, since $\max(\theta_1, \theta_2)$ is convex, and so may the payoff under a noisy selection rule. Hence, for concave priors, the comparison from Subsection 4.2 depends on how dispersed F is.

Example 10. Let $p = \frac{1}{3}$ and $\underline{\theta} = 0$, so that the noiseless selection rule gives $u_F = \frac{4}{9} \approx 0.444$ under every concave F. Under $v^*(x)$ and the median censorship of Corollary 7, the firm gets $u_F \approx 0.399$ for $F(x) = 2x - x^2$ and $u_F \approx 0.457$ for $F(x) = \frac{1+2x}{3}$.

So the noiseless selection rule is better under the first prior and worse under the second, which is a mean-preserving spread of the first; the binary prior of Sections 3–6 is the extreme case, the most dispersed prior with mean p. Overall, the binary prior is harmless for the equilibrium characterization, but not for the comparison with the noiseless rule. When $supp(F) = [0, 1]$, convex priors have $p \geq 0.5$ and concave ones $p \leq 0.5$, so they describe high and low priors: for high priors of this kind, the advantage of noise disappears, and for low ones, it depends on the prior's dispersion.

7.2 Reverse S-shaped selection rules

The well S-shaped class has a mirror image, in which the density of the selection rule is convex rather than concave. A concave density puts an agent's marginal gain from a higher signal near a tie, whereas a convex density puts it at large margins.

Definition 7. A function $v : [-1, 1] \to [0, 1]$ is called *well reverse S-shaped* if $v(x) - 0.5$ is odd and increasing, and $v'(x)$ is convex on $[-1, 1]$ and uniquely minimized at $x = 0$.

The following lemma transfers the equilibrium analysis to this class.

Lemma 7. *(i) For every well reverse S-shaped $v_2(x)$, there is a well S-shaped $v_1(x)$ with $\tilde{v}_2(x) = -\tilde{v}_1(x)$.*

(ii) Let $\tilde{v}_2(x) = -\tilde{v}_1(x)$. If $(G_1(s_1), G_2(s_2))$ is an equilibrium for $v_1(x)$ and priors (p_1, p_2), then $(\check{G}_1, \check{G}_2)$ is an equilibrium for $v_2(x)$ and priors $(1 - p_1, 1 - p_2)$, where $\check{G}_i$ is the distribution of $1 - s_i$ for $s_i \sim G_i$.

Hence, for equal priors, the unique equilibrium of a well reverse S-shaped $v(x)$ is $(\{b, 1\}; \{b, 1\})$, where b increases in p analogously to Theorem 1: each agent reveals good news rather than conceding bad news, and full revelation is reached for $p \leq \frac{1}{\mathcal{E}_{\tilde{v}}(1)}$, mirroring Proposition 1. In particular, such rules can deliver the first best for $p < 0.5$: on $[0, 1]$, $v(x) = 0.5 + (0.5 - \eta)x + \eta x|x|$ with $\eta \in (0, 0.5]$ equals $0.5 + 0.5x - \eta x(1 - x)$, so it is revelation-accurate and strictly negatively balanced (Proposition 5).

The guarantee of Proposition 2, however, fails for many such rules.

Example 11. For $v(x) = 0.5 + 0.5x^3$, $u_F(p)$ is non-monotone in p.

For these rules, informativeness often decreases in p unless the elasticity increases sharply, mirroring Lemma 3 and Corollary 2, and since $v(x)$ is convex on $[0, 1]$, this effect can outweigh the increase in the prior. Asymmetric equilibria are flipped too: under a well reverse S-shaped rule with $v'(x)$ not linear near 0, the agent with the high prior reveals fully and the other pools at his prior, the opposite of the pattern in Section 6.

7.3 Around the boundary of the well S-shaped class

This subsection describes what changes as the selection rule leaves the well S-shaped class: under a weaker condition, the equilibrium keeps the forms of Section 6, whereas further away equilibrium signals may have non-binary support.

7.3.1 Binary support equilibrium

Although $\overline{v}(s_i)$ need not stay S-shaped outside the class, the equilibrium forms of Lemma 1 and Proposition 6 survive outside it. Their proofs use the shape of $v(x)$ only through the shape of agent i's value function against agent j's equilibrium signals, $\{p_j\}$ and $\{0, c\}$. Against $\{0, c\}$, the sign of $\overline{v}''(s_i)$ on $(0, c)$ depends on $w(s_i)$ and $w(c - s_i)$, where $w(x) := -v''(x)$, so only pairs of points with sum at most 1 matter.

Proposition 12. *Let $v(x)$ be a selection rule with $v'(x)$ differentiable on $(0, 1]$, $w(x) > 0$ on $(0, 1]$, and $\frac{w(c-x)}{w(x)}$ strictly decreasing in x on $(0, c)$ for every $c \in (0, 1]$. Then for all priors, the equilibrium is unique, and each agent i plays $\{p_i\}$ or $\{0, a_i\}$ with $a_i \in (p_i, 1]$.*

These conditions hold for every well S-shaped $v(x)$ with strictly concave and differentiable $v'(x)$, but they do not require $v'(x)$ to be concave. Unlike well S-shapedness, however, they are not straightforward to verify, since the monotonicity of $\frac{w(c-x)}{w(x)}$ must hold for every $c \in (0, 1]$. For $\xi \sim N(0, \sigma^2)$, they hold if and only if $\sigma \geq \frac{1}{2}$, a quarter of the variance that well S-shapedness requires ($\sigma \geq 1$). This bound is nearly tight: in simulations, for $\sigma = 0.46$ and $(p_1, p_2) = (0.62, 0.48)$, agent 1 sends three signals in every equilibrium.

7.3.2 Non-binary support equilibrium

Outside the well S-shaped class, equilibrium signals need not have binary support, and their structure varies considerably with the selection rule.

Lemma 8. *Let $v(x)$ be real-analytic on $[-1, 1]$ with $v''(x) > 0$ for all $x \in [-1, 0)$. Then for all $p_i, p_j \in (0, 1)$ and $G_j \in \mathcal{G}(p_j)$, every optimal signal for $(\overline{v}(s_i, G_j), p_i)$ has finite support.*

These conditions hold for the selection rules generated by i.i.d. Gumbel or normal shocks of any variance, so for these natural examples, equilibrium signals are discrete.

By the identity theorem, an analytic $\overline{v}(s_i)$ coincides with a line either on a finite set or on all of $[0,1]$, and the latter is ruled out by the convexity of $v(x)$ on $[-1,0]$. Still, simulations[7] show that the number of support points grows as the noise decreases, mirroring the classical result for games on the unit square with bell-shaped kernels (see Karlin (1959)), in which optimal strategies are finitely supported and the number of support points grows without bound as the kernel sharpens.

In contrast, a selection rule that is not analytic can induce an equilibrium signal with a continuous part, already for exponential shocks.

Example 12. Let $v(x)$ be generated by i.i.d. $\varepsilon_i \sim Exp(\lambda)$, i.e. the Laplace CDF $v(x) = 0.5 + 0.5 \cdot sign(x)\left(1 - e^{-\lambda|x|}\right)$. Then for all $p \leq 0.5$, both agents playing

$$G = \frac{1}{2(1+\lambda p)}\left(\delta_{\{0\}} + \delta_{\{2p\}}\right) + \frac{\lambda p}{1+\lambda p}\, U[0,2p]$$

is a symmetric equilibrium.

In general, such equilibria are hard to find explicitly. If an equilibrium signal of agent j has an interval $[a,b]$ in its support, $\overline{v}(s_j)$ must be linear on $[a,b]$, which is an integral equation for G_i^*:

$$\overline{v}'(s_j) = \int_0^1 v'(s_j - s_i)\, dG_i^*(s_i) = const \quad \forall s_j \in [a,b].$$

Closed-form solutions are known only for special kernels,[8] such as those with a rational Fourier transform, as in Example 12, or piecewise polynomial ones, which include the triangular selection rules of the next subsection.

[7] For $p = 0.3$ and $\xi \sim N(0,\sigma^2)$, the equilibrium G_i^* has 5 support points for $\sigma = 0.1$ and 9 for $\sigma = 0.05$.

[8] This is a Fredholm equation of the first kind with a difference kernel on a finite interval; see Sakhnovich (2015) for the general theory and Polyanin and Manzhirov (2008) for solvable kernels.

7.4 Triangular selection rules

This subsection constructs the symmetric equilibrium used in Subsection 4.3 for the selection rules generated by i.i.d. $\varepsilon_i \sim U[0,u]$ with $u < 1$. Since $v(x)$ is piecewise quadratic, $v'(x)$ is piecewise linear, so the functional equation from Subsection 7.3 reduces to a difference equation, and one can find $G_i(s_i)$ s.t. $\overline{v}(s_j)$ is linear on $[a,b]$.

Definition 8. Say χ is a proper u-partition of the interval $[a,b]$ if its CDF is given by:

$$G(x) = \begin{cases} 0, x < a \\ \frac{i+1}{k+1}, x \in [a + i \cdot u, b - (k-1-i) \cdot u), i \in \{0, \dots, k-1\} \\ \frac{i+1}{k}, x \in [b - (k-1-i) \cdot u, a + (i+1) \cdot u), i \in \{0, \dots, k-2\} \\ 1, x \geq b \end{cases}, \text{with } k = \lceil \tfrac{b-a}{u} \rceil$$

Remark 5. This G solves the functional equation $G(x-u) + G(x+u) = 2G(x)$, i.e. $\mathbb{P}(x < \chi \leq x+u) = \mathbb{P}(x-u < \chi \leq x)$, for all $x \in [a,b)$, subject to $supp(\chi) \subseteq [a,b]$.

The proper u-partition of $[a,b]$ is the unique $G_i(s_i)$ supported in $[a,b]$ that makes $\overline{v}(s_j) = \int_0^1 v(s_j - s_i) dG_i(s_i)$ linear on $[a,b]$. Both agents playing it is often not an equilibrium, since $\overline{v}(s_j)$ beyond $[a,b]$ also matters, but it is the key ingredient: the next proposition constructs a symmetric equilibrium for all (u,p) from mixtures of proper u-partitions and, possibly, $\delta_{\{1\}}$.

Proposition 13. *Let $v(x)$ be the CDF of a triangular distribution on $[-u,u]$. Then a symmetric equilibrium signal strategy is:*

(i) If $p \leq 0.5$, the signal is a proper u-partition of $[0, 2p]$.

(ii) If $p > 0.5$ and $u \geq 2 - 2p$, the signal is $\{0,1\}$.

(iii) Otherwise, the signal mixes a proper u-partition of $[0,a]$ for some a with a mass point at 1 and (possibly) with a proper u-partition of $[0, 1-u]$.

So binary support survives a slight violation of well S-shapedness for extreme priors, $p \leq 0.5u$ or $p \geq 1 - 0.5u$, but fails for all $p \in (0.5u, 1 - 0.5u)$. The last case, whose exact equations are in Appendix B, shows how complex equilibria can be for general $v(x)$: as the signal 1 must be treated separately, an equilibrium may require a three-way mixture of two proper u-partitions and a mass point.

7.5 Biased selection rules

Since $v(x) - 0.5$ is odd, every selection rule in this paper treats the agents anonymously: exchanging their signals exchanges their selection probabilities. Anonymity is natural when the agents are ex-ante identical, and it is often required by equal-treatment norms, but it rules out a simple alternative to noise – favoring one of the agents. Head starts of this kind are a standard instrument of contest design (Kirkegaard, 2012), and a designer who wants to select a high-ability contestant can benefit from biasing the contest even between ex-ante identical contestants (Kawamura and Moreno de Barreda, 2014). To compare this instrument with noise, suppose that the firm uses the noiseless rule with a head start $b \in (0, 1)$ for agent 1: agent 1 is selected if and only if $s_1 + b \geq s_2$, so that ties are broken in his favor.[9] I consider symmetric priors $p \leq 0.5$, since for $p > 0.5$ the well S-shaped rule from Corollary 4 already attains the first best.

Proposition 14. *Let the priors be (p, p) with $p \leq 0.5$, and let agent 1 be selected if and only if $s_1 + b \geq s_2$.*

(i) If $b \leq 1 - \sqrt{2p}$, let $M = L$, $c = \frac{2p}{L(2b+L)}$ and $d = \frac{1}{b+L}$, where $L = p + \sqrt{p^2 + 2pb}$.

(ii) If $1 - \sqrt{2p} < b < 1 - p$, let $M = 1 - b$, $c = \frac{2p}{1-b^2}$ and $d = \frac{2(1-b-p)}{1-b^2}$.

(iii) If $b \geq 1 - p$, agent 1 is selected with probability one in every equilibrium, and $u_F = p$.

[9] Tie-breaking matters only in case (ii) of Proposition 14: with ties broken equally, no exact equilibrium exists, but ε-equilibria with agent 1's atom just above $1 - b$ converge to it, as does the firm's utility.

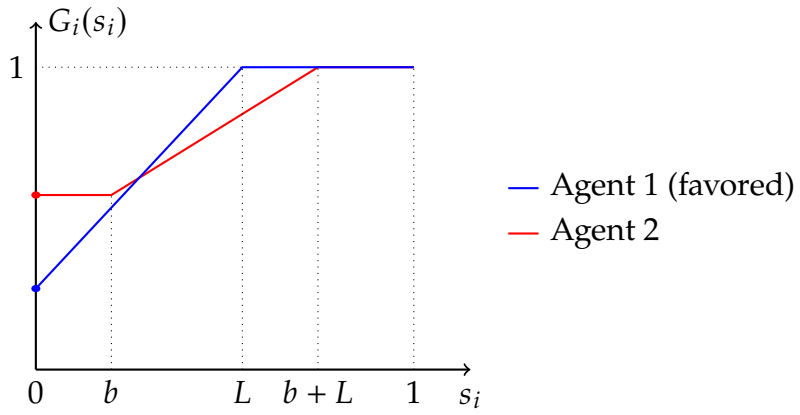


Figure 4: Equilibrium signals under the biased noiseless rule for $p = b = 0.2$

In cases (i) and (ii), the unique equilibrium is: agent 1 sends $s_1 = 0$ *with probability* bd, $s_1 = M$ *with probability* $1 - d(b + M)$, *which is zero in case (i), and otherwise draws* s_1 *from* $U[0, M]$; *agent 2 sends* $s_2 = 0$ *with probability* $1 - cM$ *and otherwise draws* s_2 *from* $U[b, b + M]$. *The firm's utility is* $u_F = p + cdM\left(\frac{M^2}{6} + bM + b^2\right)$.

The equilibrium keeps the structure of the noiseless equilibrium of Boleslavsky and Cotton (2015): on the range where the agents compete, the signal of agent 1 and the handicapped signal of agent 2, $s_2 - b$, are both uniformly distributed on $[0, M]$, so that each agent's value function is linear on the support of his signal. What the head start changes is what lies outside this range, see Figure 4. Agent 2 never sends a signal in $(0, b]$, since such a signal never wins, and concedes bad news instead. In turn, agent 1 concedes bad news too, since the signal $s_1 = 0$ still wins against every concession of agent 2. So, in contrast to the noiseless rule, under which neither agent ever reveals the bad type for $p \leq 0.5$, both agents now reveal it with positive probability. This additional information comes at the cost of selecting agent 1 whenever $s_1 < s_2 \leq s_1 + b$. The two instruments work through different channels: noise changes the curvature of the agents' value functions, whereas a head start keeps them linear and instead makes a range of signals useless for one of the agents.

Corollary 9. *Let $p \leq 0.5$.*

(i) A small head start strictly improves on the noiseless rule: the right derivative of u_F with respect to b at $b = 0$ is positive.

(ii) As $p \to 0$, the share of the first best $\frac{u_F}{u_F^*}$ under the optimal head start converges to 1, whereas under every well S-shaped rule it is at most $\frac{u_F^{ws}}{u_F^*}$, which converges to 0.5.

Hence, the information gain dominates for small head starts, and for sufficiently low priors, a biased noiseless rule outperforms every well S-shaped selection rule, and even the best selection rule within the triangular family of Proposition 3, whose share of the first best converges to 0.75. This ranking reverses as p approaches 0.5, where the best noisy selection rules attain the first best and the biased rule does not. Both instruments are weakly dominated by revelation-accurate negatively balanced rules, which attain the first best for all $p < 0.5$ by Proposition 5 but, by Corollary 6, cannot be generated by i.i.d. shocks. Whether combining a head start with noise improves further on both is left for future research.

7.6 A single agent

Without a rival, the firm compares the agent with an outside option of known value $c \in (p, 1)$: she hires him if his score exceeds c, i.e., with probability $a(s) := \mathbb{P}(s + \varepsilon \geq c)$, where ε has a continuous CDF F with $F(0) = 0.5$. Her gain over not hiring is $W := \mathbb{E}_G[a(s)(s - c)]$, and her first best is $p(1 - c)$. The agent maximizes $\mathbb{E}_G[a(s)]$ over $G \in \mathcal{G}(p)$.

Proposition 15. *Let $c \in (p, 1)$.*

(i) Without noise, $W = 0$. With noise, $W \leq p \min(0.5, 1 - c)$.

(ii) If $\varepsilon \sim U[-u, u]$ with $u < c$ and $c + u \leq 1$, the agent's unique best reply is $\{0, c + u\}$, and $W = \frac{pu}{c+u}$. In particular, if $c > 0.5$ and $u = 1 - c$, he reveals fully, and the firm attains her first best.

(iii) The firm can attain her first best only if $c \geq 0.5$.

Without noise, the agent splits the prior into $\{0, c\}$, and the firm hires him exactly when she is indifferent, so she gains nothing from the evidence. Noise devalues a narrow lead over c, as it devalues a narrow lead over a rival in the main model, so the agent must prove more. The bound in (i) holds because the agent can always split into $\{0, c\}$, which wins with probability at least $\frac{p}{c}a(c) = \frac{p}{2c}$: at the outside option, the noise makes hiring a coin flip. The firm thus hires with probability at least $\frac{p}{2c}$, each hire costs her c, and she can hire at most the mass p of good types. By (ii), uniform noise of half-width $\min(c, 1 - c)$ attains this bound, strictly for $c > 0.5$ and in the limit for $c \leq 0.5$: it is the widest noise that never hires a revealed bad agent and never rejects a revealed good one. As in Corollary 1, a tie is a coin flip, and this is why full revelation requires $c \geq 0.5$.[10] A privately known threshold can likewise force a single sender to reveal fully (Kolotilin et al., 2017); here, the threshold is random to both sides, and the noise is unbiased.

More generally, if the density of ε is unimodal with its peak at 0 and is not constant on $[0, c]$, then $a(s)$ is convex on $[0, c]$ and concave on $[c, 1]$, and the agent splits the prior into $\{0, y\}$ with $y > c$. The firm then gains if and only if p exceeds

$$\hat{p} := \frac{y\, c\, (1 - F(c))}{a(y)(y - c) + c\, (1 - F(c))}$$

Here $1 - F(c)$ is the probability that the noise hires a revealed bad agent, so $\hat{p} = 0$ if $\varepsilon \leq c$,

[10] The bound uses only $a(c) \geq 0.5$, so it holds for any noise with median 0. With mean-zero noise instead, rare shocks far beyond the posterior scale can move the mean without affecting decisions, and the firm can approach her first best for every c.

and for normal noise, $\hat{p}$ is close to 0 for small variances and rises with the variance. Two differences from the main model stand out. Without a rival, the noiseless firm learns nothing useful, whereas with two agents, she already learns much (Boleslavsky and Cotton, 2015). And a single agent's threshold is fixed, so a unimodal density suffices for bad news signal, whereas a rival's randomization makes the threshold random, and the well S-shaped class requires a concave density. The comparison also shows why noise is the relevant instrument: a firm facing a single agent that could commit to any rule would hire only on conclusive evidence and attain her first best without noise, but a firm comparing two agents has no such absolute standard.

References

Abdulkadiroğlu, A. and T. Sönmez (2003). School choice: a mechanism design approach. *American Economic Review 93*(3), 729–747.

Au, P. H. and K. Kawai (2020). Competitive information disclosure by multiple senders. *Games and Economic Behavior 119*, 56–78.

Au, P. H. and M. Whitmeyer (2023). Attraction versus persuasion: Information provision in search markets. *Journal of Political Economy 131*(1), 202–245.

Ball, I. (2025). Scoring strategic agents. *American Economic Journal: Microeconomics 17*(1), 97–129.

Bar, T., V. Kadiyali, and A. Zussman (2012). Putting grades in context. *Journal of Labor Economics 30*(2), 445–478.

Bergemann, D., T. Gan, and Y. Li (2023). Managing persuasion robustly: The optimality of quota rules. Working paper, arXiv:2310.10024.

Blackwell, D. (1953). Equivalent comparisons of experiments. *The Annals of Mathematical Statistics 24*(2), 265–272.

Blume, A., O. J. Board, and K. Kawamura (2007). Noisy talk. *Theoretical Economics 2*(4), 395–440.

Board, S. and J. Lu (2018). Competitive information disclosure in search markets. *Journal of Political Economy 126*(5), 1965–2010.

Boleslavsky, R. and C. Cotton (2015). Grading standards and education quality. *American Economic Journal: Microeconomics 7*(2), 248–279.

Boleslavsky, R. and C. Cotton (2018). Limited capacity in project selection: Competition through evidence production. *Economic Theory 65*(2), 385–421.

Boleslavsky, R. and C. Li (2026). Evaluating university graduates: Prestige versus merit. *Journal of Economics & Management Strategy*. Forthcoming.

Border, K. C. and J. Sobel (1987). Samurai accountant: a theory of auditing and plunder. *Review of Economic Studies 54*(4), 525–540.

Chan, W., H. Li, and W. Suen (2007). A signaling theory of grade inflation. *International Economic Review 48*(3), 1065–1090.

Che, Y.-K. and I. Gale (2000). Difference-form contests and the robustness of all-pay auctions. *Games and Economic Behavior 30*(1), 22–43.

Curello, G. and L. Sinander (2025). The comparative statics of persuasion. Working paper, arXiv:2204.07474.

Drugov, M. and D. Ryvkin (2020). How noise affects effort in tournaments. *Journal of Economic Theory 188*, 105065.

Ederer, F., R. Holden, and M. Meyer (2018). Gaming and strategic opacity in incentive provision. *RAND Journal of Economics 49*(4), 819–854.

Erdil, A. and H. Ergin (2008). What's the matter with tie-breaking? Improving efficiency in school choice. *American Economic Review 98*(3), 669–689.

Fang, D. and T. Noe (2022). Less competition, more meritocracy? *Journal of Labor Economics 40*(3), 669–701.

Fang, F. C. and A. Casadevall (2016). Research funding: the case for a modified lottery. *mBio 7*(2), e00422–16.

Frankel, A. and N. Kartik (2022). Improving information from manipulable data. *Journal of the European Economic Association 20*(1), 79–115.

Fu, Q. and Z. Wu (2019). Contests: Theory and topics. In *Oxford Research Encyclopedia of Economics and Finance*. Oxford University Press.

Gao, E. and D. Luo (2026). (Robust) information acquisition design. Working paper, arXiv:2312.02465.

Gentzkow, M. and E. Kamenica (2016). A Rothschild-Stiglitz approach to Bayesian persuasion. *American Economic Review 106*(5), 597–601.

Gentzkow, M. and E. Kamenica (2017). Competition in persuasion. *The Review of Economic Studies 84*(1), 300–322.

Grossman, S. J. and J. E. Stiglitz (1980). On the impossibility of informationally efficient markets. *American Economic Review 70*(3), 393–408.

Harbaugh, R. and E. Rasmusen (2018). Coarse grades: Informing the public by withholding information. *American Economic Journal: Microeconomics 10*(1), 210–235.

He, W. and J. Li (2023). Competitive information disclosure in random search markets. *Games and Economic Behavior 140*, 132–153.

Heyard, R., M. Ott, G. Salanti, and M. Egger (2022). Rethinking the funding line at the Swiss National Science Foundation: Bayesian ranking and lottery. *Statistics and Public Policy 9*(1), 110–121.

Hirshleifer, J. (1989). Conflict and rent-seeking success functions: Ratio vs. difference models of relative success. *Public Choice 63*(2), 101–112.

Hoppe, H. C., B. Moldovanu, and A. Sela (2009). The theory of assortative matching based on costly signals. *Review of Economic Studies 76*(1), 253–281.

Hvide, H. K. and E. G. Kristiansen (2003). Risk taking in selection contests. *Games and Economic Behavior 42*(1), 172–179.

Hwang, I., K. Kim, and R. Boleslavsky (2026). Competitive advertising and pricing. *The Review of Economic Studies*. Forthcoming.

Jain, V. and M. Whitmeyer (2026). Competitive disclosure of information to a rationally inattentive agent. *Journal of Economic Behavior & Organization 242*, 107410.

Kamenica, E. (2019). Bayesian persuasion and information design. *Annual Review of Economics 11*, 249–272.

Kamenica, E. and M. Gentzkow (2011). Bayesian persuasion. *American Economic Review 101*(6), 2590–2615.

Karlin, S. (1959). *Mathematical Methods and Theory in Games, Programming, and Economics, Vol. II: The Theory of Infinite Games*. Reading, MA: Addison-Wesley.

Kawamura, K. and I. Moreno de Barreda (2014). Biasing selection contests with ex-ante identical agents. *Economics Letters 123*(2), 240–243.

Kirkegaard, R. (2012). Favoritism in asymmetric contests: Head starts and handicaps. *Games and Economic Behavior 76*(1), 226–248.

Kleiner, A., B. Moldovanu, and P. Strack (2021). Extreme points and majorization: Economic applications. *Econometrica 89*(4), 1557–1593.

Kolotilin, A., T. Mylovanov, and A. Zapechelnyuk (2022). Censorship as optimal persuasion. *Theoretical Economics 17*(2), 561–585.

Kolotilin, A., T. Mylovanov, A. Zapechelnyuk, and M. Li (2017). Persuasion of a privately informed receiver. *Econometrica 85*(6), 1949–1964.

Lazear, E. P. and S. Rosen (1981). Rank-order tournaments as optimum labor contracts. *Journal of Political Economy 89*(5), 841–864.

Letina, I., S. Liu, and N. Netzer (2023). Optimal contest design: Tuning the heat. *Journal of Economic Theory 213*, 105616.

Morgan, J., J. Tumlinson, and F. Várdy (2022). The limits of meritocracy. *Journal of Economic Theory 201*, 105414.

Niculescu, C. P. and L.-E. Persson (2003). Old and new on the Hermite–Hadamard inequality. *Real Analysis Exchange 29*(2), 663–686.

Ostrovsky, M. and M. Schwarz (2010). Information disclosure and unraveling in matching markets. *American Economic Journal: Microeconomics 2*(2), 34–63.

Perez-Richet, E. and V. Skreta (2022). Test design under falsification. *Econometrica 90*(3), 1109–1142.

Pier, E. L., M. Brauer, A. Filut, A. Kaatz, J. Raclaw, M. J. Nathan, C. E. Ford, and M. Carnes (2018). Low agreement among reviewers evaluating the same NIH grant applications. *Proceedings of the National Academy of Sciences 115*(12), 2952–2957.

Polyanin, A. D. and A. V. Manzhirov (2008). *Handbook of Integral Equations* (Second ed.). Boca Raton, FL: Chapman & Hall/CRC.

Popov, S. V. and D. Bernhardt (2013). University competition, grading standards, and grade inflation. *Economic Inquiry 51*(3), 1764–1778.

Prat, A. (2005). The wrong kind of transparency. *American Economic Review 95*(3), 862–877.

Prendergast, C. and R. H. Topel (1996). Favoritism in organizations. *Journal of Political Economy 104*(5), 958–978.

Röbbecke, M. and D. Simon (2023). Riskante Forschung und teilrandomisierte Begutachtungsverfahren: Neue Wege der Förderlinie "Experiment!" der VolkswagenStiftung. *Beiträge zur Hochschulforschung 45*(2), 8–30.

Sakhnovich, L. A. (2015). *Integral Equations with Difference Kernels on Finite Intervals* (Second ed.), Volume 84 of *Operator Theory: Advances and Applications*. Cham: Birkhäuser.

Scharfstein, D. S. and J. C. Stein (2000). The dark side of internal capital markets: Divisional rent-seeking and inefficient investment. *The Journal of Finance 55*(6), 2537–2564.

Stone, P. (2011). *The Luck of the Draw: The Role of Lotteries in Decision Making*. New York: Oxford University Press.

Tsakas, E. and N. Tsakas (2021). Noisy persuasion. *Games and Economic Behavior 130*, 44–61.

Tsakas, E., N. Tsakas, and D. Xefteris (2021). Resisting persuasion. *Economic Theory 72*(3), 723–742.

Tullock, G. (1980). Efficient rent seeking. In J. M. Buchanan, R. D. Tollison, and G. Tullock (Eds.), *Toward a Theory of the Rent-Seeking Society*, pp. 97–112. College Station: Texas A&M University Press.

Wang, Z. (2010). The optimal accuracy level in asymmetric contests. *The B.E. Journal of Theoretical Economics 10*(1), Article 13.

A. Proofs of Sections 3–5

A.1 Properties of well S-shaped selection rules

Throughout the proofs, I use the following properties of a well S-shaped $v(x)$.

1) *Regularity.* Since $v'(x)$ is concave, it is continuous on $(-1,1)$ and has one-sided derivatives there; I set $v'(\pm 1)$ equal to its one-sided limits. I write $v''(x)$ and $\tilde{v}''(x)$ for either one-sided derivative, and every statement involving them holds for both.

2) *Shape.* Since $v(x)$ is increasing, $v'(1) \geq 0$, and since $v'(x)$ is even, concave and uniquely maximized at 0, $v'(x) \geq (1-|x|)v'(0) + |x|v'(1) > v'(1) \geq 0$ on $(-1,1)$, so $v(x)$ is strictly increasing. For $0 \leq x < y \leq 1$, the same argument gives $v'(x) \geq \frac{y-x}{y}v'(0) + \frac{x}{y}v'(y) > v'(y)$, so $v'(x)$ is strictly decreasing on $[0,1]$. Since $\tilde{v}'(x) = \frac{v'(x)-v'(0)}{|v(1)-0.5-v'(0)|}$, it is even, concave, and strictly decreasing on $[0,1]$ with $\tilde{v}'(0) = 0$. So $\tilde{v}(x)$ is negative, strictly decreasing and strictly concave on $(0,1]$. By the concavity of $\tilde{v}'(x)$, $\tilde{v}''(x) \leq \frac{\tilde{v}'(x)}{x} < 0$ on $(0,1)$, and $-\tilde{v}''(x)$ is non-decreasing on $(0,1)$.

3) *Elasticity.* The following lemma bounds the elasticity of $\tilde{v}(x)$ from below.

Lemma 9. *Let $v(x)$ be well S-shaped. Then $\mathcal{E}_{\tilde{v}}(x) \geq 2$ for all $x \in (0,1]$, with equality at x only if $\tilde{v}'(t)$ is linear on $[0,x]$. Hence $\frac{\tilde{v}(x)}{x^2}$ is non-increasing on $(0,1]$, so $\tilde{v}(x) \geq -x^2$ on $[0,1]$; if $v'(x)$ is not linear near 0, the ratio is strictly decreasing.*

Proof. Let $x \in (0,1]$ and $\phi := -\tilde{v}'$, which is convex on $[0,x]$ with $\phi(0) = 0$ by 2), so $\phi(t) \leq \frac{t}{x}\phi(x)$ for $t \in [0,x]$. Hence $-\tilde{v}(x) = \int_0^x \phi(t)dt \leq \frac{x\phi(x)}{2}$, i.e., $\mathcal{E}_{\tilde{v}}(x) \geq 2$, with

equality only if ϕ is linear on $[0, x]$. Finally, $\left(\frac{\tilde{v}(x)}{x^2}\right)' = \frac{\tilde{v}(x)(\mathcal{E}_{\tilde{v}}(x)-2)}{x^3} \leq 0$, with strict inequality if $v'(x)$ is not linear near 0, and $\tilde{v}(1) = -1$. □

4) *Non-linearity near* 0. If $v'(x)$ is not linear near 0, then it is not affine on any interval whose closure contains 0: on an interval with 0 in its interior, an affine $v'(x)$ would be constant, contradicting the unique maximum at 0, and an affine piece ending at 0 gives, by evenness, one starting at 0. Hence, for every $G \in \mathcal{G}(p)$, $\overline{v}'_G(s) := \int_0^1 v'(s-t)dG(t)$ is not affine on any interval $[\alpha, \beta]$ with $G([\alpha, \beta]) > 0$. Indeed, $\overline{v}'_G$ is a mixture of the concave functions $v'(s-t)$, so it is affine on $[\alpha, \beta]$ only if $v'(x)$ is affine on $[\alpha - t, \beta - t]$ for G-almost every t, and the closure of this interval contains 0 whenever $t \in [\alpha, \beta]$.

A.2 Proof of Lemma 1

(i) As shown in the text, $\overline{v}'(s_i) = \int_0^1 v'(s_i - s_j)dG_j(s_j)$ is concave on $[0, 1]$. For a maximizer c of $\overline{v}'$, $\overline{v}$ is convex on $[0, c]$ and concave on $[c, 1]$.

(ii) With $slp(0) = \overline{v}'(0)$, slp is continuous on $[0, 1]$, so a exists. For $s \in (0, 1]$, differentiating gives $slp'(s) = \frac{N(s)}{s^2}$, where

$$N(s) := s\overline{v}'(s) - \overline{v}(s) + \overline{v}(0) = \int_0^s \left(\overline{v}'(s) - \overline{v}'(t)\right) dt.$$

By (i), $N(s) \geq 0$ on $[0, c]$, and $N(s)$ is non-increasing on $[c, 1]$, since for $c \leq s_1 < s_2 \leq 1$,

$$N(s_2) - N(s_1) = s_1 \left(\overline{v}'(s_2) - \overline{v}'(s_1)\right) + \int_{s_1}^{s_2} \left(\overline{v}'(s_2) - \overline{v}'(t)\right) dt \leq 0.$$

Hence slp is non-decreasing on $[0, c]$ and changes its monotonicity on $[c, 1]$ at most once, from non-decreasing to non-increasing. So $a \geq c$, and $\overline{v}$ is concave on $[a, 1]$. Let $l(s) := \overline{v}(0) + s \cdot slp(a)$. Then $l \geq \overline{v}$ on $[0, a]$ with equality at 0 and a, and if $a < 1$, $N(a) = 0$, i.e., l is tangent to $\overline{v}$ at a. Therefore $\overline{v}^{cv}$ equals l on $[0, a]$ and $\overline{v}$ on $[a, 1]$. By Kamenica and Gentzkow (2011), a signal is optimal if and only if it is supported where

$\overline{v}$ touches a line supporting $\overline{v}^{cv}$ at p_i. So $\{0, a\}$ is optimal if $p_i < a$, and $\{p_i\}$ is optimal if $p_i \geq a$, where $\overline{v}^{cv}(p_i) = \overline{v}(p_i)$.

If $v'(x)$ is not linear near 0, the concave $\overline{v}'$ is strictly increasing on $[0, z_1]$, constant on $[z_1, z_2]$ and strictly decreasing on $[z_2, 1]$ for some $z_1 \leq z_2$, and $z_1 > 0$ unless $z_2 = 0$: if $\overline{v}'$ were constant on $[0, z_2]$ with $z_2 > 0$, then $G_j([0, z_2]) = 0$ by property 4) in Appendix A.1, so $\overline{v}''(s) = \int_{z_2}^{1} v''(s - s_j) dG_j(s_j) > 0$ on $(0, z_2)$, as $v'' > 0$ on $[-1, 0)$ by property 2), a contradiction. Hence $N > 0$ on $(0, z_2]$ and N is strictly decreasing on $[z_2, 1]$, so $\overline{v} < l$ on $(0, a)$, and $\overline{v}$ is strictly concave on $[a, 1]$. Hence the supporting line at p_i touches $\overline{v}$ only on $\{0, a\}$ if $p_i \leq a$ and only at p_i if $p_i > a$, so the optimum is unique.

A.3 Proof of Proposition 1

Let $\overline{v}(s_i)$ denote the value function of agent i for $\tilde{v}(x)$ rather than $v(x)$, which does not change best replies. If agent j plays $\{0, 1\}$, then $\overline{v}(s_i) = (1 - p_j)\tilde{v}(s_i) - p_j\tilde{v}(1 - s_i)$, which is S-shaped by Lemma 1, with $\overline{v}(1) - \overline{v}(0) = \tilde{v}(1) = -1$ and $\overline{v}'(1) = (1 - p_j)\tilde{v}'(1)$. By Lemma 1, $\{0, 1\}$ is a best reply if and only if $a = 1$, i.e., if and only if $\overline{v}$ lies below its secant $L(s) := \overline{v}(0) - s$ on $[0, 1]$. This implies $\overline{v}'(1) \geq -1$. Conversely, if $a < 1$, then $\overline{v}$ is concave on $[a, 1]$ by the proof of Lemma 1, so $\overline{v}(a) \leq \overline{v}(1) - (1 - a)\overline{v}'(1)$, while $slp(a) > slp(1)$ gives $\overline{v}(a) > \overline{v}(0) - a$; together, $\overline{v}'(1) < -1$. Hence

$$\{0, 1\} \text{ is a best reply } \Leftrightarrow \ (1 - p_j)\tilde{v}'(1) \geq -1 \ \Leftrightarrow \ p_j \geq 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)} = \overline{p},$$

and $(\{0, 1\}; \{0, 1\})$ is an equilibrium if and only if $p_1, p_2 \geq \overline{p}$.

For uniqueness, since all equilibria form a product set, it suffices that $\{0, 1\}$ is the unique best reply to $\{0, 1\}$ for both agents. Another best reply exists only if $D := \overline{v} - L \leq 0$ vanishes at some $s_0 \in (0, 1)$. Since D is convex on $[0, c]$ and concave on $[c, 1]$ for some c, it then vanishes on $[s_0, 1]$: a convex D with an interior maximum at $s_0 < c$ vanishes on $[0, c]$, and a concave D that vanishes at both ends of $[\max(s_0, c), 1]$

is non-negative there. Hence $\overline{v}'(1) = -1$, i.e., $p_j = \overline{p}$, and $\overline{v}$ is affine on $[s_0, 1]$, which property 4) in Appendix A.1 rules out if $v'(x)$ is not linear near 0, as $\{0, 1\}$ puts mass at 1. This proves uniqueness if $v'(x)$ is not linear near 0 or $p_1, p_2 > \overline{p}$. For $p = \overline{p}$, the uniqueness follows from the uniqueness part of the proof of Theorem 1, which covers $p = \overline{p}$.

A.4 Proof of Theorem 1

If $p \geq \overline{p}$, full revelation is an equilibrium by Proposition 1, and it is unique for $p > \overline{p}$ by Proposition 1 and for $p = \overline{p}$ by the uniqueness part of this proof below. So let $p < \overline{p}$, and suppose agent j plays a signal $\{0, a\}$. Then for i we have

$$\overline{v}(s_i) = \frac{a-p}{a}\tilde{v}(s_i) + \frac{p}{a}\tilde{v}(s_i - a).$$

Then, for $\{0, a\}$ to be a best reply, it should be true that the slope of the secant is equal to the slope of the tangent:

$$\overline{v}'(a) = \frac{a-p}{a}\tilde{v}'(a) = \frac{a-p}{a}\frac{\tilde{v}(a)}{a} + \frac{p}{a}\frac{\tilde{v}(a)}{a},$$

hence $(a-p)\tilde{v}'(a) = \tilde{v}(a)$.

Then, since $\tilde{v}'(a) < 0 \; \forall a \in (0, 1]$, a should be a solution of the equation $p = a - \frac{\tilde{v}(a)}{\tilde{v}'(a)}$. The RHS is the zero of the tangent $T_a(x) := \tilde{v}(a) + (x-a)\tilde{v}'(a)$, which is strictly decreasing in x, and it lies in $[0, a)$. For $a_1 < a_2$ and all $x \leq a_1$,

$$T_{a_2}(x) - T_{a_1}(x) = \int_{a_1}^{a_2} \left(\tilde{v}'(t) - \tilde{v}'(a_2)\right) dt + (a_1 - x)\left(\tilde{v}'(a_1) - \tilde{v}'(a_2)\right) > 0,$$

since $\tilde{v}'(x)$ is strictly decreasing on $[0, 1]$. So T_{a_2} is positive at the zero of T_{a_1}, and its own zero lies strictly to the right. Hence the RHS is strictly increasing in a, and the

solution is unique and monotone in p, if it exists.

To prove existence, notice that the RHS is continuous in a and its value at $a = 1$ is $\overline{p}$. By the definition of $\mathcal{E}_{\tilde{v}}(x)$ and Lemma 9,

$$a - \frac{\tilde{v}(a)}{\tilde{v}'(a)} = a\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right) \in \left[\frac{a}{2}, a\right),$$

hence the RHS converges to 0 as $a \to 0_+$. So $\forall p \in (0, \overline{p})$ there always exists a unique monotone solution to this equation, which is exactly the equation from the statement. Moreover, as the inverse of a continuous and strictly increasing function, $a(p)$ is continuous and strictly increasing.

Best reply. Let $p \in (0, \overline{p}]$, where for $p = \overline{p}$, $a = 1$ satisfies $(a - p)\tilde{v}'(a) = \tilde{v}(a)$ by the definition of $\overline{p}$. Let $l(s)$ be the secant from $(0, \overline{v}(0))$ to $(a, \overline{v}(a))$ and $d(s) := l(s) - \overline{v}(s)$, so that $d(0) = d(a) = 0$, and $d'(a) = 0$ by this equation. As $\tilde{v}'(x)$ is even,

$$d'(s) = \frac{\tilde{v}(a)}{a} - \frac{a - p}{a}\tilde{v}'(s) - \frac{p}{a}\tilde{v}'(a - s),$$

which is convex on $[0, a]$, since $\tilde{v}'(x)$ is concave. By the condition above, $d'(0) = \frac{\tilde{v}(a)}{a} - \frac{p}{a}\tilde{v}'(a) = \frac{a-2p}{a}\tilde{v}'(a) \geq 0$, since $\tilde{v}'(a) < 0$ and $\frac{p}{a} = 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)} \in [\frac{1}{2}, 1)$ by Lemma 9, i.e., $p < a \leq 2p$. So $d' \geq 0$ on some $[0, z]$ and $d' \leq 0$ on $[z, a]$, and hence $d \geq 0$ on $[0, a]$. For $s \in (a, 1]$, both s and $s - a$ lie in $(0, 1]$, where $\tilde{v}'(x)$ is strictly decreasing, so

$$\overline{v}'(s) = \frac{a - p}{a}\tilde{v}'(s) + \frac{p}{a}\tilde{v}'(s - a) < \frac{a - p}{a}\tilde{v}'(a) + \frac{p}{a}\tilde{v}'(0) = \overline{v}'(a),$$

and $\overline{v}(s)$ lies strictly below its tangent $l(s)$ on $(a, 1]$. Hence $l(s)$ supports $\overline{v}(s)$ on $[0, 1]$ and touches it at 0 and $a > p$, so $\{0, a\}$ is a best reply to itself, and every best reply to $\{0, a\}$ is supported on $\{s \in [0, a] : d(s) = 0\}$.

Uniqueness. By the constant-sum property, all equilibria form a symmetric product

set $E \times E$ with $\{0, a\} \in E$. Hence, if $G \in E$, then both $(G, \{0, a\})$ and $(\{0, a\}, G)$ are equilibria, i.e., G is a best reply to $\{0, a\}$, and $\{0, a\}$ is a best reply to G.

Step 1: best replies to $\{0, a\}$. Suppose that $d(c) = 0$ for some $c \in (0, a)$. Then c minimizes d on $[0, a]$, so $d'(c) = 0$. By convexity, $d'(s) \leq 0$ on $[c, a]$, and since $\int_c^a d'(s)ds = d(a) - d(c) = 0$, $d' \equiv 0$ on $[c, a]$. Again by convexity, $d'(s) \geq 0$ on $[0, c]$, and since $d(0) = d(c) = 0$, $d' \equiv 0$ on $[0, c]$. Hence $\frac{a-p}{a}\tilde{v}'(s) + \frac{p}{a}\tilde{v}'(a - s)$ is constant on $[0, a]$. A sum of two concave functions is affine only if both are affine, so $\tilde{v}'(s)$ is affine on $[0, a]$, and as $\tilde{v}'(0) = 0 > \tilde{v}'(a)$, $\tilde{v}'(s) = -Ks$ on $[0, a]$ for some $K > 0$. Then

$$\frac{a-p}{a}\tilde{v}'(s) + \frac{p}{a}\tilde{v}'(a-s) = -pK - \frac{K(a-2p)}{a}s$$

is constant only if $a = 2p$.

So, either $d(s) > 0$ on $(0, a)$, in which case $\{0, a\}$ is the only best reply to $\{0, a\}$, and the equilibrium is unique, or $\tilde{v}(x) = -\frac{K}{2}x^2$ on $[0, a]$ and $a = 2p$.

Step 2: the quadratic case. Let $\tilde{v}(x) = -\frac{K}{2}x|x|$ on $[-2p, 2p]$ and $a = 2p$, so that $d \equiv 0$ on $[0, 2p]$ and every G with support in $[0, 2p]$ and mean p is a best reply to $\{0, 2p\}$. Take such a $G \neq \{0, 2p\}$ and let $w(s) := \int_0^{2p} \tilde{v}(s - t)dG(t)$ be the normalized value function against it. For $s \in [0, 2p]$, $w'(s) = -K\int_0^{2p} |s - t|dG(t)$, hence $w'(2p) = -Kp$, while $w(0) = \frac{K}{2}\mathbb{E}_G[t^2]$ and $w(2p) = -\frac{K}{2}\mathbb{E}_G[(2p - t)^2] = -\frac{K}{2}\mathbb{E}_G[t^2]$. So the slope of the secant from $(0, w(0))$ to $(2p, w(2p))$ exceeds $w'(2p)$ by

$$-\frac{K\mathbb{E}_G[t^2]}{2p} + Kp = \frac{K}{2p}\left(2p^2 - \mathbb{E}_G[t^2]\right) > 0,$$

since $t^2 \leq 2pt$ on $[0, 2p]$, with equality only at $t \in \{0, 2p\}$. Hence $w(s)$ lies strictly above the secant for $s < 2p$ close enough to $2p$, so, for such $s > p$, the signal $\{0, s\}$ yields a strictly higher payoff against G than $\{0, 2p\}$. Thus $\{0, 2p\}$ is not a best reply to G, $G \notin E$, and the equilibrium is unique.

Let me also introduce a bound on a that is used in later proofs.

Lemma 10. *Let $v(x)$ be well S-shaped and $p \in (0, \overline{p})$. In the symmetric equilibrium $(\{0, a\}; \{0, a\})$, $p < a \leq 2p$, and $a < 2p$ if $v'(x)$ is not linear near 0.*

Proof. Since $\frac{p}{a} = 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}$, and $\mathcal{E}_{\tilde{v}}(a) \geq 2$ by Lemma 9, with strict inequality if $v'(x)$ is not linear near 0, $p < a \leq 2p$, and $a < 2p$ in the latter case. □

A.5 Proof of Lemma 2

Against $\{p_j\}$, $\overline{v}(s) = v(s - p_j)$, so $\overline{v}'(s) = v'(s - p_j)$ is strictly increasing on $[0, p_j]$, since $v'(x)$ is even and strictly decreasing on $[0, 1]$ by property 2) in Appendix A.1. As in the proof of Lemma 1, let $slp(s) := \frac{\overline{v}(s) - \overline{v}(0)}{s}$, so that $slp'(s) = \frac{N(s)}{s^2}$ with $N(s) = \int_0^s (\overline{v}'(s) - \overline{v}'(t))dt$. Then $N > 0$ on $(0, p_j]$, and, by continuity, slp is strictly increasing on $(0, s]$ for some $s \in (p_j, 1]$. If $p_i \leq p_j$, the signal $\{0, s\}$ with mean p_i yields $\overline{v}(0) + p_i\, slp(s) > \overline{v}(0) + p_i\, slp(p_i) = \overline{v}(p_i)$, so $\{p_i\}$ is not a best reply. Hence $\{p_i\}$ is a best reply only if $p_i > p_j$, which cannot hold for both agents.

A.6 Proof of Lemma 11

Lemma 11. *Let $p \in (0, 1)$, let $v_n(x)$ be well S-shaped for all $n \in \mathbb{N}$, and let $(\{0, a_n\}; \{0, a_n\})$ be the equilibrium for v_n and priors (p, p). If $\mathcal{E}_n(x) \to \infty$ uniformly on $(0, 1)$, then $a_n \to p$.*

Since $\mathcal{E}_n(x) \to \infty$ uniformly on $(0, 1)$ and $\mathcal{E}_n(x)$ is continuous on $(0, 1]$, $\varepsilon_n := \sup_{x \in (0,1]} \frac{1}{\mathcal{E}_n(x)} \to 0$. Hence $p < \overline{p}_n = 1 - \frac{1}{\mathcal{E}_n(1)}$ for large n, and then, by Theorem 1, $a_n = \frac{p}{1 - 1/\mathcal{E}_n(a_n)} \in \left[p, \frac{p}{1-\varepsilon_n}\right]$, so $a_n \to p$.

A.7 Proof of Lemma 3

By Theorem 1, $\frac{\mathbb{P}(s_i = 0)}{1-p} = \frac{1}{1 + (\mathcal{E}_{\tilde{v}}(a) - 1)(1-a)}$, where, by its proof, $a(p)$ is continuous and strictly increasing, so it maps $(p, p + \varepsilon)$ and $(p - \varepsilon, p)$ onto a right and a left neighbourhood of

$a(p)$. Therefore agents reveal more for all $p_h \in (p, p+\varepsilon)$ and less for all $p_l \in (p-\varepsilon, p)$ if and only if $(\mathcal{E}_{\tilde{v}}(x)-1)(1-x)$ is strictly decreasing at $x = a$.

A.8 Proof of Proposition 2

By Theorem 1 and Proposition 1, the equilibrium is $(\{0,a\};\{0,a\})$ with $a = a(p) > p$, where $a(p)$ is non-decreasing on $(0,1)$, as $a(p) = 1$ for $p \geq \overline{p}$. When both agents play $\{0,a\}$ with prior $p < a$, the firm gets

$$U(p,a) := \frac{p^2}{a} + \frac{2p(a-p)}{a}v(a) = p + p\Big(1-\frac{p}{a}\Big)\big(2v(a)-1\big).$$

For a fixed p, U is non-decreasing in a, as $1-\frac{p}{a}$ and $2v(a)-1$ are non-negative and non-decreasing. For a fixed a, $\partial_p U = 1 + \big(1-\frac{2p}{a}\big)\big(2v(a)-1\big) > 2\big(1-v(a)\big) \geq 0$, as $1-\frac{2p}{a} > -1$ and $0 < 2v(a)-1 \leq 1$. Hence, for $p_1 < p_2$,

$$u_F(p_1) = U\big(p_1, a(p_1)\big) \leq U\big(p_1, a(p_2)\big) < U\big(p_2, a(p_2)\big) = u_F(p_2).$$

A.9 Proof of Theorem 2

Lemma 12. *Let $v(x)$ be well S-shaped and $\hat{v}(x) := v(x) - 0.5$. Then $\hat{v}(x) \leq x(2-x)\hat{v}(1) \leq x - \frac{x^2}{2}$ for all $x \in [0,1]$, and if $\hat{v}(x) = x - \frac{x^2}{2}$ for some $x \in (0,1)$, then $v = v^*$.*

Proof. For $0 \leq t < s < 1$, s is a convex combination of t and 1 with weight $\frac{1-s}{1-t}$ on t, so the concavity of $v'(x)$ and $v'(1) \geq 0$ give $v'(s) \geq \frac{1-s}{1-t}v'(t)$. Hence $r(t) := \frac{v'(t)}{1-t}$ is non-decreasing on $[0,1)$: the density $v'(t)$ has a monotone likelihood ratio relative to the triangular density $1-t$. Let $T(x) := \int_0^x (1-t)dt = x - \frac{x^2}{2}$. For $x \in (0,1)$,

$$\frac{\hat{v}(x)}{T(x)} = \frac{\int_0^x r(t)(1-t)dt}{\int_0^x (1-t)dt} \leq r(x) \leq \frac{\int_x^1 r(t)(1-t)dt}{\int_x^1 (1-t)dt} = \frac{\hat{v}(1)-\hat{v}(x)}{T(1)-T(x)},$$

so $\hat{v}(x) \leq \frac{T(x)}{T(1)}\hat{v}(1) = x(2-x)\hat{v}(1) \leq x - \frac{x^2}{2}$, as $\hat{v}(1) \leq 0.5$. If $\hat{v}(x) = x - \frac{x^2}{2}$ for some $x \in (0,1)$, then $\hat{v}(1) = 0.5$ and both inequalities in the display are equalities, so r, which is continuous on $[0,1)$ by property 1) in Appendix A.1, is constant there and equal to $2\hat{v}(1) = 1$. Hence $v'(t) = 1 - t$ on $[0,1]$, and $v = v^*$, as $v'(x)$ is even. □

By Theorem 1 and Proposition 1, the equilibrium is $(\{0,a\};\{0,a\})$, where $a = 1$ if and only if $p \geq \overline{p}$, and $p < a \leq a^* := \min(2p, 1)$ by Lemma 10 and $\overline{p} \geq 0.5$ (Lemma 9). As in the proof of Proposition 2, $u_F(p) = p + 2p\left(1 - \frac{p}{a}\right)\hat{v}(a)$, so, by Lemma 12,

$$u_F(p) \leq p + 2p\left(1 - \frac{p}{a}\right)\left(a - \frac{a^2}{2}\right) \leq p + 2p\left(1 - \frac{p}{a^*}\right)\left(a^* - \frac{a^{*2}}{2}\right) = u_F^{ws}(p),$$

where the second inequality holds as both factors are positive and strictly increasing in a on $(p,1]$. This proves (i), and equality holds if and only if $a = a^*$ and $\hat{v}(a) = a - \frac{a^2}{2}$.

(ii) If $p \geq 0.5$, then $a^* = 1$, so equality means $a = 1$, i.e., $p \geq \overline{p}$, and $v(1) = 1$.

(iii) If $p < 0.5$, then $a^* = 2p \in (0,1)$, so equality requires $\hat{v}(2p) = 2p - 2p^2$, hence $v = v^*$ by Lemma 12. Conversely, for v^*, $\tilde{v}(x) = -x^2$ on $[0,1]$, so $a = 2p$ by Theorem 1, and both inequalities hold with equality.

A.10 Proof of Corollary 4

(i) Here $\tilde{v}(x) = -x^2$ on $[0,1]$, so $\overline{p} = 0.5$ and $v(1) = 1$; apply Theorem 2(ii) for $p \geq 0.5$ and (iii) for $p < 0.5$.

(ii) By the Fact in Appendix D, the utility of the firm under the noiseless selection rule is $\frac{4p}{3}$ for $p \leq 0.5$ and $1 - \frac{(1-p)^2(4p-1)}{3p^2}$ for $p > 0.5$. For $p \leq 0.5$, by (i),

$$p + 2p^2(1-p) - \frac{4p}{3} = -\frac{p}{3}\left(6p^2 - 6p + 1\right) > 0 \Leftrightarrow p \in \left(\frac{3-\sqrt{3}}{6}, \frac{3+\sqrt{3}}{6}\right).$$

For $p > 0.5$, by (i),

$$2p - p^2 - 1 + \frac{(1-p)^2(4p-1)}{3p^2} = \frac{(1-p)^2(4p-1-3p^2)}{3p^2} = \frac{(1-p)^3(3p-1)}{3p^2} > 0.$$

A.11 Proof of Proposition 3

Throughout, G_u is the equilibrium signal constructed in Proposition 13, and proper u-partitions are as defined in Subsection 7.4.

(i) Let $p \leq 0.5$, $u < 2p$, $k = \lceil \frac{2p}{u} \rceil \geq 2$ and $r = 2p - (k-1)u \in (0, u]$. The proper u-partition of $[0, 2p]$ has atoms iu with masses $\frac{k-i}{k(k+1)}$ and atoms $iu + r = 2p - (k-1-i)u$ with masses $\frac{i+1}{k(k+1)}$, $i = 0, \ldots, k-1$. For fixed k, as u increases, iu moves at rate i and $iu + r$ at rate $-(k-1-i)$, so $\Psi_u(t) := \int_0^t G_u(x)dx$ satisfies $\partial_u \Psi_u(t) = -M(t)$, where $M(t)$ sums rate times mass over the atoms below t. The pair $\{iu, iu + r\}$ contributes $\frac{2i+1-k}{k(k+1)}$ and the atom iu alone $\frac{i(k-i)}{k(k+1)}$, hence $M(t) = 0$ for $t \in (ju, ju + r]$ and $M(t) = \frac{(j+1)(j+1-k)}{k(k+1)} < 0$ for $t \in (ju + r, (j+1)u]$, $j \leq k-2$. So $\Psi_u(t)$ is non-decreasing in u, and strictly increasing for t in a set of positive measure, i.e. G_u is a strict mean-preserving spread of $G_{u'}$ for $u' < u$ with the same k. At $u = \frac{2p}{k}$, $r = u$ and G_u is the uniform distribution on $\{0, \frac{2p}{k}, \ldots, 2p\}$, which is also its limit as $u \uparrow \frac{2p}{k}$, where $k+1$ is used and $r \to 0$; so G_u is continuous in u and the ranking extends to $(0, 2p]$. By Proposition 13, $G_u = \{0, \min(2p, 1)\}$ if $u \geq 2p$ and $p \leq 0.5$, or $u \geq 2 - 2p$ and $p > 0.5$.

For the example after Proposition 3, let $p \geq \frac{5}{9}$ and $u \in [1-p, 2-2p)$. Then $k = 1$ in case (iii) of the full statement in the proof of Proposition 13, and $a = 2u - \frac{u^2}{1-p} \leq 1 - u$, as $3u - \frac{u^2}{1-p}$ is maximized at $u = \frac{3(1-p)}{2}$, where it equals $\frac{9(1-p)}{4} \leq 1$. The proper u-partition of $[0, a]$ is $\{0, a\}$ with equal probabilities, so G_u puts probabilities $\frac{1-p}{2-a}$, $\frac{1-p}{2-a}$ and $\frac{2p-a}{2-a}$ on $\{0, a, 1\}$. As $\frac{da}{du} = 2 - \frac{2u}{1-p} < 0$ for $u > 1 - p$, a higher u lowers the probability of 0 and raises that of 1, so no two of these signals are ranked in the Blackwell order.

(ii) Since $v(-x) = 1 - v(x)$, with $d = s_1 - s_2$,

$$u_F = \mathbb{E}[s_1 v(d) + s_2 v(-d)] = \mathbb{E}[\max(s_1, s_2)] - \mathbb{E}\big[|d|\big(1 - v(|d|)\big)\big],$$

and $v(x) = 1 - \frac{(u-x)^2}{2u^2}$ for $x \in [0, u]$. Let $p \leq 0.5$. For $u \geq 2p$, $G_u = \{0, 2p\}$ and

$$u_F = \frac{p}{2} + p\, v(2p) = \frac{3p}{2} - \frac{p(u-2p)^2}{2u^2},$$

which equals $\frac{3p}{2}$ at $u = 2p$ and is strictly decreasing on $[2p, 1]$. For $u < 2p$, $u_F \leq \mathbb{E}[\max(s_1, s_2)] < \frac{3p}{2}$: for $s_2 \in [0, 2p]$, the convex function $\max(\cdot, s_2)$ lies below its chord on $[0, 2p]$, strictly on $(0, 2p)$ if $s_2 \in (0, 2p)$, so $\mathbb{E}_{s_1}[\max(s_1, s_2)] \leq p + \frac{s_2}{2}$, strictly if $s_2 \in (0, 2p)$, as G_u has mass in $(0, 2p)$; averaging over s_2 gives the bound $\frac{3p}{2}$, which is strict for the same reason.

Let $p > 0.5$. For $u \in [2 - 2p, 1]$, $G_u = \{0, 1\}$ and $v(1) = 1$, so $u_F = u_F^*(p)$. For $u < 2 - 2p$, $v(x)$ does not implement a fully revealing equilibrium by the proof of Proposition 13 (ii), so $u_F < u_F^*(p)$ by Lemma 4.

(iii) The proper u-partition of $[0, a]$ with $k = \lceil \frac{a}{u} \rceil$ puts mass $\frac{1}{k}$ on each pair $\{iu, iu + r\} \subset [iu, (i+1)u]$, $i = 0, \ldots, k-1$, so it converges weakly to $U[0, a]$ as $u \to 0$. If $p \leq 0.5$, G_u is the proper u-partition of $[0, 2p]$ and converges to $U[0, 2p]$. If $p > 0.5$ and $u < \min(2p - 1, 2 - 2p)$, then with k from Proposition 13 (iii), we have $a \in ((k-1)u, ku]$ and $ku < 2 - 2p \leq (k+1)u$, so $a \in (2 - 2p - 2u, 2 - 2p)$ and $a < 1 - u$. Hence case (iii) applies, $a \to 2 - 2p$, the weight $\frac{2(1-p)}{2-a} \to \frac{1-p}{p}$, and G_u converges to the $(\frac{1-p}{p}, \frac{2p-1}{p})$ mix of $U[0, 2 - 2p]$ and $\delta_{\{1\}}$ — exactly the distributions from Appendix D.

A.12 Proof of Lemma 4

Since $v(-x) = 1 - v(x)$, for $s_1 \geq s_2$,

$$\begin{aligned} s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1) &= s_2 + (s_1 - s_2) v(s_1 - s_2) \leq s_2 + (s_1 - s_2) v(1) \\ &= v(1) \max(s_1, s_2) + (1 - v(1)) \min(s_1, s_2) \leq \max(s_1, s_2), \end{aligned}$$

where the first inequality holds as $v(x)$ is increasing and $s_1 - s_2 \leq 1$, and the second as $v(1) \leq 1$, strictly if $s_1 \neq s_2$ and $v(1) < 1$. Hence $u_F^*(p) \leq \max_{G_1, G_2 \in \mathcal{G}(p)} \mathbb{E}[\max(s_1, s_2)]$. For $s_2 \in [0, 1]$, $\max(s, s_2)$ lies below its secant $s_2 + s(1 - s_2)$ on $[0, 1]$, strictly on $(0, 1)$ if $s_2 \in (0, 1)$. Hence $\mathbb{E}_{G_1}[\max(s_1, s_2)] \leq s_2 + p(1 - s_2)$, with equality for all s_2 if G_1 is full revelation, and for $s_2 \in (0, 1)$ only then. Averaging over s_2, $\mathbb{E}[\max(s_1, s_2)] \leq 2p - p^2$, with equality if and only if at least one agent plays full revelation. If both agents play full revelation and $v(1) = 1$, then $s_1 - s_2 \in \{-1, 0, 1\}$ and both inequalities above are equalities, so $u_F^*(p) = 2p - p^2$.

By definition, $\overline{u}_F(p) \leq u_F^*(p)$. Let (G_1^*, G_2^*) be an equilibrium with $\mathbb{E}[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)] = u_F^*(p)$. Then all inequalities above are equalities, so some agent plays full revelation, and since then $s_1 \neq s_2$ with positive probability, $v(1) = 1$. As all equilibria form a symmetric product set, $(\{0, 1\}; \{0, 1\})$ is then an equilibrium too. Conversely, if $(\{0, 1\}; \{0, 1\})$ is an equilibrium and $v(1) = 1$, then $s_1 - s_2 \in \{-1, 0, 1\}$ and both inequalities are equalities, so $\overline{u}_F(p) = u_F^*(p)$.

A.13 Proof of Lemma 5

Let agent j with prior p_j play full revelation. Then $\overline{v}(s_i) - 0.5 = (1 - p_j)\hat{v}(s_i) - p_j \hat{v}(1 - s_i)$ does not depend on p_i, and full revelation is a best reply of agent i for any $p_i \in (0, 1)$ if

and only if $\overline{v}$ lies below its secant from 0 to 1:

$$(1-p_j)\hat{v}(x) - p_j\hat{v}(1-x) \leq \hat{v}(1)(x-p_j) \quad \forall x \in (0,1).$$

Hence $(\{0,1\};\{0,1\})$ is an equilibrium for (p_1,p_2) if and only if this condition holds for $p_j \in \{p_1,p_2\}$, so it is an equilibrium for all $(p_1,p_2) \in S^2$ if and only if the condition holds for all $p_j \in S$, i.e., if and only if it is an equilibrium for all (p,p), $p \in S$.

A.14 Proof of Proposition 4

Let $a := \hat{v}(1)$; since $v(x) \leq 1$ is increasing, $a \in [0, 0.5]$. By the proof of Lemma 5, $v(x)$ implements a fully revealing equilibrium for (p,p) if and only if $(1-p)\hat{v}(x) - p\hat{v}(1-x) \leq a(x-p)$ for all $x \in (0,1)$. Applying it also at $1-x$ and summing the two inequalities with weights $1-p$ and p gives $(1-2p)\big(\hat{v}(x) - ax\big) \leq 0$. Taking $p \in S$ below and above 0.5 yields $\hat{v}(x) = ax$, which proves (ii); then the condition holds with equality for all $p \in (0,1)$, which proves (i).

A.15 Proof of Proposition 5

(i) Let a and $g(x)$ be as in Definition 6. Then the condition $(1-p)\hat{v}(x) - p\hat{v}(1-x) \leq a(x-p)$ $\forall x \in (0,1)$ from the proof of Lemma 5 simplifies to

$$(1-p)g(x) - pg(1-x) \leq 0 \quad \forall x \in (0,1).$$

Since this condition is linear in p, it holds $\forall p \in [0.5,1)$ if and only if it holds for $p = 0.5$ and $p = 1$; the latter means $g(1-x) \geq 0$ $\forall x \in (0,1)$.

$p = 0.5$ implies $0.5g(x) - 0.5g(1-x) \leq 0$ $\forall x \in (0,1)$. Replacing x with $1-x$ gives $0.5g(x) - 0.5g(1-x) \geq 0$ $\forall x \in (0,1)$, so $v(x)$ is balanced.

Finally, since $\hat{v}(x)$ should be increasing, on the one hand, for $x < y$, $g(x) + ax \leq$

$g(y) + ay$; on the other hand, for $1 - y < 1 - x$,

$$g(1-y) + a(1-y) \leq g(1-x) + a(1-x) \;\Leftrightarrow\; g(y) - ay \leq g(x) - ax,$$

hence $|g(x) - g(y)| \leq a(y - x) = a|x - y|$, and the converse is immediate.

The proof for $p \in (0, 0.5]$ is done analogously.

(ii) Since all equilibria form a product set, if full revelation is each agent's unique best reply to full revelation, it is the unique equilibrium.

The uniqueness of the best reply is equivalent to $(1-p)\hat{v}(x) - p\hat{v}(1-x) < a(x-p)\ \forall x \in (0,1)$, and it can be rewritten as $(1-2p)g(x) < 0\ \forall x \in (0,1)$, if $v(x)$ is balanced. Hence, if $g(x) < 0\ \forall x \in (0,1)$, this condition holds $\forall p < 0.5$, and vice versa.

A.16 Proof of Corollary 6

Suppose that agent j plays $\{0,1\}$. Then $\overline{v}(s_i) = (1-p_j)v(s_i) + p_j v(s_i - 1)$, and $\{0,1\}$ is a best reply of agent i only if the secant from 0 to 1 lies weakly above $\overline{v}(s_i)$ on $[0,1]$, which requires $\overline{v}'(0) \leq \overline{v}(1) - \overline{v}(0) \leq \overline{v}'(1)$. Since $v'(x)$ is even, $\overline{v}'(0) = (1-p_j)v'(0) + p_j v'(1)$ and $\overline{v}'(1) = (1-p_j)v'(1) + p_j v'(0)$, so this requires $(2p_j - 1)\big(v'(0) - v'(1)\big) \geq 0$, i.e., $p_j \geq 0.5$. Hence in a fully revealing equilibrium both priors are at least 0.5.

If $v(x)$ is generated by i.i.d. shocks with a square-integrable density f, then $v'(x) = \int f(t)f(t+x)dt$ is continuous, and by the Cauchy–Schwarz inequality,

$$v'(1) = \int f(t)f(t+1)dt \leq \int f^2(t)dt = v'(0),$$

with equality only if $f(t+1) = f(t)$ almost everywhere, which is impossible for a probability density. Hence $v'(0) > v'(1)$.

B. Proofs of Sections 6 and 7

B.1 Proof of Proposition 6

If $p_2 \geq \overline{p}$, the claim follows from Proposition 1, so let $p_2 < \overline{p}$.

Preliminaries. For a signal of agent j let $N(s) := s\overline{v}'(s) - \overline{v}(s) + \overline{v}(0)$, so that $slp'(s) = N(s)/s^2$. By the proof of Lemma 1 (ii), $N > 0$ on $(0, a)$ and $N < 0$ on $(a, 1]$, with a from Lemma 1. Hence agent i plays $\{p_i\}$ if and only if $N(p_i) \leq 0$, and $\{0, 1\}$ if and only if $N(1) \geq 0$. For $y, c \in [0, 1]$, let

$$A(y) := \tilde{v}(y) - y\tilde{v}'(y) = -\int_0^y x\tilde{v}''(x)dx,$$
$$B(y; c) := y\tilde{v}'(c - y) + \tilde{v}(c - y) - \tilde{v}(c) = -\int_0^y x\tilde{v}''(c - x)dx.$$

Since $-\tilde{v}'' > 0$ on $(0, 1]$, $A(y) > 0$, and $B(y; c) > 0$ for $0 < y \leq c$. If agent j plays $\{0, c\}$, so that $\mathbb{P}(s_j = c) = q := \frac{p_j}{c}$, then $\overline{v}(s) = (1 - q)\tilde{v}(s) + q\tilde{v}(s - c)$ and

$$N(s) = -(1 - q)A(s) + qB(s; c); \tag{2}$$

pooling is the case $c = p_j$, $q = 1$.

Step 1: $b(\cdot)$ and p^*. Against $\{p_1\}$, $N(s) = B(s; p_1)$. Hence agent 2 never plays $\{p_2\}$, as $p_2 \leq p_1$, and plays $\{0, b(p_1)\}$, where $b(p_1)$ solves $B(b; p_1) = 0$, i.e. $b\tilde{v}'(b - p_1) = \tilde{v}(b - p_1) + \tilde{v}(p_1)$, or $b(p_1) = 1$ if $B(1; p_1) \geq 0$. With $b = p_1 + \tilde{b}$ this equation reads

$$\varphi(\tilde{b}) := \tilde{v}(\tilde{b}) - (p_1 + \tilde{b})\tilde{v}'(\tilde{b}) + \tilde{v}(p_1) = 0,$$

where $\varphi'(\tilde{b}) = -(p_1 + \tilde{b})\tilde{v}''(\tilde{b}) > 0$, $\varphi(0) = \tilde{v}(p_1) < 0 < 2A(p_1) = \varphi(p_1)$ and $\partial_{p_1}\varphi = \tilde{v}'(p_1) - \tilde{v}'(\tilde{b}) < 0$ for $\tilde{b} < p_1$. So the root is unique, $p_1 < b(p_1) < 2p_1$, and b is continuous, strictly increasing while $b < 1$. Let p^* be the smallest p_1 with $b(p_1) = 1$,

solving $\tilde{v}(1-p^*)+\tilde{v}(p^*)=\tilde{v}'(1-p^*)$; it exists and $p^*\in(\frac{1}{2},1)$, since $b(\frac{1}{2})<1$ and $B(1;1)=1>0$. Thus against $\{x\}$ the other agent plays $\{0,1\}$ if and only if $B(1;x)\geq 0$, i.e. $x\geq p^*$, as b is non-decreasing.

Step 2: $a(\cdot)$. Against $\{0,1\}$, by (2),

$$N(s)=\big(A(s)+B(s;1)\big)\big(p_2-\Phi(s)\big),\qquad \Phi:=\frac{A}{A+B(\cdot\,;1)}.$$

Φ is strictly increasing: writing $w=-\tilde{v}''$,

$$\partial_y B(y;1)\,A(y)-B(y;1)\,A'(y)=y\int_0^y x\Big[w(1-y)w(x)-w(1-x)w(y)\Big]dx<0,$$

since $w(x)\leq w(y)$ and $w(1-y)\leq w(1-x)$ for $x<y$, and $w(x)<w(y)$ for x close to 0, as $v'(x)$ is not linear near 0. Moreover, $\Phi(1)=\frac{|\tilde{v}'(1)|-1}{|\tilde{v}'(1)|}=\overline{p}$. Hence agent 1 plays $\{p_1\}$ if $p_1\geq a(p_2)$ and $\{0,a(p_2)\}$ otherwise, where $a(p_2)=0$ if $p_2\leq\Phi(0^+)$, $a(p_2)=\Phi^{-1}(p_2)$ if $\Phi(0^+)<p_2<\overline{p}$, and $a(p_2)=1$ if $p_2\geq\overline{p}$; a is continuous and non-decreasing, since Φ is continuous and strictly increasing.

Step 3: $\check{p}_2(\cdot)$ and $\hat{p}_1(\cdot)$. By (2), against $\{0,b(p_1)\}$, agent 1 plays $\{p_1\}$ if and only if

$$p_2\leq\check{p}_2(p_1):=\frac{b(p_1)A(p_1)}{A(p_1)+B\big(p_1;b(p_1)\big)},$$

and against $\{0,a\}$ with $a>p_1$, agent 2 plays $\{0,1\}$ if and only if $-(a-p_1)A(1)+p_1B(1;a)\geq 0$. By Step 1, $B(1;a)\geq 0$ if and only if $a\geq p^*$, so for $a<p^*$ this fails, and for $a\geq p^*$ it reads $p_1\geq\frac{aA(1)}{A(1)+B(1;a)}$. For $a=a(p_2)\geq p^*$ denote this bound by $\hat{p}_1(p_2)$.

Step 4: cases (i)–(iii). A pair of signals is an equilibrium if and only if each is a best reply to the other. By Steps 1 and 2, $(\{p_1\};\{0,1\})$ is an equilibrium if and only if $p_1\geq p^*$ and $p_1\geq a(p_2)$; by Steps 1 and 3, $(\{p_1\};\{0,b\})$ with $b<1$ is an equilibrium if and only if $b=b(p_1)$, $p_1<p^*$ and $p_2\leq\check{p}_2(p_1)$; by Steps 2 and 3, $(\{0,a\};\{0,1\})$ is an equilibrium

if and only if $a = a(p_2) \geq p^*$ and $\hat{p}_1(p_2) \leq p_1 < a(p_2)$. An equilibrium exists by the minimax theorem, and as the game is constant-sum, all equilibria form a product set. By Lemma 1 every best reply is unique, hence so is the equilibrium.

Step 5: case (iv). For equal priors the game is symmetric, so the unique equilibrium is symmetric and none of the forms in (i)–(iii) is an equilibrium. By Step 4 this gives: $a(p) > p$ if $p \geq p^*$; $\check{p}_2(p) < p$ if $p < p^*$; and $\hat{p}_1(p) > p$ if $a(p) \geq p^*$ and $p < \overline{p}$, since then $p < a(p)$. Now let $p_2 \leq p_1$ and suppose that none of (i)–(iii) holds. By Lemma 1 each agent plays $\{p_i\}$, $\{0, 1\}$ or $\{0, c\}$ with $p_i < c < 1$, and by Lemma 2 not both pool at their priors. Apart from (i)–(iv), this leaves the following forms, and each requires, by Steps 1–3 with the roles of the agents exchanged (the functions b, a, $\check{p}_2$ and $\hat{p}_1$ stay the same, only their arguments change), a condition incompatible with $p_2 \leq p_1$:

- $(\{0, b(p_2)\}; \{p_2\})$ requires $p_2 < p^*$ and $p_1 \leq \check{p}_2(p_2) < p_2$;
- $(\{0, 1\}; \{p_2\})$ requires $p_2 \geq p^*$ and $p_2 \geq a(p_1) \geq a(p_2) > p_2$;
- $(\{0, 1\}; \{0, a(p_1)\})$ with $a(p_1) < 1$ requires $a(p_1) \geq p^*$, $p_1 < \overline{p}$ and $p_2 \geq \hat{p}_1(p_1) > p_1$;
- $(\{0, 1\}; \{0, 1\})$ requires $p_2 \geq \overline{p}$ by Proposition 1.

Hence the equilibrium is $(\{0, a\}; \{0, b\})$ with $p_1 < a < 1$ and $p_2 < b < 1$.

Step 6: $a \leq b$ and continuity. The correspondence $p \mapsto \mathcal{G}(p)$ is continuous: upper hemicontinuous since the mean is weakly continuous, and lower hemicontinuous since any $G \in \mathcal{G}(p)$ is approximated by mixtures of G with δ_0 or δ_1. As payoffs are continuous, the equilibrium correspondence has a closed graph, and being single-valued with values in a compact set, it is continuous. If $a = b = c$ in (iv), then, since $B(c; c) = -\tilde{v}(c)$, the indifference conditions $N(c) = 0$ of the two agents, with $q = \frac{p_2}{c}$ and $q = \frac{p_1}{c}$, read $\frac{c}{p_2} - 1 = -\frac{\tilde{v}(c)}{A(c)} = \frac{c}{p_1} - 1$, so $p_1 = p_2$; conversely $a = b$ on the diagonal by symmetry. On the convex set $U = \{0 < p_2 < p_1 < 1,\ p_2 < \overline{p}\}$, let R be the set where (iv) holds. R is open in U, as the strategies in (iv) are not weak limits of $\{0, 1\}$ or of point

masses, and $a - b \neq 0$ is continuous on R. Since U is connected and (i) holds for p_1 close to 1, each connected component of R has a boundary point in $U \setminus R$. There one of (i)–(iii) holds, and continuity gives $(a, b) \to (p_1, 1)$, $(p_1, b(p_1))$ or $(a(p_2), 1)$ respectively, with $a(p_2) < 1$ as $p_2 < \overline{p}$. Each limit has $a < b$, hence $a < b$ on R.

B.2 Proof of Proposition 7

Describe a signal of agent i by its high signal $c_i \in [p_i, 1]$, where $c_i = p_i$ stands for $\{p_i\}$. For a fixed p_i, $\frac{\mathbb{P}(s_i=0)}{1-p_i} = \frac{c_i - p_i}{c_i(1-p_i)}$ is increasing in c_i, so it suffices to show that c_i is non-decreasing in p_j. By Lemma 1, the best reply to c_j is $c_i = \max\{r(c_j, \frac{p_j}{c_j}), p_i\}$, where $r(c, q)$ is the point at which N from (2) changes sign; it is continuous in (c, q). With $w = -\tilde{v}''$, which is odd and non-decreasing,

$$\partial_q N(s) = A(s) + B(s; c) = \int_0^s x\big[w(x) + w(c - x)\big]dx > 0,$$

since $w(x) + w(c - x)$ is at least $w(x) > 0$ if $x \leq c$ and equals $w(x) - w(x - c) \geq 0$ if $x > c$. Hence $r(c, q)$ is non-decreasing in q, and strictly increasing where $r(c, q) \in (0, 1)$, and $r(c, 1) = b(c)$ from Step 1 of the proof of Proposition 6.

Let $\rho(a) := r(a, \frac{p_1}{a})$, so that agent 2's best reply to a is $\max\{\rho(a), p_2\}$, and let $\Gamma(a) := \max\{r(c, \frac{p_2}{c}), p_1\}$ with $c = \max\{\rho(a), p_2\}$, for $a \in [p_1, 1]$. Γ is continuous and its fixed points are exactly the equilibria, so by Step 4 of the proof of Proposition 6 it has a unique fixed point a^*; since $\Gamma(p_1) \geq p_1$, $\Gamma(a) > a$ on $[p_1, a^*)$. As p_2 increases to p_2', $\Gamma(a)$ does not decrease for any a: if $\rho(a) \geq p_2'$, only the weight $\frac{p_2}{\rho(a)}$ increases; if $\rho(a) \leq p_2$, agent 2 pools and $b(p_2) \leq b(p_2')$; and if $p_2 < \rho(a) < p_2'$, then $r(\rho, \frac{p_2}{\rho}) \leq r(\rho, 1) = b(\rho) \leq b(p_2')$. Hence the fixed point does not decrease, and if $p_1 < a^* < 1$, $\Gamma(a^*)$ strictly increases, and so does the fixed point. As the argument does not use $p_2 \leq p_1$, it applies to both agents.

B.3 Proof of Proposition 8

Since $v(-x) = 1-v(x)$, $s_1v(s_1-s_2)+s_2v(s_2-s_1) = \frac{s_1+s_2}{2}+H(s_1-s_2)$, where $H(x) := x\hat{v}(x)$ is even, so the firm's expected utility is

$$u_F = \frac{p_1+p_2}{2} + \mathbb{E}[H(s_1-s_2)].$$

For $x \in (0,1]$, $0 < h(x) := H'(x) = \hat{v}(x) + xv'(x) \le 2\hat{v}(x) \le 2x$, since $v'(x)$ is decreasing on $[0,1]$ and $v'(0) \le 1$, as $0.5 \ge \hat{v}(1) \ge \frac{v'(0)}{2}$ by the concavity of $v'(x)$. Also, $H(x) \le \frac{x}{2}$.

Full revelation. $u_F = \frac{p_1+p_2}{2} + H(1)(p_1+p_2-2p_1p_2)$, so $\partial_{p_1}u_F = 0.5 + H(1)(1-2p_2) > 0.5 - H(1) \ge 0$, and similarly for p_2.

Region (i). $u_F = \frac{p_1+p_2}{2}+(1-p_2)H(p_1)+p_2H(1-p_1)$, so $\partial_{p_2}u_F = 0.5-H(p_1)+H(1-p_1) > 0$ and, as $p_2 \le p_1$ and $h(x) \le 2x$,

$$\partial_{p_1}u_F = 0.5 + (1-p_2)h(p_1) - p_2h(1-p_1) > 0.5 - 2p_1(1-p_1) \ge 0.$$

Region (ii). Let $b = b(p_1)$ and $y := b - p_1$. By Step 1 of the proof of Proposition 6, $0 < y < p_1$ and $y' = -\frac{\partial_{p_1}\varphi}{\varphi'} > 0$. Then

$$u_F = \frac{p_1+p_2}{2} + H(p_1) - \frac{p_2}{b}\big(H(p_1) - H(y)\big),$$

where $0 < H(p_1) - H(y) \le \frac{p_1}{2} < \frac{b}{2}$. Hence $\partial_{p_2}u_F > 0$ and

$$\partial_{p_1}u_F = 0.5 + \Big(1 - \frac{p_2}{b}\Big)h(p_1) + \frac{p_2}{b}h(y)y' + \frac{p_2(1+y')}{b^2}\big(H(p_1) - H(y)\big) > 0.$$

Region (iii). Let $a = a(p_2)$, which is non-decreasing by Step 2 of the proof of

Proposition 6, with $p_2 \le p_1 < a$ and $a \ge p^* > 0.5$. Then

$$u_F = \frac{p_1 + p_2}{2} + p_2 H(1) + \frac{p_1}{a} K, \quad \text{where} \quad K := (1 - p_2)H(a) + p_2\big(H(1-a) - H(1)\big),$$

so $\partial_{p_1} u_F = 0.5 + \frac{K}{a} \ge 0.5 - \frac{p_2 H(1)}{a} > 0$ and, with $K_a := \partial_a K$,

$$\partial_{p_2} u_F = 0.5 + H(1) + \frac{p_1}{a}\big(H(1-a) - H(1) - H(a)\big) + \frac{p_1 a'}{a^2}(aK_a - K).$$

The first three terms sum to at least $0.5 - H(a) > 0$, and

$$aK_a - K = (1 - p_2)a^2 v'(a) + p_2\big(\hat{v}(1) - \hat{v}(1-a) - a(1-a)v'(1-a)\big) \ge 0,$$

since, by the concavity of $v'(x)$ and $a \ge 0.5$,

$$\hat{v}(1) - \hat{v}(1-a) = \int_{1-a}^{1} v'(x)dx \ge \frac{a}{2} v'(1-a) \ge a(1-a)v'(1-a).$$

Example 9. Both signals solve the indifference conditions $N = 0$ from (2), which I solved numerically. For instance, for $p_2 = 0.84, 0.86, 0.873$, $(a, b) \approx (0.9375, 0.9996)$, $(0.9406, 0.9925)$ and $(0.9435, 0.9876)$, and $u_F \approx 0.96145$, 0.96092 and 0.96076.

B.4 Proof of Lemma 6

Write $u_i(p_i, p_j)$ for agent i's equilibrium payoff when his prior is p_i and agent j's prior is p_j. For $G_j \in \mathcal{G}(p_j)$, let $\overline{v}_{G_j}(x)$ be agent i's value function against G_j and $\overline{v}^{cv}_{G_j}$ its concavification on $[0, 1]$. By Kamenica and Gentzkow (2011), $\overline{v}^{cv}_{G_j}(p_i) = \max_{H \in \mathcal{G}(p_i)} \mathbb{E}_H[\overline{v}_{G_j}]$ is agent i's best payoff against G_j, where the maximum is attained as $\overline{v}_{G_j}$ is continuous.

The game is constant-sum, so agent j's equilibrium signal G_j^* minimizes i's payoff:

$$u_i(p_i, p_j) = \min_{G_j \in \mathcal{G}(p_j)} \overline{v}^{cv}_{G_j}(p_i) = \overline{v}^{cv}_{G_j^*}(p_i).$$

Step 1: each $\overline{v}^{cv}_{G_j}$ is non-decreasing on $[0,1]$, and strictly increasing if $v(x)$ is. As $v(x)$ is increasing, so is $\overline{v}_{G_j}$, hence $\overline{v}_{G_j} \leq \overline{v}_{G_j}(1)$ on $[0,1]$. The constant $\overline{v}_{G_j}(1)$ is then a concave majorant of $\overline{v}_{G_j}$, so $\overline{v}^{cv}_{G_j} \leq \overline{v}_{G_j}(1)$, while $\overline{v}^{cv}_{G_j}(1) = \overline{v}_{G_j}(1)$, since $\{1\}$ is the only signal with mean 1. So the concave function $\overline{v}^{cv}_{G_j}$ attains its maximum at $x = 1$. For $0 \leq x < y \leq 1$, write $y = \lambda x + (1-\lambda)\cdot 1$ with $\lambda = \frac{1-y}{1-x} \in [0,1)$; by concavity,

$$\overline{v}^{cv}_{G_j}(y) \geq \lambda\, \overline{v}^{cv}_{G_j}(x) + (1-\lambda)\, \overline{v}^{cv}_{G_j}(1) \geq \overline{v}^{cv}_{G_j}(x).$$

If $v(x)$ is strictly increasing, so is $\overline{v}_{G_j}$. Then every $H \in \mathcal{G}(x)$ with $x < 1$ puts positive mass on $[0,1)$, where $\overline{v}_{G_j} < \overline{v}_{G_j}(1)$, so $\overline{v}^{cv}_{G_j}(x) = \max_{H \in \mathcal{G}(x)} \mathbb{E}_H[\overline{v}_{G_j}] < \overline{v}_{G_j}(1) = \overline{v}^{cv}_{G_j}(1)$, and the last inequality is strict.

Step 2: monotonicity of u_i. Let $p_i < p_i'$, and let G_j^* be agent j's equilibrium signal for priors (p_i', p_j). By Step 1,

$$u_i(p_i, p_j) \leq \overline{v}^{cv}_{G_j^*}(p_i) \leq (<)\ \overline{v}^{cv}_{G_j^*}(p_i') = u_i(p_i', p_j).$$

Finally, since $v(x) + v(-x) = 1$, $u_i(p_i, p_j) = 1 - u_j(p_j, p_i)$, so, by the above applied to agent j, u_i is non-increasing (strictly decreasing) in p_j.

B.5 Proof of Proposition 9

For $t \in (0,1)$, let G_t be the upper censorship with cutoff t, which pools the types above t at $m(t) := \mathbb{E}[\theta|\theta \geq t] > t$, and let

$$w_t(x) := \int_0^t \tilde{v}(x-s)f(s)ds + (1-F(t))\tilde{v}(x-m(t))$$

be the normalized value function against G_t, which has the same best replies as the value function itself. Let L_t be the tangent to w_t at $m(t)$. Since $\tilde{v}$ is odd with $\tilde{v}'(0) = 0$, the left-hand side of (1) at $\theta^* = t$ equals $E(t) := L_t(t) - w_t(t)$. So (1) is the first-order condition of Kolotilin et al. (2022) for the cutoff of an upper censorship against G_t, evaluated at the cutoff t itself; Steps 2 and 3 verify directly that its solutions are equilibria, which avoids any regularity condition on f.

Step 1: (1) *has a solution.* E is continuous on $(0,1)$, and $E(t) \to \tilde{v}(\mathbb{E}[\theta]) < 0$ as $t \to 0$. Let $t \geq \frac{3}{4}$ and $m = m(t)$. For $s \leq t$, the integrand in (1) is non-negative by the concavity of $\tilde{v}$ on $[0,1]$, and for $s \leq \frac{1}{2}$ it is at least $\frac{\kappa}{2}(m-t)^2$, where $\kappa := -\tilde{v}'(\frac{1}{4}) > 0$, since $\tilde{v}''(x) \leq \frac{\tilde{v}'(x)}{x} \leq -\kappa$ on $[\frac{1}{4}, 1]$ by property 2) in Appendix A.1. As $\tilde{v}(m-t) \geq -(m-t)^2$ by Lemma 9, the last term of $E(t)$ is at least $-(m-t)^2(1-F(t))$, so

$$E(t) \geq (m-t)^2\Big(\frac{\kappa}{2}F\big(\tfrac{1}{2}\big) - \big(1-F(t)\big)\Big) > 0$$

for t close to 1. Hence E has a zero in $(0,1)$.

Step 2: every solution gives an equilibrium. Let $E(t) = 0$, $m = m(t)$, $w = w_t$, $L = L_t$ and $D := L - w$, so that $D(t) = D(m) = D'(m) = 0$ and $D' = w'(m) - w'$, where $w'(x) = \int \tilde{v}'(x-s)dG_t(s)$ is concave, as a mixture of concave functions. If $w'(x) > w'(m)$ for some $x > m$, then $w' < w'(m)$ on $[0,m)$ by concavity, so D is strictly increasing on $[0,m]$, contradicting $D(t) = D(m)$. Hence D is non-decreasing, and

non-negative, on $[m, 1]$. The set of $x \leq m$ with $w'(x) \geq w'(m)$ is an interval $[\alpha, m]$, so D is non-decreasing on $[t, \alpha]$ and non-increasing on $[\max(\alpha, t), m]$, and $D(t) = D(m) = 0$ gives $D \geq 0$ on $[t, m]$. If $\alpha > t$, then $w'(t) < w'(\alpha)$, so w' is increasing on $[0, t]$ and $w'(t) < w'(m)$. If $\alpha \leq t$, then $D = 0$ on $[t, m]$, so w' is constant on $[t, m]$, hence maximal there and non-decreasing on $[0, t]$. In both cases

$$p(x) := \begin{cases} w(x), & x \leq t, \\ L(x), & x \geq t, \end{cases}$$

is convex, $p \geq w$, and $p = w$ on $[0, t] \cup \{m\} \supseteq supp(G_t)$. Since p is affine on $[t, 1]$ and G_t pools $[t, 1]$ at its conditional mean, $\int p \, dG_t = \int p \, dF$. So every mean-preserving contraction H of F satisfies

$$\int w \, dH \leq \int p \, dH \leq \int p \, dF = \int p \, dG_t = \int w \, dG_t,$$

i.e., G_t is a best reply to itself, and (G_t, G_t) is an equilibrium.

Step 3: uniqueness. Now suppose that $v'(x)$ is not linear near 0. By property 4) in Appendix A.1, w' is affine on no interval that G_t charges. Hence $\alpha > t$ in Step 2, as otherwise w' is constant on $[t, m]$; $D > 0$ on $(t, 1] \setminus \{m\}$, as otherwise D, and hence $w' - w'(m)$, vanishes on an interval with endpoint m; and w' is strictly increasing on $[0, t]$, so p is strictly convex on $[0, t]$ with a kink at t. Let H be a best reply to G_t, so that both inequalities above are equalities. The first one gives $supp(H) \subseteq [0, t] \cup \{m\}$. Writing $p(x) = p(0) + p'(0)x + \int_0^1 (x - y)^+ dp'(y)$, the second one gives $\int_0^1 \big(\Psi_F(y) - \Psi_H(y)\big) dp'(y) = 0$, where $\Psi_K(y) := \int_0^y K(x) dx \leq \Psi_F(y)$ for every mean-preserving contraction K of F. As dp' charges every subinterval of $[0, t]$, $\Psi_H = \Psi_F$ on $[0, t]$, so $H = F$ on $[0, t)$. The remaining mass $1 - F(t)$ of H lies on $\{t, m\}$ with mean m, so it is all at m: $H = G_t$. So G_t is the unique best reply to itself. As

all equilibria form a product set, every equilibrium strategy is a best reply to G_t, so (G_t, G_t) is the unique equilibrium, and by Step 2, (1) has no other solution.

B.6 Proof of Corollary 7

Here $\tilde{v}(x) = -x|x|$, so for $0 \le s \le \theta^* \le m^*$, the integrand in (1) is

$$-(m^* - s)^2 + (\theta^* - s)^2 + 2(m^* - \theta^*)(m^* - s) = (m^* - \theta^*)^2,$$

and (1) becomes $(m^* - \theta^*)^2(2F(\theta^*) - 1) = 0$. Since $m^* > \theta^*$, $F(\theta^*) = 0.5$.

The same computation, with $dF(s)$ in place of $f(s)ds$, applies to any prior F that is continuous at a median θ^*, and Step 2 of the proof of Proposition 9 does not use the density of F. Hence, for every such prior, the median censorship is a symmetric equilibrium under $v(x) = 0.5 + x - 0.5x|x|$.

B.7 A convexity lemma

The following lemma extends the converse of the Hermite–Hadamard inequality (see Niculescu and Persson (2003)), which is the case of a uniform K.

Lemma 13. *Let a CDF K and a function h be continuous on an interval J, with K strictly increasing. If $h\big(\mathbb{E}_K[\theta|\theta \in J']\big) \le \mathbb{E}_K[h(\theta)|\theta \in J']$ for every interval $J' \subseteq J$, then h is convex.*

Proof. Let $x < z$ in J, let ℓ be the secant of h through x and z, and suppose that $g := h - \ell$ has a positive maximum on $[x, z]$, first attained at $c \in (x, z)$. For $\delta > 0$ small enough that $\mathbb{E}_K[\theta|\theta \in [c - \delta, z]] > c$, there is $\delta' \in (0, z - c)$ such that $J' := [c - \delta, c + \delta']$ satisfies $\mathbb{E}_K[\theta|\theta \in J'] = c$, as K is continuous and strictly increasing. Since $g < g(c)$ on $[c - \delta, c)$, which has positive K-mass, and ℓ is affine, $\mathbb{E}_K[h(\theta)|\theta \in J'] = \ell(c) + \mathbb{E}_K[g(\theta)|\theta \in J'] < h(c)$, a contradiction. Hence $h \le \ell$ on $[x, z]$. □

B.8 Proof of Proposition 10

Let $I = [\underline{\theta}, \overline{\theta}]$. For a CDF K let $\Psi_K(y) := \int_{\underline{\theta}}^{y} K(x)dx$, so that a signal G is feasible if and only if $\Psi_G \leq \Psi_F$ on I with equality at $\overline{\theta}$, and for convex ϕ and feasible G,

$$\int \phi \, dF - \int \phi \, dG = \int \big(\Psi_F(y) - \Psi_G(y)\big) d\phi'(y) \geq 0.$$

Against an atomless distribution H, the payoff of a signal s is $H(s)$. In particular, against full revelation, a deviation to G yields $\mathbb{E}_G[F]$, and full revelation yields $\mathbb{E}_F[F] = 0.5$.

Sufficiency. If F is convex on I, the display with $\phi = F$ gives $\mathbb{E}_G[F] \leq \mathbb{E}_F[F] = 0.5$ for every feasible G, so (F, F) is an equilibrium.

Necessity. Pooling the types in an interval $J \subseteq I$ into their mean m_J is feasible and changes the deviator's payoff by $\mathbb{P}(\theta \in J)\big(F(m_J) - \mathbb{E}_F[F(\theta)|\theta \in J]\big)$, so in a fully revealing equilibrium $F(m_J) \leq \mathbb{E}_F[F(\theta)|\theta \in J]$ for every J. First, F is strictly increasing on I: otherwise $F = \gamma \in (0, 1)$ on a maximal interval $[L, R] \subset I$ with $L < R$, and for $J = \{\theta : \gamma - r \leq F(\theta) \leq \gamma + \frac{r}{2}\}$, $\mathbb{E}_F[F(\theta)|\theta \in J] = \gamma - \frac{r}{4}$, while $m_J \to \frac{2L+R}{3} > L$ as $r \to 0$, so $F(m_J) \geq \gamma$ for small r. Then F is convex on I by Lemma 13 with $K = h = F$.

Uniqueness. Let F be convex, so that it is strictly increasing on I, and let (F, H) be an equilibrium. As all equilibria form a symmetric product set, (H, H) and (H, F) are equilibria too, and each agent gets 0.5 in all of them. Against H, the payoff of a signal s is $H(s^-) + \frac{1}{2}\big(H(s) - H(s^-)\big)$. First, H is atomless. It has no atom at $\underline{\theta}$ or $\overline{\theta}$, as $\Psi_H \leq \Psi_F$ with equality at $\overline{\theta}$ and F is atomless. If H had an atom $a > 0$ at $m \in (\underline{\theta}, \overline{\theta})$, then $\Psi_H(m) < \Psi_F(m)$, since otherwise $\Psi_F - \Psi_H \geq 0$ would have a minimum at m with left derivative $F(m) - H(m^-)$ above its right derivative $F(m) - H(m)$. So replacing the atom by masses $\frac{2a}{3}$ at $m + \varepsilon$ and $\frac{a}{3}$ at $m - 2\varepsilon$ is feasible for small ε, as $\Psi_F - \Psi_H > 0$ near m, and against H it raises the payoff by $a^2(\frac{2}{3} - \frac{1}{2}) + o(1) > 0$, contradicting that H is a best reply to H. Second, H is convex on I: as F is a best reply to H, pooling the

types in an interval $J \subseteq I$ does not pay, i.e., $H(m_J) \leq \mathbb{E}_F[H(\theta)|\theta \in J]$, and Lemma 13 applies with $K = F$ and $h = H$. Finally, as F and H are convex, $\int F\,dH \leq \int F\,dF$ and $\int H\,dH \leq \int H\,dF$, and all four integrals equal 0.5. By the identity above, $\Psi_F = \Psi_H$ on the supports of dF' and dH', and there $F = H$, since each such point in $(\underline{\theta}, \overline{\theta})$ is a minimum of the differentiable function $\Psi_F - \Psi_H \geq 0$, while $F = H$ at $\underline{\theta}$ and $\overline{\theta}$. On each open interval off these supports, F and H are affine and agree at the endpoints. Hence $H = F$, and (F, F) is the unique equilibrium.

B.9 Proof of Corollary 8

Let $d = s_1 - s_2$. For any selection rule and feasible G_1, G_2, since $v(-x) = 1 - v(x)$,

$$u_F = \mathbb{E}[\max(s_1, s_2)] - \mathbb{E}\big[|d|\big(1 - v(|d|)\big)\big] \leq \mathbb{E}[\max(s_1, s_2)] \leq \mathbb{E}[\max(\theta_1, \theta_2)],$$

where, as $\max(s, s') = s' + (s - s')^+$, the last gap equals $\mathbb{E}_F\big[(\Psi_F - \Psi_{G_2})(\theta)\big] + \mathbb{E}_{G_2}\big[(\Psi_F - \Psi_{G_1})(s)\big]$, with Ψ as in the proof of Proposition 10. The first term vanishes only if $\Psi_{G_2} = \Psi_F$ on $supp(F)$. Then $G_2 = F$: each point of $supp(F)$ in $(\underline{\theta}, \overline{\theta})$ is a minimum of $\Psi_F - \Psi_{G_2} \geq 0$, so $G_2 = F$ there, G_2 has no atom at $\underline{\theta}$, and on each gap of $supp(F)$, G_2 is constant, as it equals F at both ends. Likewise, the second term then vanishes only if $G_1 = F$. So the first best requires full revelation by both agents, and then also $|d|\big(1 - v(|d|)\big) = 0$ almost surely. As F is atomless, $0 < |d| < \varepsilon$ with positive probability for every $\varepsilon > 0$, so $v(x) = 1$ for arbitrarily small $x > 0$, i.e., $v(x)$ is the noiseless selection rule, as it is increasing and $v(x) \leq 1$. By Proposition 10, (F, F) is then an equilibrium if and only if F is convex, and it attains the first best. Finally, a well S-shaped $v(x)$ is strictly increasing, so it is not the noiseless rule.

B.10 Proof of Proposition 11

Let $c = 2p - \underline{\theta}$ and $G = U[\underline{\theta}, c]$. As F is concave, it lies above the uniform CDF on $[\underline{\theta}, \overline{\theta}]$, so $p \leq \frac{\underline{\theta}+\overline{\theta}}{2}$, i.e. $c \leq \overline{\theta}$. Moreover, $F - G$ is concave on $[\underline{\theta}, c]$ and non-negative at $\underline{\theta}$, while $F - G = F - 1 \leq 0$ on $[c, \overline{\theta}]$, so $F - G$ changes sign at most once, from positive to negative. Since F and G have the same mean, G is a mean-preserving contraction of F by the single-crossing condition, i.e. it is feasible. Against G, $\overline{v}(s) = G(s)$, which is concave on $[\underline{\theta}, \overline{\theta}]$, so by Jensen's inequality every feasible H gives $\mathbb{E}_H[G] \leq G(p) = 0.5 = \mathbb{E}_G[G]$, and (G, G) is an equilibrium. Finally,

$$u_F = \mathbb{E}[\max(s_1, s_2)] = \underline{\theta} + \frac{2}{3}(c - \underline{\theta}) = \frac{4p - \underline{\theta}}{3}.$$

For uniqueness, let (G, H) be an equilibrium, so that (H, H) and (H, G) are equilibria too. As in the proof of Proposition 10, H has no atom in $(\underline{\theta}, \overline{\theta}]$, and an atom $a > 0$ at $\underline{\theta}$ would not be a best reply to H either: pooling it with a small mass ε of H above $\underline{\theta}$ raises the payoff per unit of the atom from $\frac{a}{2}$ to at least a, at a total cost of at most $\varepsilon < \frac{a^2}{2}$. So H is atomless. As H is a best reply to G and $G(s) = \min\{\frac{s-\underline{\theta}}{c-\underline{\theta}}, 1\}$ is concave, $\mathbb{E}_H[G] \leq G(p) = 0.5$ with equality only if $supp(H) \subseteq [\underline{\theta}, c]$. As G is a best reply to H and every mean-preserving contraction of G is feasible, pooling the types of G in any interval $J \subseteq [\underline{\theta}, c]$ does not pay, so H is convex on $[\underline{\theta}, c]$ by Lemma 13 with $K = G$. Hence $H \leq G$ on $[\underline{\theta}, c]$, as G is the secant of H there, and $\int_{\underline{\theta}}^{c}(G - H)dx = \mathbb{E}_H[s] - \mathbb{E}_G[s] = 0$, so $H = G$.

B.11 Proof of Lemma 7

(i) Consider $v_1(x) = 0.5 + 0.5x - a\tilde{v}_2(x)$. The convexity of $\tilde{v}_2'(x)$ and the uniqueness of its minimum at 0 are simply inverted into the concavity of $v_1'(x)$ and the uniqueness of its maximum at 0, and $v_1(x) - 0.5$ is trivially odd and increasing for sufficiently small

a, since $\tilde{v}_2'(x)$ is bounded. So $v_1(x)$ is well S-shaped for such a, while

$$\tilde{v}_1(x) = \frac{0.5x - a\tilde{v}_2(x) - x(0.5 - a\tilde{v}_2'(0))}{|0.5 - a - (0.5 - a\tilde{v}_2'(0))|} = -\tilde{v}_2(x).$$

(ii) Since a normalization changes payoffs by a positive factor and terms independent of the signals, $v_k(x)$ and $\tilde{v}_k(x)$ have the same equilibria, $k \in \{1, 2\}$. The map $G \mapsto \check{G}$, where $\check{G}$ is the distribution of $1 - s$ for $s \sim G$, is a bijection between $\mathcal{G}(p)$ and $\mathcal{G}(1 - p)$. Since $\tilde{v}_1(x)$ is odd, $\tilde{v}_2\big((1 - s_i) - (1 - s_j)\big) = -\tilde{v}_1(s_j - s_i) = \tilde{v}_1(s_i - s_j)$, so each agent's payoff from $(\check{H}_1, \check{H}_2)$ under $\tilde{v}_2$ equals that from (H_1, H_2) under $\tilde{v}_1$ for all H_1, H_2. Hence $(\check{G}_1, \check{G}_2)$ is an equilibrium for $\tilde{v}_2$ and priors $(1 - p_1, 1 - p_2)$.

B.12 Proof of Proposition 12

Let $G_j = \{0, c\}$ with $c \in (p_j, 1]$, so that $\mathbb{P}(s_j = c) = q := \frac{p_j}{c} \in (0, 1)$. Since $v''(x)$ is odd, $\overline{v}''(s) = (1 - q)v''(s) + q\, v''(s - c) = -(1 - q)w(s) - q\, w(s - c) < 0$ on $(c, 1]$, and on $(0, c)$,

$$\overline{v}''(s) = -(1 - q)w(s) + q\, w(c - s) = w(s)\left(q\, \frac{w(c - s)}{w(s)} - (1 - q)\right),$$

which, as the ratio is strictly decreasing, is first positive and then negative, with at most one zero. If $G_j = \{p_j\}$, then $\overline{v}''(s) = v''(s - p_j)$ is positive on $[0, p_j)$ and negative on $(p_j, 1]$. In both cases, there is $c^* \in [0, 1]$ such that $\overline{v}$ is strictly convex on $[0, c^*]$ and strictly concave on $[c^*, 1]$, so, as in the proof of Lemma 1, agent i's best reply is unique and is either $\{p_i\}$ or $\{0, a_i\}$ with $a_i \in (p_i, 1]$.

Now restrict each agent i to these signals, indexed by $a_i \in [p_i, 1]$, where $a_i = p_i$ stands for $\{p_i\}$. Payoffs are continuous in (a_1, a_2), and, by the above, the best reply within this family is unique and is also a best reply among all signals. By the maximum theorem, each agent's best reply is continuous in the other agent's index, so Brouwer's theorem gives a pair of mutual best replies, which is an equilibrium. Since all equilibria

form a product set and each agent's best reply to the other's equilibrium signal is unique, this equilibrium is unique.

B.13 Proof of Lemma 8

Since $v(x)$ is real-analytic on $[-1,1]$, it extends analytically to $\{z \in \mathbb{C} : dist(z,[-1,1]) < \delta\}$ for some $\delta > 0$, hence $\overline{v}(s_i) = \int_0^1 v(s_i - s_j)dG_j(s_j)$ extends analytically to $\{z \in \mathbb{C} : dist(z,[0,1]) < \delta\}$. Let $l(s_i)$ be a line supporting $\overline{v}^{cv}(s_i)$ at p_i. Then $l(s_i) \geq \overline{v}(s_i)$ on $[0,1]$, and every optimal signal is supported on $\{s_i \in [0,1] : \overline{v}(s_i) = l(s_i)\}$. If this set were infinite, it would have an accumulation point in $[0,1]$, and by the identity theorem $\overline{v}(s_i) = l(s_i)$ on $[0,1]$, so $\overline{v}''(0) = 0$. However, $\overline{v}''(0) = \int_0^1 v''(-s_j)dG_j(s_j) > 0$, since $v''(-s_j) > 0$ for $s_j \in (0,1]$ and G_j puts positive mass on $(0,1]$ as $p_j > 0$.

B.14 Proof of Proposition 13

Full statement: let $v(x)$ be the CDF of a triangular distribution on $[-u,u]$; then there exists a symmetric equilibrium in which each agent's signal G is:

(i) Proper u-partition of $[0,2p]$ if $p \leq 0.5$

(ii) $\{0,1\}$, if $p > 0.5, u \geq 2-2p$.

(iii) $(\frac{2(1-p)}{2-a}, \frac{2p-a}{2-a})$ mix of the proper u-partition of $[0,a]$ with $\delta_{\{1\}}$, if (i) $p > 0.5, u < 2-2p$, and (ii) $a \leq 1-u$, where $a = 2ku - \frac{k(k+1)u^2}{2(1-p)}$, $k = \lceil \frac{2-2p}{u} \rceil - 1$.

(iv) If, in the solution of the equations in (iii), $a > 1-u$, the true equilibrium G is constructed as follows: let $k = \lceil \frac{1}{u} \rceil$, and let $\tilde{G}$ be a $(\frac{k-1}{k}, \frac{1}{k})$ mix of the proper u-partition of $[0,1-u]$ and $\delta_{\{1\}}$. Then let G be the

$$\left(\frac{(2p-a)k}{(k-1)(1-u)+2-ka}, \frac{(k-1)(1-u)+2-2pk}{(k-1)(1-u)+2-ka} \right)$$

mix of $\tilde{G}$ and the proper u-partition of $[0, a]$, where $a \in (1-u, (k-1)u)$ is the unique solution of $(1-a)\overline{v}(0) + a\overline{v}(1) = \overline{v}(a)$.

If $u \geq 1$, then $v(x)$ is quadratic and $\tilde{v}(x) = -x^2$ on $[0, 1]$; hence I refer the reader to Proposition 1 and Theorem 1. Consider now $u < 1$, where $v(x)$ is the CDF of the triangular distribution on $[-u, u]$:

$$v(x) = \begin{cases} 0, & x < -u, \\ \frac{(u+x)^2}{2u^2}, & x \in [-u, 0), \\ 1 - \frac{(u-x)^2}{2u^2}, & x \in [0, u), \\ 1, & x \geq u, \end{cases} \qquad v'(x) = \begin{cases} 0, & x < -u, \\ \frac{u+x}{u^2}, & x \in [-u, 0), \\ \frac{u-x}{u^2}, & x \in [0, u), \\ 0, & x \geq u, \end{cases}$$

i.e. $v'(x) = \frac{\max(u-|x|,0)}{u^2}$.

Here I use the continuity of $v'(x)$, which makes $\overline{v}'(x)$ continuous.

(i) Suppose that $p \leq 0.5, u < 1$, and let $G(x)$ be the CDF of the proper u-partition χ

on $[0, 2p]$ with $k = \lceil \frac{2p}{u} \rceil$. Let's analyze $\overline{v}'(x) = \int_0^1 v'(x-s)dG(s) = \int_0^{2p} v'(x-s)dG(s)$:

$$\begin{aligned}
\int_0^{2p} v'(x-s)dG(s) &= \sum_{i=0}^{k-1} v'(x-iu)\mathbb{P}(\chi = iu) + \sum_{i=0}^{k-1} v'(x-2p+iu)\mathbb{P}(\chi = 2p-iu) \\
&= \sum_{i=0}^{k-1} v'(x-iu)\frac{k-i}{k(k+1)} + \sum_{i=0}^{k-1} v'(x-2p+iu)\frac{k-i}{k(k+1)} \\
&= \sum_{i=0}^{k-1} \frac{\max(u-|x-iu|,0)}{u^2}\frac{k-i}{k(k+1)} \\
&\quad + \sum_{i=0}^{k-1} \frac{\max(u-|x-2p+iu|,0)}{u^2}\frac{k-i}{k(k+1)} \\
&= \underbrace{\sum_{i=0}^{k} \frac{\max(u-|x-iu|,0)}{u^2}\frac{k-i}{k(k+1)}}_{S_L} \\
&\quad + \underbrace{\sum_{i=0}^{k} \frac{\max(u-|x-2p+iu|,0)}{u^2}\frac{k-i}{k(k+1)}}_{S_R}.
\end{aligned}$$

Consider S_L and say that $a_i = \max(u - |x - iu|, 0)$ is *active* if $a_i > 0$. Then, as long as $x \in (iu, (i+1)u)$, only a_i and a_{i+1} are active, so

$$S_L = \frac{(u-(x-iu))(k-i) + (u-((i+1)u-x))(k-i-1)}{u^2k(k+1)} = \frac{ku-x}{u^2k(k+1)}.$$

By the continuity of S_L at $x = iu$, $S_L = \frac{ku-x}{u^2k(k+1)}$ on $[0, ku]$, and hence on $[0, 2p]$, since $ku \geq 2p$. Analogously, $S_R = \frac{ku+x-2p}{u^2k(k+1)}$ on $[0, 2p]$, hence $\overline{v}'(x) = \frac{2ku-2p}{u^2k(k+1)}$ on $[0, 2p]$.

For $x > 2p$, $S_L = \frac{(ku-x)^+}{u^2k(k+1)}$ and $S_R = \frac{k(u-x+2p)^+}{u^2k(k+1)}$.

$$\overline{v}'(x) = \begin{cases} \frac{2ku-2p}{u^2k(k+1)}, & x \in [0, 2p], \\ \frac{2ku-x(1+k)+2kp}{u^2k(k+1)}, & x \in (2p, ku], \\ \frac{k(u-x+2p)}{u^2k(k+1)}, & x \in (ku, 2p+u], \\ 0, & x \geq 2p+u. \end{cases}$$

$\overline{v}'(x)$ is continuous, decreasing on $[2p, 1]$ and constant on $[0, 2p]$, hence $\overline{v}(x)$ is concave on $[0, 1]$, and linear on $[0, 2p]$, so indeed a pair of proper u-partitions of $[0, 2p]$ is an equilibrium. Also, $\overline{v}'(x)$ is convex on $[2p, 1]$.

(ii) Suppose that $p > 0.5, 2 - 2p \leq u < 1$; then $\overline{v}(x) = (1-p)v(x) - pv(1-x)$. Then

$$\overline{v}'(x) = \frac{(1-p)\max(u-x, 0) + p\max(u+x-1, 0)}{u^2},$$

where the numerator is a piecewise linear convex function, hence $\overline{v}(x)$ is concave and then convex, and we need to check that $\overline{v}'(0) \leq \overline{v}(1) - \overline{v}(0)$.

$\overline{v}'(0) = \frac{1-p}{u}$, $\overline{v}(1) - \overline{v}(0) = v(1) - v(0) = 0.5$, so $\{0, 1\}$ is a best reply iff $2 - 2p \leq u$.

(iii) Suppose $p > 0.5, u < 2 - 2p$, and $a \leq 1 - u$, where $a = 2ku - \frac{k(k+1)u^2}{2(1-p)}$ for $k = \lceil \frac{2-2p}{u} \rceil - 1$. Then consider the $(\frac{2(1-p)}{2-a}, 1 - \frac{2(1-p)}{2-a})$ mix of the proper u-partition of $[0, a]$ with $\delta_{\{1\}}$; notice that this is a proper G, since the proper u-partition of $[0, a]$ has a mean $0.5a$, and $\frac{2(1-p)}{2-a} \cdot 0.5a + 1 - \frac{2(1-p)}{2-a} = p$.

First, $\lceil \frac{2-2p}{u} \rceil - 1 = \lceil \frac{a}{u} \rceil$, so k is the same in the proper u-partition of $[0, a]$:

$$\left\lceil \frac{a}{u} \right\rceil = \left\lceil 2k - \frac{k(k+1)u}{2(1-p)} \right\rceil = k,$$

since $k < \frac{2(1-p)}{u} \leq k+1$, hence $k \leq \frac{k(k+1)u}{2(1-p)} < k+1$.

Since $\overline{v}(x)$ is linear in the strategy of the other agent,

$$\overline{v}'(x) = \frac{2(1-p)}{2-a} S(x) + \frac{2p-a}{2-a} \cdot \frac{\max(u+x-1, 0)}{u^2},$$

where $S(x)$ is $\overline{v}'(x)$ from (i) against the proper u-partition χ of $[0, a]$. Hence $\overline{v}(x)$ is linear on $[0, a]$ with slope $\overline{v}'(0) = \frac{2(1-p)}{2-a} \cdot \frac{2ku-a}{u^2 k(k+1)}$. Since $v(1-s) = 1$ for $s \le a \le 1-u$,

$$\overline{v}(1) - \overline{v}(0) = \frac{2(1-p)}{2-a} \mathbb{E}[v(\chi)] + \frac{2p-a}{2(2-a)}, \qquad \mathbb{E}[v(\chi)] = 1 - \frac{1}{2(k+1)} - \frac{(ku-a)^2}{2u^2 k(k+1)},$$

and the equation $\overline{v}(1) - \overline{v}(0) = \overline{v}'(0)$ reduces to a linear equation in a with the root $a = 2ku - \frac{k(k+1)u^2}{2(1-p)}$. Hence $\int_a^1 (\overline{v}'(x) - \overline{v}'(a))dx = 0$.

On $[a, 1]$ both summands of $\overline{v}'(x)$ are convex: the first by (i), the second as a maximum of linear functions. Hence $\overline{v}'(x) - \overline{v}'(a)$ is convex on $[a, 1]$, vanishes at a and integrates to zero, so $\exists c : \overline{v}'(x) \le \overline{v}'(a)\ \forall x \in [a, c]$ and $\overline{v}'(x) \ge \overline{v}'(a)\ \forall x \in [c, 1]$. Therefore $\overline{v}(x) \le \overline{v}(a) + (x-a)\overline{v}'(a)$ on $[a, 1]$, with equality at $x = 1$, so $[0, a] \cup \{1\}$ is the concavification support, and (G, G) is an equilibrium.

(iv) Notice that G is a feasible distribution, since the mean of $\tilde{G}$ is $\frac{k-1}{k}\frac{1-u}{2} + \frac{1}{k} = \frac{Q}{2k}$ with $Q := (k-1)(1-u) + 2$, the mean of the proper u-partition of $[0, a]$ is $\frac{a}{2}$, and $\beta := \frac{(2p-a)k}{Q-ka}$ solves $\beta\frac{Q}{2k} + (1-\beta)\frac{a}{2} = p$.

Suppose that a from (iii) satisfies $a > 1-u$. I first show that if the other agent plays the $(\frac{k-1}{k}, \frac{1}{k})$ mix of the proper u-partition of $[0, 1-u]$ and $\delta_{\{1\}}$, where $k = \lceil \frac{1}{u} \rceil$, then

$\overline{v}(x)$ is linear on $[0,(k-1)u]$. Again, consider $\overline{v}'(x)$, noting that $\lceil \frac{1-u}{u} \rceil = k-1$:

$$\overline{v}'(x) = \frac{k-1}{k}\Bigg[\underbrace{\sum_{i=0}^{k-1} \frac{\max(u-|x-iu|,0)}{u^2}\frac{k-1-i}{k(k-1)}}_{S_L} + \underbrace{\sum_{i=0}^{k-1} \frac{\max(u-|x-1+(i+1)u|,0)}{u^2}\frac{k-1-i}{k(k-1)}}_{S_R}\Bigg] + \frac{1}{k}\frac{\max(u-|x-1|,0)}{u^2}.$$

By the proof of (i), $\overline{v}'(x)$ is constant on $[0,1-u]$ (since the last term is zero on that interval) and equal to $\frac{k-1}{k}\frac{2(k-1)u-(1-u)}{u^2k(k-1)} = \frac{(2k-1)u-1}{u^2k^2}$. On $[1-u,(k-1)u]$, the only active terms are a_{k-2} in S_L, a_0 in S_R and the last term.

$$\begin{aligned}\overline{v}'(x) &= \frac{k-1}{k}\left[\frac{((k-1)u-x)}{u^2}\frac{1}{k(k-1)} + \frac{1-x}{u^2}\frac{k-1}{k(k-1)}\right] + \frac{1}{k}\frac{x+u-1}{u^2} \\ &= \frac{((k-1)u-x)+(k-1)(1-x)+k(x+u-1)}{k^2u^2} = \frac{(2k-1)u-1}{k^2u^2}.\end{aligned}$$

Finally, on $[(k-1)u,1]$, only a_0 in S_R and the last term are active.

$$\overline{v}'(x) = \frac{k-1}{k}\frac{1-x}{u^2}\frac{k-1}{k(k-1)} + \frac{1}{k}\frac{x+u-1}{u^2} = \frac{(k-1)(1-x)+k(x+u-1)}{k^2u^2} = \frac{x+ku-1}{k^2u^2}.$$

So against $\tilde{G}$, $\overline{v}'(x)$ is constant on $[0,(k-1)u]$ and increasing linear on $[(k-1)u,1]$, while against the proper u-partition of $[0,a]$ it is constant on $[0,a]$ and convex on $[a,1]$ by (i). Hence, if the other agent plays G, $\overline{v}(x)$ is linear on $[0,a]$ and $\overline{v}'(x)$ is convex on $[a,1]$, and as in (iii), if a solves $(1-a)\overline{v}(0)+a\overline{v}(1) = \overline{v}(a)$, then $[0,a]\cup\{1\}$ is the concavification support, and (G,G) is an equilibrium.

It remains to show that such an $a \in (1-u,(k-1)u)$ exists and is unique, and that $\beta \in (0,1)$. Let a_3 and $k_3 = \lceil \frac{2-2p}{u} \rceil - 1$ be a and k from (iii). Since $a_3 \leq k_3 u$ and

$k_3 \leq k-1$, $a_3 > 1-u > (k-2)u$ requires $k_3 = k-1$, and then $a_3 > 1-u$ is equivalent to $p < p_u := 1 - \frac{k(k-1)u^2}{2((2k-1)u-1)}$. Let χ be the proper u-partition of $[0,a]$ and, for a signal H of the other agent, let s_H be the slope of $\overline{v}(x)$ on $[0,a]$ and $C_H := \overline{v}(1) - \overline{v}(0)$, so that the condition on a is $s_G = C_G$. A direct computation gives

$$s_{\tilde{G}} - C_{\tilde{G}} = -\frac{((k-1)u-1)^2}{2k^2u^2}, \qquad \frac{d}{da}(s_\chi - C_\chi) = -\frac{1-a}{(k-1)u^2},$$

and $s_\chi = \frac{2(k-1)u-a}{u^2k(k-1)}$ by (i). Hence

$$\Lambda(a) := (Q-ka)(s_G - C_G) = -(2p-a)\frac{((k-1)u-1)^2}{2ku^2} + (Q-2kp)(s_\chi - C_\chi)$$

has the sign of $s_G - C_G$, as $Q - ka \geq (k+1)(1-(k-1)u) > 0$, and $\Lambda(a)$ is quadratic in a with the leading coefficient $\frac{Q-2kp}{2(k-1)u^2}$. Since $p < p_u$ and $\frac{Q}{2k} - p_u = \frac{(k-1)((k-1)u-1)^2}{2k((2k-1)u-1)} > 0$, $2kp < Q$ and Λ is strictly convex. Moreover,

$$\Lambda(1-u) = \frac{(1+u)((2k-1)u-1)}{k(k-1)u^2}(p_u - p) > 0,$$
$$\Lambda((k-1)u) = -\frac{(k+1)(2p-1)((k-1)u-1)^2}{2ku^2} < 0,$$

so Λ has a unique root in $(1-u,(k-1)u)$. And $a < 1 < 2p$ and $2kp < Q$ give $\beta \in (0,1)$.

B.15 Proof of Proposition 14

Let $X = s_1$ and $Y = s_2$. The value functions are $\overline{v}_1(x) = \mathbb{P}(Y \leq x+b)$ and $\overline{v}_2(y) = \mathbb{P}(X+b < y)$. In cases (i) and (ii), let M, c and d be as in the statement.

Equilibrium. Against Y, $\overline{v}_1(x) = 1 - cM + cx$ on $[0,M]$ and $\overline{v}_1(x) = 1$ on $[M,1]$, so $\overline{v}_1$ is concave and linear on $[0,M] \supseteq supp(X)$, and X is a best reply. Against X, $\overline{v}_2(y) = 0$ on $[0,b]$ and $\overline{v}_2(y) = bd + d(y-b) = dy$ on $(b,b+M]$, where in case (ii) the atom of X at $1-b$ is not beaten by $y=1$. In case (i), $d(b+L) = 1$, so $\overline{v}_2(y) = 1 \leq dy$ on $[b+L,1]$. Hence

$\overline{v}_2$ lies weakly below the line dy and coincides with it on $\{0\} \cup (b, b+M] \supseteq supp(Y)$, so Y is a best reply. By construction, both signals have mean p, and are well defined: in case (i), $1 - cL = 1 - \frac{2p}{2b+L} \geq 0$ as $L \geq 2p$, and $b + L \leq 1$ if and only if $b \leq 1 - \sqrt{2p}$; in case (ii), $1 - \frac{2p}{1+b} \geq 0$ as $p \leq 0.5$, and $d \in [0,1]$ if and only if $1 - \sqrt{2p} \leq b \leq 1 - p$.

Uniqueness. As the game is constant-sum, all equilibria form a product set. Let X' be an equilibrium signal of agent 1. Then Y is a best reply to X', and since $supp(Y) \supseteq \{0\} \cup [b, b+M]$, $\mathbb{P}(X' < y - b) = \kappa y$ on $(b, b+M]$ for some κ, i.e. X' has an atom κb at 0 and density κ on $(0, M)$. Since X' is also a best reply to Y and $p < M$, $supp(X') \subseteq [0, M]$, and the mass and the mean of X' pin down κ and its atom at M, hence $X' = X$. Let Y' be an equilibrium signal of agent 2. Since Y' is a best reply to X, it puts no mass outside $\{0\} \cup (b, b+M]$, and since X is a best reply to Y' and $supp(X) \supseteq [0, M]$, $\mathbb{P}(Y' \leq x + b)$ is linear on $[0, M]$. Hence Y' has an atom at 0 and constant density on $(b, b+M]$, both pinned down by its mass and mean, so $Y' = Y$.

Utility of the firm. The firm receives X unless $Y > X + b$, so

$$\begin{aligned} u_F &= p + \mathbb{E}[(Y - X)\mathbb{1}\{Y > X + b\}] = p + c \int_b^{b+M} \int_{[0, y-b)} (y - x) dX \, dy \\ &= p + cd \int_0^M \left(\frac{t^2}{2} + 2bt + b^2 \right) dt, \end{aligned}$$

which gives the expression in the statement. Finally, if $b \geq 1 - p$, agent 1 is selected with probability one by pooling, since $p + b \geq 1 \geq s_2$, so he is selected with probability one in every equilibrium, and $u_F = \mathbb{E}[s_1] = p$.

B.16 Proof of Corollary 9

(i) If $p < 0.5$, case (i) of Proposition 14 applies for small b, and

$$u_F = p + \frac{2p}{(2b + L)(b + L)} \left(\frac{L^2}{6} + bL + b^2 \right).$$

At $b = 0$, $L = 2p$ and $\frac{\partial L}{\partial b} = 1$, which gives $\frac{\partial u_F}{\partial b} = 0.5$. If $p = 0.5$, case (ii) of Proposition 14 applies for all $b > 0$, and

$$u_F = p + \frac{2p(1-b-p)(1+4b+b^2)}{3(1-b)(1+b)^2},$$

which gives $\frac{\partial u_F}{\partial b} = \frac{1}{6}$ at $b = 0$.

(ii) For $b = 1 - \sqrt{2p}$, $L = 1 - b = \sqrt{2p}$, and

$$u_F = p + \frac{2p}{2-\sqrt{2p}}\left(\frac{p}{3} + (1-\sqrt{2p})\sqrt{2p} + (1-\sqrt{2p})^2\right) = 2p + o(p),$$

while $u_F^*(p) = 2p - p^2$, so the share of the first best under the optimal head start converges to 1. For every well S-shaped selection rule, Theorem 2 gives $\frac{u_F}{u_F^*} \leq \frac{1+2p(1-p)}{2-p} \to 0.5$.

B.17 Proof of Proposition 15

(i) Without noise, $a(s) = \mathbb{1}_{s \geq c}$ lies below $\frac{s}{c}$ on $[0,1]$, touching it only at 0 and c, so the agent's unique best reply is $\{0, c\}$, and $W = \frac{p}{c}(c - c) = 0$. With noise, the split $\{0, c\}$ is feasible since $p < c$, and it gives the agent $(1-\frac{p}{c})a(0) + \frac{p}{c}a(c) \geq \frac{p}{2c}$, as $a(c) = 1 - F(0) = 0.5$ by continuity. Hence $\mathbb{E}_G[a(s)] \geq \frac{p}{2c}$ in equilibrium, and since $a \leq 1$ and $s \geq 0$,

$$W = \mathbb{E}_G[a(s)s] - c\,\mathbb{E}_G[a(s)] \leq p - c \cdot \frac{p}{2c} = \frac{p}{2}.$$

Also $W \leq \mathbb{E}_G[(s-c)^+] \leq p(1-c)$, since $(s-c)^+ \leq s(1-c)$ on $[0,1]$.

(ii) Here $a(s) = 0$ on $[0, c-u]$, $a(s) = \frac{s-c+u}{2u}$ on $[c-u, c+u]$, and $a(s) = 1$ on $[c+u, 1]$. Let $l(s) := \frac{s}{c+u}$. Then $l > a$ on $(0, c-u]$, $l - a$ is affine on $[c-u, c+u]$, positive at $c-u$ and zero at $c+u$, and $l > a$ on $(c+u, 1]$. So l supports the concavification of a and touches a only at 0 and $c+u$, and by Kamenica and Gentzkow (2011), the agent's unique best reply is $\{0, c+u\}$, with $\mathbb{P}(s = c+u) = \frac{p}{c+u}$. As $a(0) = 0$ and

$a(c+u) = 1$, $W = \frac{p}{c+u}(c+u-c) = \frac{pu}{c+u}$. If $c > 0.5$ and $u = 1-c$, then $u < c$, $c+u = 1$, and $W = p(1-c)$.

(iii) Since $a \leq 1$ and $(s-c)^+ \leq s(1-c)$ with equality only at $s \in \{0,1\}$, $W = p(1-c)$ requires $G = \{0,1\}$, $a(0) = 0$ and $a(1) = 1$. Full revelation is a best reply only if a lies below its chord from $(0,0)$ to $(1,1)$, i.e., $a(s) \leq s$ on $[0,1]$, and at $s = c$ this gives $0.5 \leq c$.

C. Costly signals

Suppose that signals are costly: agent i pays $\kappa\, c(G_i)$, where $\kappa > 0$ and $c(G_i) = \int_0^1 \phi\, dG_i - \phi(p_i)$ for a strictly convex ϕ, continuous on $[0,1]$, so that his payoff is $\mathbb{E}[v(s_i - s_j)] - \kappa\, c(G_i)$. The firm does not bear the cost, and her utility is unchanged.

Lemma 14. *For any priors and selection rule, equilibria exist and form a product set.*

Proof. Since agent i's cost does not depend on G_j, both agents have the same best replies as in the zero-sum game in which agent 1 receives $W(G_1, G_2) := \mathbb{E}[v(s_1 - s_2)] - \kappa c(G_1) + \kappa c(G_2)$ and agent 2 receives $-W(G_1, G_2)$. W is affine in each G_i and continuous in the weak* topology, as v and ϕ are continuous, and each $\mathcal{G}(p_i)$ is convex and weak* compact. By Sion's minimax theorem, this game has a value and optimal strategies, and its equilibria are exactly the pairs of optimal strategies. □

Unlike the value of this zero-sum game, the agents' own payoffs need not be the same in all equilibria.

C.1 The variance cost

Let $\phi(x) = x^2$, so that $c(G_i) = \mathrm{Var}_{G_i}[s_i]$ is the expected reduction in the residual variance of θ_i. As the linear part of $v(x)$ adds only a linear function of s_i to agent i's value function, best replies depend on $v(x)$ only through $\tilde{v}(x)$ and

$$\lambda := \frac{\kappa}{|v(1) - 0.5 - v'(0)|},$$

and the normalized value function against G_j is $f(s_i) := \overline{v}(s_i) - \lambda s_i^2$.

Proposition 16. *Let $v(x)$ be well S-shaped, the priors (p, p), and $c(G_i) = Var_{G_i}[s_i]$.*

(i) If $p \geq \overline{p}_\lambda := 1 - \frac{1-\lambda}{\mathcal{E}_{\tilde{v}}(1)}$, the unique equilibrium is $(\{0,1\};\{0,1\})$.

(ii) If $\lambda \geq \frac{|\tilde{v}(p)|}{p^2}$, the unique equilibrium is $(\{p\};\{p\})$.

(iii) Otherwise, the unique equilibrium is $(\{0,a\};\{0,a\})$, where $a \in (p,1)$ solves $p = a - \frac{|\tilde{v}(a)|-\lambda a^2}{|\tilde{v}'(a)|}$, and $\mathbb{P}(s_i = 0) = \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\left(1 - \frac{\lambda a^2}{|\tilde{v}(a)|}\right)$.

The conditions in (i) and (ii) are mutually exclusive.

For $\lambda = 0$, (ii) never applies, and Proposition 16 reduces to Theorem 1. Since λ is the cost relative to the curvature of $v(x)$, noise that flattens $v(x)$ makes the same cost weigh more. For instance, for $\varepsilon_i \sim U[0,u]$ with $u \geq 1$, $\tilde{v}(x) = -x|x|$ and $\lambda = 2\kappa u^2$: agents reveal fully if $p \geq \frac{1+\lambda}{2}$, pool if $\lambda \geq 1$, and otherwise play $\{0, \frac{2p}{1+\lambda}\}$ with $\mathbb{P}(s_i = 0) = \frac{1-\lambda}{2}$. Without the cost, informativeness does not depend on $u \geq 1$; with it, it decreases in u.

Proof. Exclusivity. By the concavity of $\tilde{v}(x)$ on $[0,1]$ (property 2 in Appendix A.1), $\tilde{v}(p) \leq \tilde{v}(1) + (p-1)\tilde{v}'(1) = -1 + (1-p)\mathcal{E}_{\tilde{v}}(1)$. So if (i) holds, $\lambda \leq 1 - (1-p)\mathcal{E}_{\tilde{v}}(1) \leq |\tilde{v}(p)| < \frac{|\tilde{v}(p)|}{p^2}$.

Uniqueness. By Lemma 14 and symmetry, all equilibria form a product set $E \times E$. So if $G^* \in E$ is the unique best reply to itself, then $E = \{G^*\}$, as every $G \in E$ is a best reply to G^*.

Best reply to $\{0,a\}$. Let agent j play $\{0,a\}$ with $a \in (p,1]$, and let $\gamma := \frac{a-p}{a} \in (0,1)$. Then

$$f(s) = \gamma\tilde{v}(s) + (1-\gamma)\tilde{v}(s-a) - \lambda s^2.$$

Let $l(s)$ be the line through $(0, f(0))$ and $(a, f(a))$, whose slope is $\frac{\tilde{v}(a)-\lambda a^2}{a}$, and $d(s) :=$

$l(s) - f(s)$ on $[0, a]$, so that $d(0) = d(a) = 0$. As $\tilde{v}'(x)$ is even,

$$d'(s) = \frac{\tilde{v}(a) - \lambda a^2}{a} - \gamma \tilde{v}'(s) - (1-\gamma)\tilde{v}'(a-s) + 2\lambda s,$$

which is convex on $[0, a]$, since $\tilde{v}'(x)$ is concave. If $d'(a) \leq 0$, then $d \geq 0$ on $[0, a]$: with $z := \inf\{s \in [0, a] : d'(s) \leq 0\}$, convexity gives $d' \leq 0$ on $[z, a]$, while $d' > 0$ on $[0, z)$. If $a < 1$ and $d'(a) = 0$, then for $s \in (a, 1]$,

$$f'(s) = \gamma \tilde{v}'(s) + (1-\gamma)\tilde{v}'(s-a) - 2\lambda s < \gamma \tilde{v}'(a) + (1-\gamma)\tilde{v}'(0) - 2\lambda a = f'(a),$$

as $\tilde{v}'(x)$ is strictly decreasing on $[0, 1]$, so f lies strictly below its tangent l on $(a, 1]$. Hence, if $d'(a) \leq 0$ for $a = 1$, or $d'(a) = 0$ for $a < 1$, then $\{0, a\}$ is a best reply to itself. Since $f'(a) = \gamma \tilde{v}'(a) - 2\lambda a$,

$$d'(a) = 0 \;\Leftrightarrow\; \gamma \tilde{v}'(a) = \frac{\tilde{v}(a)}{a} + \lambda a \;\Leftrightarrow\; \gamma = \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\left(1 - \frac{\lambda a^2}{|\tilde{v}(a)|}\right) \;\Leftrightarrow\; p = a - \frac{|\tilde{v}(a)| - \lambda a^2}{|\tilde{v}'(a)|},$$

where the last step uses $\gamma a = a - p$.

Step 1: other best replies. Suppose that $d'(a) = 0$ and $d(c) = 0$ for some $c \in (0, a)$. Then c minimizes d on $[0, a]$, so $d'(c) = 0$, and as in the proof of Theorem 1, convexity and $d(0) = d(c) = d(a) = 0$ give $d' \equiv 0$ on $[0, a]$. So $\gamma \tilde{v}'(s) + (1-\gamma)\tilde{v}'(a-s)$ is affine on $[0, a]$; as a sum of two concave functions, it is affine only if both are, so $\tilde{v}'(s) = -Ks$ on $[0, a]$ for some $K > 0$, and $d' \equiv 0$ then requires $K(1-2\gamma) = 2\lambda$, i.e., $\frac{p}{a} = 1 - \gamma = \frac{1}{2} + \frac{\lambda}{K}$. Hence, either $d > 0$ on $(0, a)$, and $\{0, a\}$ is the unique best reply to itself, or $\tilde{v}(x) = -\frac{K}{2}x|x|$ on $[-a, a]$ and $p = a\left(\frac{1}{2} + \frac{\lambda}{K}\right)$, and every $G \in \mathcal{G}(p)$ supported on $[0, a]$ is a best reply to $\{0, a\}$.

Step 2: the quadratic case. Take such a $G \neq \{0, a\}$ and let $w(s) := \int_0^a \tilde{v}(s-t)dG(t) - \lambda s^2$ be the normalized value function against it. For $s \in [0, a]$, $w'(s) = -K\int_0^a |s -$

$t|dG(t) - 2\lambda s$, hence $w'(a) = -K(a-p) - 2\lambda a$, while $w(0) = \frac{K}{2}\mathbb{E}_G[t^2]$ and $w(a) = -\frac{K}{2}\mathbb{E}_G[(a-t)^2] - \lambda a^2$. So the slope of the secant from $(0, w(0))$ to $(a, w(a))$ exceeds $w'(a)$ by

$$\frac{K}{a}\Big(\frac{a^2}{2} - \mathbb{E}_G[t^2]\Big) + \lambda a > \frac{K}{a}\Big(\frac{a^2}{2} - ap\Big) + \lambda a = 0,$$

since $t^2 \le at$ on $[0, a]$, with equality only at $t \in \{0, a\}$, and $p = a\big(\frac{1}{2} + \frac{\lambda}{K}\big)$. Hence $w(s)$ lies strictly above the secant for $s < a$ close enough to a, so, for such $s > p$, $\{0, s\}$ yields a strictly higher payoff against G than $\{0, a\}$. Thus $\{0, a\}$ is not a best reply to G, $G \notin E$, and the equilibrium is unique.

(i) For $a = 1$, $\gamma = 1 - p$ and $d'(1) = -1 + \lambda + (1-p)\mathcal{E}_{\tilde{v}}(1)$, so $d'(1) \le 0$ if and only if $p \ge \overline{p}_\lambda$, and then $\{0, 1\}$ is a best reply to itself. If $p > \overline{p}_\lambda$, then $d > 0$ on $(0, 1)$: otherwise, $d(c) = 0$ for some $c \in (0, 1)$ gives $d'(c) = 0$, so $d' \le 0$ on $[c, 1]$ by convexity, and $\int_c^1 d'(s)ds = 0$ gives $d' \equiv 0$ on $[c, 1]$, contradicting $d'(1) < 0$. So $\{0, 1\}$ is the unique best reply to itself, and the equilibrium is unique. If $p = \overline{p}_\lambda$, then $d'(1) = 0$, and Steps 1 and 2 with $a = 1$ give uniqueness.

(ii) Against $\{p\}$, $f(s) = \tilde{v}(s-p) - \lambda s^2$, and $f'(p) = -2\lambda p$. So $\{p\}$ is a best reply if and only if f lies below its tangent at p, i.e., if and only if $\tilde{v}(s-p) \le \lambda(s-p)^2$ on $[0, 1]$. For $s > p$, this holds strictly, as $\tilde{v} < 0$ on $(0, 1]$. For $s < p$, with $t = p - s$, it reads $\frac{|\tilde{v}(t)|}{t^2} \le \lambda$, and as this ratio is non-decreasing on $(0, 1]$ by Lemma 9, it holds for all $t \in (0, p]$ if and only if $\lambda \ge \frac{|\tilde{v}(p)|}{p^2}$. Then every best reply to $\{p\}$ is supported on $[0, p]$ and has mean p, so it is $\{p\}$, and the equilibrium is unique.

(iii) Let $p < \overline{p}_\lambda$ and $\lambda < r(p)$, where $r(t) := \frac{|\tilde{v}(t)|}{t^2}$ is continuous and non-decreasing on $(0, 1]$. Let $h(a) := a - \frac{|\tilde{v}(a)| - \lambda a^2}{|\tilde{v}'(a)|} = a\big(1 - \frac{1 - \lambda/r(a)}{\mathcal{E}_{\tilde{v}}(a)}\big)$, continuous on $(0, 1]$, with $h(1) = \overline{p}_\lambda > p$. Let $a_0 := \sup\{t \in (0, 1] : r(t) \le \lambda\}$, and $a_0 := 0$ if this set is empty; by continuity and monotonicity of r, $a_0 < p$. If $a_0 > 0$, then $r(a_0) = \lambda$ and $h(a_0) = a_0 < p$. If $a_0 = 0$, then $\lambda < r(a)$ for all $a \in (0, 1]$, so $h(a) \in [\frac{a}{2}, a)$ by Lemma 9, and $h(a) \to 0$ as $a \to 0$. In both cases, by continuity, $h(a) = p$ for some $a \in (a_0, 1)$. As $r(a) > \lambda$ there, $p = h(a) < a$, so

$\gamma \in (0,1)$, and $d'(a) = 0$ by the equivalence above. Hence $\{0, a\}$ is a best reply to itself, with $\mathbb{P}(s_i = 0) = \gamma$ as stated, and Steps 1 and 2 give uniqueness. □

C.2 Robust implementation

For the rules of Section 5, let ϕ be continuously differentiable on $[0,1]$, including the endpoints, and let $\psi(s) := (1-s)\phi(0) + s\phi(1) - \phi(s)$, which is positive on $(0,1)$ by strict convexity, with $\lim_{s\to 0} \frac{\psi(s)}{s} = \phi(1) - \phi(0) - \phi'(0) > 0$ and $\lim_{s\to 1} \frac{\psi(s)}{1-s} = \phi'(1) - \phi(1) + \phi(0) > 0$.

Proposition 17. *Let ϕ be continuously differentiable on $[0,1]$.*

(i) If $v(x) = 0.5 + ax$, the unique equilibrium is $(\{p_1\}; \{p_2\})$ for all (p_1, p_2).

(ii) Let $v(x)$ be strictly positively [negatively] balanced with $\liminf_{s\to 0_+} \frac{|g(s)|}{s} > 0$. Then $m := \inf_{s\in(0,1)} \frac{|g(s)|}{\psi(s)} > 0$, and full revelation is the unique equilibrium for all (p_1, p_2) with $p_1, p_2 > 0.5 + \frac{\kappa}{2m}$ $\left[p_1, p_2 < 0.5 - \frac{\kappa}{2m}\right]$.

(iii) Let $v(x)$ be positively or negatively balanced with $\liminf_{s\to 0_+} \frac{|g(s)|}{s} = 0$. Then $(\{0,1\}; \{0,1\})$ is not an equilibrium for any (p_1, p_2).

For the variance cost, $\psi(s) = s(1-s)$; for $v^*(x)$, $g(s) = \frac{s(1-s)}{2}$, so $m = \frac{1}{2}$ and (ii) gives full revelation for $p > 0.5 + \kappa$, in line with Proposition 16(i), as $\lambda = 2\kappa$. If instead $\phi'(0) = -\infty$, as for entropy, then $\frac{\psi(s)}{s} \to \infty$ as $s \to 0$, while $\frac{|g(s)|}{s} \le a$ for every balanced rule, so by $(*)$ below, no balanced rule implements full revelation for any $\kappa > 0$.

Proof. (i) Agent i's payoff is $0.5 + a(p_i - p_j) - \kappa c(G_i)$, and by strict convexity, $c(G_i) > 0$ unless $G_i = \{p_i\}$.

(ii)–(iii) Let agent j play $\{0,1\}$. With $\hat{v}(x) = ax + g(x)$ on $[0,1]$ and $g(x) = g(1-x)$, agent i's value function is, up to a constant,

$$(1-p_j)\hat{v}(s) - p_j\hat{v}(1-s) = a(s-p_j) + (1-2p_j)g(s),$$

and he maximizes the expectation of this function minus $\kappa\phi(s)$. By Kamenica and Gentzkow (2011), $\{0,1\}$ is a best reply if and only if this function lies below its chord on $[0,1]$. As $g(0) = g(1) = 0$, this reads

$$(2p_j - 1)g(s) \geq \kappa\psi(s) \quad \forall s \in (0,1), \tag{$*$}$$

which does not depend on p_i; if $(*)$ is strict on $(0,1)$, $\{0,1\}$ is the unique best reply.

(ii) Consider the positive case. The ratio $\frac{g}{\psi}$ is continuous and positive on $(0,1)$, and $\liminf_{s\to 0} \frac{g(s)}{\psi(s)} > 0$ by the limit of $\frac{\psi(s)}{s}$ above. As $g(s) = g(1-s)$, $\liminf_{s\to 1} \frac{g(s)}{1-s} > 0$ too, so $\liminf_{s\to 1} \frac{g(s)}{\psi(s)} > 0$, and hence $m > 0$. If $p_j > 0.5 + \frac{\kappa}{2m}$, then $(2p_j - 1)g(s) \geq (2p_j - 1)m\psi(s) > \kappa\psi(s)$ on $(0,1)$, so $(*)$ is strict. Hence, if both priors exceed $0.5 + \frac{\kappa}{2m}$, full revelation is each agent's unique best reply to itself, and by Lemma 14, it is the unique equilibrium. The negative case is symmetric.

(iii) Consider the positive case. If $p_j \leq 0.5$, the left side of $(*)$ is non-positive, and the right side is positive. If $p_j > 0.5$, take $s_n \to 0$ with $\frac{g(s_n)}{s_n} \to 0$; then $(2p_j - 1)\frac{g(s_n)}{s_n} \to 0$, while $\kappa\frac{\psi(s_n)}{s_n} \to \kappa\big(\phi(1) - \phi(0) - \phi'(0)\big) > 0$, so $(*)$ fails. The negative case is symmetric. □

D. Equations for the noiseless selection rule

Fact (Boleslavsky and Cotton (2015) Lemma 3)**.** *The unique symmetric equilibrium for* $v(x) = 0.5 + 0.5 \cdot sign(x)$ *has the form* (G, G)*, where for* $p \geq 0.5$

$$G(x) = \begin{cases} \frac{x}{2p}, & x \leq 2 - 2p, \\ \frac{1}{p} - 1, & 2 - 2p \leq x < 1, \\ 1, & x = 1, \end{cases}$$

while for $p \leq 0.5$

$$G(x) = \begin{cases} \frac{x}{2p}, & x \leq 2p, \\ 1, & x \in [2p, 1]. \end{cases}$$

Hence the firm's equilibrium utility is

$$u_F^0(p) = \mathbb{E}_G[\max(s_1, s_2)] = \begin{cases} \frac{4p}{3}, & 0 < p \leq 0.5, \\ 1 - \frac{(1-p)^2(4p-1)}{3p^2}, & 0.5 < p < 1. \end{cases}$$